RAJIV GANDHI COLLEGE OF ENGINEERING AND TECHNOLOGY
KIRUMAMPAKKAM, PUDUCHERRY-607403.
(Approved by AICTE and Affiliated to Pondicherry University)
(ISO 9001:2015)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

COURSE FILE

Staff Name	**:**	S.SURIA
Designation	**:**	ASSISTANT PROFESSOR
Academic Year	**:**	2023-2024 (EVEN SEM)
Course	**:**	B. TECH (CSE)
Theory Subject & Code	**:**	ENTERPRISE SOLUTIONS(CS T61)

COLLEGE VISION- MISSION

Vision

To be in the forefront of higher education in order to give India the high caliber manpower she needs.

Mission

- **To provide quality collegiate education from under graduate to post doctoral programmes.**
- **To ensure high standard of behavior and discipline amongst our students community.**
- **To create a climate of joyful learning.**
- **To impart skills in students which will make them successful in their endeavors.**
- **To provide meaningful industrial education, research and training at all levels.**
- **To offer a wide range and flexibility of options especially in the areas of non-formal and continuing education.**
- **To set a high standard of professional conduct and ethics for staff and students alike**

Rajiv Gandhi College of Engineering and Technology

(Approved by AICTE & Affliated to Pondicherry University)

Pondy - Cuddalore main road, Kirumampakkam, Puducherry - 607 403

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

COMPUTER SCIENCE & ENGINEERING

VISION AND MISSION

Vision

Cultivate student as technocrats, researchers and entrepreneurs in the field of computer science and engineering.

Mission

1. To impart quality education in Computer Science and Engineering by means of learning techniques and value-added courses.

2. To inculcate professional excellence and research culture by encouraging projects in cutting-edge technologies through industry interactions.

3. To build leadership skills, ethical values and teamwork among the students.

4. To strengthen the collaboration of department and industry through internships, mentorships and professional body activities.

VI SEMESTER

Subject Code	Subject Name	Lectures (Periods)	Tutorial (Periods)	Practical (Periods)
CS T61	**ENTERPRISE SOLUTIONS**	3	1	-

Course Objective:

1. To make the students to get familiar with the industry project platforms and to write codes.

Course Outcomes:

On successful completion of the subject students will be able to :

1. Understand basic concepts of SAP, Oracle, PeopleSoft and Siebel.
2. Write code in SAP, Oracle, PeopleSoft, and Siebel.
3. Ready to cope up with industrial application development.

UNIT – I

Introduction : ERP - Definition – Concept – Fundamentals – Need for ERP - Advantages of ERP – Implementation of ERP – Key issues and Characteristics of ERP Typical architecture components of ERP – ERP system Architecture. ERP and related technologies: Business Process RE-engineering – Management Information System– Decision Support System - Executive Support System – On-Line Analytical Processing, Supply Chain Management, Customer Relationship Management.

UNIT – II

SAP : History – SAP R/2 – SAP R/3 – Characteristics of SAP R/3 – Architecture of SAP R/3 - SAP Modules, NetWeaver, Customer Relationship Management, Business Warehouse, Advanced Planner and Optimiser. ABAP/4: Workbench - Workbench Tools - ABAP/4 Data Dictionary - ABAP/4 Repository Information – Structure of ABAP/4 program - ABAP/4 syntax – Data types – Constants and Variables – Statements : DATA, PARAMETERS, TABLE, MOVE, MOVE-CORRESPONDING, CLEAR, WRITE, CHECK, FORMAT. LOOP STRUCTURES. Sample programs.

UNIT – III

Oracle Suite : Oracle Apps 11i - Application Framework - File System - Workflow Analysis - SQL / PLSQL fundamentals - Creating Forms - Oracle Reports. Oracle Electronic Data Interchange – functions of EDI – Data File Structure - Oracle Data, Oracle Database - Oracle Database - DW vs OLTP - DW Connectors.

UNIT – IV

PeopleSoft: Basic PeopleSoft Functionality – Opening Multiple Windows - Database structure – Understanding People Soft Data Mover – Records - Pages vs. Forms. PeopleSoft HRMS: Introduction to PeopleSoft HRMS database - PeopleSoft products - Functional PeopleSoft - financial management system - PeopleSoft Enterprise HRMS.

UNIT – V

Siebel Enterprise Applications - Siebel eBusiness Applications – Siebel Tools – Tables and Columns – Business Component – Business Objects – Applets – Joins – Links – Views – Screens – Configuring applications.

TOTAL PERIODS: 60
Text Books: 1. V.K. Garg and N.K. Venkatkrishnan, ERP Concepts and Planning, PHI, 2004. 2. SAP ABAP/4, Black Book, DreamTech Press, 2012. 3. Oracle EDI Gateway User guise, Oracle Corporation. 4. Jim J. Marion, PeopleSoft PeopleTools: Tips and Techniques, Oracle Press, 2010. 5. Rishi Kumar Shrivastava, Siebel CRM 8.1: Navigation and Configuration, TMH, 2012.
Reference Books: 1. Chrispopher Allen, Oracle Database PL/SQL, TMH, 2004. 2. Paula Dean and Jim J. Marion, PeopleSoft PeopleTools: Data Management and Upgrade Handbook, Oracle Press, 2013.

Subject Name ENTERPRISE SOLUTIONS
Subject Code: CS T61

Program Outcomes (POs)

The **Enterprise Solutions** course focuses on introducing students to various enterprise software solutions, including ERP systems like SAP, Oracle, PeopleSoft, and Siebel. Students will gain a comprehensive understanding of the concepts, architecture, and development processes involved in these solutions. They will also develop practical skills in writing code for these platforms, preparing them for real-world industrial application development.

Course Outcomes (COs)

1. **CO1**: Understand basic concepts of SAP, Oracle, PeopleSoft, and Siebel, and their significance in enterprise solutions.
2. **CO2**: Write code in SAP, Oracle, PeopleSoft, and Siebel, demonstrating an ability to work with these enterprise platforms.
3. **CO3**: Be prepared to cope with industrial application development and apply enterprise solutions in real-world scenarios.

Alignment of COs with Pos and PSOs:

The Program Outcomes (POs), Program Specific Outcomes (PSOs) and their corresponding achievements through Course Outcomes (COs) are detailed below.

Program Outcomes (POs) Based on NBA Guidelines:

1. **Engineering Knowledge**
2. **Problem Analysis**
3. **Design/Development of Solutions**
4. **Conduct Investigations of Complex Problems**
5. **Modern Tool Usage**
6. **The Engineer and Society**
7. **Environment and Sustainability**
8. **Ethics**
9. **Individual and Team Work**

10. **Communication**
11. **Project Management and Finance**
12. **Lifelong Learning**

Program Specific Outcomes (PSOs):

- **PSO1:** Apply foundational principles of computer science, engineering, and mathematics to develop computational systems and solve complex problems.
- **PSO2**: Design and implement software solutions for diverse real-world problems using modern technologies and programming practices.
- **PSO3**: Demonstrate expertise in specialized areas such as artificial intelligence, data science, cybersecurity, and software development to contribute to research and industry advancements.

Explanation of How POs Are Achieved

1. **CO1**: Understand basic concepts of SAP, Oracle, PeopleSoft, and Siebel, and their significance in enterprise solutions.
    - **PO1 (Engineering Knowledge)**: Achieved through gaining foundational knowledge of ERP systems and their architectures.
    - **PO5 (Modern Tool Usage)**: Supported by using and understanding the tools and components of ERP systems like SAP and Oracle.
    - **PSO1**: Learning and understanding ERP systems prepares students for working with enterprise solutions.
2. **CO2**: Write code in SAP, Oracle, PeopleSoft, and Siebel, demonstrating an ability to work with these enterprise platforms.
    - **PO3 (Design/Development of Solutions)**: Supported by coding exercises where students develop solutions using various ERP platforms.
    - **PO5 (Modern Tool Usage)**: Achieved by using tools like ABAP/4 for SAP and SQL/PLSQL for Oracle to write code for enterprise applications.
    - **PSO2**: Writing code for ERP platforms demonstrates the ability to apply theoretical knowledge to develop enterprise solutions.
3. **CO3**: Be prepared to cope with industrial application development and apply enterprise solutions in real-world scenarios.

- **PO3 (Design/Development of Solutions)**: Supported by project-based learning where students design, implement, and deploy enterprise applications.
- **PO6 (The Engineer and Society)**: Achieved through understanding and solving real-world problems using ERP systems that improve business processes.
- **PSO3**: This CO directly supports applying ERP knowledge in industrial settings, making students industry-ready.

CO-PO-PSO Matrix

Course Outcomes (COs)	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	2	2	1	0	0	0	0	0	0	3	1	-
CO2	3	2	3	1	2	2	0	0	0	0	0	2	3	2	-
CO3	3	3	3	2	3	3	0	0	0	0	0	3	3	3	-

Explanation of the CO-PO-PSO Matrix

- **PO1 (Engineering Knowledge)**: All three COs require foundational knowledge of ERP systems, particularly SAP, Oracle, PeopleSoft, and Siebel, with CO1 and CO2 strongly aligning to this outcome.
- **PO3 (Design/Development of Solutions)**: Strongly supported by CO2 and CO3, where students design and implement real-world enterprise solutions.
- **PO5 (Modern Tool Usage)**: Supported by CO1 and CO2, which involve using modern ERP tools and platforms like SAP ABAP/4, Oracle SQL/PLSQL, and Siebel for practical coding.
- **PO6 (The Engineer and Society)**: Achieved through CO3, where students apply ERP knowledge to solve real-world business challenges in industry.
- **PSO1**: Learning ERP systems and their applications is central to this outcome, helping students design enterprise solutions using SAP, Oracle, PeopleSoft, and Siebel.
- **PSO2**: Writing code and understanding various ERP tools aligns with students' development in integrating ERP systems in enterprise solutions.

- **PSO3**: CO3 ensures that students are ready for industry, applying the ERP system concepts and tools to real-world applications.

RAJIV GANDHI COLLEGE OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE AND TECHNOLOGY
LESSON PLAN - EVEN SEMESTER
III YEAR/VI SEM - 2023-2024

SUBJECT : ENTERPRISE SOLUTIONS
HANDLED BY : S.SURIA **SEC - A**

WEEK	UNIT	TOPIC	TEXT BOOK	PROPOSED DATE	ACTUAL DATE	REMARKS
		INTRODUCTION				
1st week	I	ERP - Definition - Concept - Fundamentals - Need for ERP - Advantages of ERP - Implementation of ERP - Key issues and Characteristics of ERP Typical architecture components of ERP ERP system Architecture	T1	22.01.2024 to 25.01.2024	23.1.24 to 1.2.24	
2nd week		ERP and related technologies Business Process RE-engineering – Management Information System– Decision Support System - Executive Support System – On-Line Analytical Processing, Supply Chain Management, Customer Relationship Management	T1	29.01.2024 to 02.02.2024	2.2.24 to 13.2.24	13/02/24
		SAP				
3rd week	II	SAP History – SAP R/2 – SAP R/3 – Characteristics of SAP R/3 – Architecture of SAP R/3 - SAP Modules, NetWeaver, Customer Relationship Management, Business Warehouse, Advanced Planner and Optimiser	T2	05.02.2024 to 10.02.2024	14.2.24 to 19.2.24	
4th week		ABAP/4 Workbench - Workbench Tools - ABAP/4 Data Dictionary - ABAP/4 Repository Information – Structure of ABAP/4 program - ABAP/4 syntax – Data types	T2	12.02.2024 to 16.02.2024	21.2.24 to 9.3.24	19/3/24
5th week		Constants and Variables - Statements DATA, PARAMETERS, TABLE, MOVE, MOVE-CORRESPONDING, CLEAR, WRITE, CHECK, FORMAT LOOP STRUCTURES Sample programs	T2	19.02.2024 to 24.02.2024	11.3.24 to 18.3.24	
		ORACLE SUITE				
6th week	III	Oracle Suite Oracle Apps 11i - Application Framework - File System - Workflow Analysis - SQL / PLSQL fundamentals - Creating Forms - Oracle Reports	T3	26.02.2024 to 01.03.2024	19.3.24 to 23.3.24	
7th week		Oracle Electronic Data Interchange – functions of EDI – Data File Structure - Oracle Data, Oracle Database - Oracle Database - DW vs OLTP - DW Connectors	T3	04.03.2024 to 07.03.2024	26.3.24 to 13.4.24	13/04/24
		PEOPLE SOFT				
8th week	IV	PeopleSoft Basic PeopleSoft Functionality – Opening Multiple Windows - Database structure – Understanding People Soft Data Mover – Records - Pages vs Forms	T4	09.03.2024 to 15.03.2024	16.4.24 to 22.4.24	
9th week		PeopleSoft HRMS Introduction to PeopleSoft HRMS database - PeopleSoft products - Functional PeopleSoft - financial management system - PeopleSoft Enterprise HRMS	T4	18.03.2024 to 29.03.2024	23.4.24 to 26.4.24	27/4/24
		APPLICATIONS				
10th week	V	Siebel Enterprise Applications - Siebel eBusiness Applications – Siebel Tools – Tables and Columns – Business Component	T5	01.04.2024 to 05.04.2024	27.4.24 to 2.5.24	
11th week		Business Objects – Applets – Joins – Links – Views – Screens – Configuring applications	T5	08.04.2024 to 19.04.2024	3.5.24 to 4.5.24	04/05/24

S.NO	TEXT BOOK	AUTHOR NAME	PUBLICATION
1	ERP Concepts and Planning	V K Garg and N K	ERP Concepts and Planning, PHI, 2004
2	SAP ABAP 4, Black Book		DreamTech Press, 2012
3	Oracle EDI Gateway User guise		Oracle Corporation
4	PeopleSoft PeopleTools Tips and Techniques	Jim J Marion	1 Oracle Press, 2010
5	Siebel CRM 8.1 Navigation and Configuration,	Rishi Kumar Shrivastava	TMH, 2012

S. Suria
SIGNATURE OF STAFF

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DEPARTMENT OF CSE

CS T61 ENTERPRISE SOLUTIONS

Unit-I – ERP AND ITS RELATED TECHNOLOGIES

Introduction : ERP - Definition – Concept – Fundamentals – Need for ERP - Advantages of ERP – Implementation of ERP – Key issues and Characteristics of ERP Typical architecture components of ERP – ERP system Architecture. **ERP and related technologies**: Business Process RE-engineering – Management Information System– Decision Support System - Executive Support System – On-Line Analytical Processing, Supply Chain Management, Customer Relationship Management.

PART A

1. **Define ERP.**
    - Enterprise Resource Planning (ERP) is business management software.
    - It typically a suite of integrated applications that a company can use to collect, store, manage and interpret data from many business activities, including product planning, cost , Manufacturing or service delivery, Marketing and sales, Inventory.
2. **What is enterprise solutions?**
    - Enterprise Solutions provide for a scalable, easy to manage programming solution to providing business management and information accessibility for internal and external clients.
    - Enterprise solutions deal with the problem of providing information to clients both externally and internally. It deals with programming and databases.
    - The main problem being how to most efficiently get our data available to those we want to access it. The solution have the following characteristics and more: Security, Scalability, Cost, and Management Portable.
3. **Describe about Business Intelligence.**
    - Business Intelligence (BI) has become a standard component of most ERP packages.
    - In general, BI tools allow user to share and analyze the data collected the enterprise and centralized in the ERP database.
    - BI can come in the form of dashboards, automated reporting and analysisis tools used to monitor the organization as business performance. BI supports informed decision making by everyone, from executives to line managers and accountants.

4. List out the Characteristics of ERP.

The characteristics of ERP are

- Adaptability
- Openness
- Integration
- Completeness
- Homogenization
- Transversality
- Best practices
- Real time
- Simulation

5. What is the need of ERP?

- Separate systems were being maintained during 1960/70 for traditional business functions like Sales & Marketing, Finance, Human Resources, Manufacturing, and Supply Chain Management.
- These systems were often incongruent, hosted in different databases and required batch updates. It was difficult to manage business processes across business functions e.g. procurement to pay and sales to cash functions.
- ERP system grew to replace the islands of information by integrating these traditional business functions.

6. List the advantages of ERP.

- Complete **visibility** into all the important processes, across various departments of an organization (especially for senior management personnel).
- Automatic and coherent **workflow** from one department/function to another, to ensure a smooth transition and quicker completion of processes. This also ensures that all the inter-departmental activities are properly tracked and none of them is 'missed out'.
- A unified and single **reporting** system to analyze the statistics/status etc. in real-time, across all functions/departments.

7. What are the fundamental components of ERP?

The following are the fundamental components of the ERP.

- Financial Management
- Business Intelligence
- Supply Chain Management

- Human Resource Management
- Manufacturing Operations
- Integration

8. List out the elements of ERP.

The following are the elements of ERP.

- Production planning
- Integrated logistics
- Human resources
- Accounting and financials
- Sales, distribution (order entry)

9. Why companies are forced to have ERP suite?

The forces driving ERP are;

- The need to create a framework that will, improve customer order processing.
- The need to consolidate and unify business functions such as manufacturing finance distributions/logistics, & human resources.
- The need to integrate a broad range of desperate technologies, along with the processes they support.
- The need to create a new foundations and which next-generation applications can be developed.

10. In what way ERP is used in real world?

- Microsoft in the software industry
- Owens-Corning in the building supplies industry.
- Colgate-Palmolive in the consumer products industry

11. List the implementation methodology phases of accelerated ERP approach.

The Project Preparation Phase

- The Blueprint Phase
- The Pilot Phase
- The Assessment Phase
- The Final Phase

12. What are the capabilities of effective co-ordination management?

The capabilities of effective co-ordination managements are,

- Strategic Thinking
- Process Reengineering
- Managing Implementation Complexity
- Transition Management

13. Mention the elements required in ERP to achieve flexibility.

Four crucial elements are required to achieve flexibility:

1. Components, not modules.
2. Incremental migration, rather than massive reengineering
3. Dynamic, rather than static, configuration of ERP systems.
4. Management of multiple strategic sourcing and partnership relationships.

14. What is Two Tier architecture?

- In typical two-tier architecture, the server handles both application and database duties.
- The clients are responsible for presenting the data and passing user input back to the server.
- While there may be multiple servers and the clients may be distributed across several types of local and wide area links, this distribution of processing responsibilities remains the same.

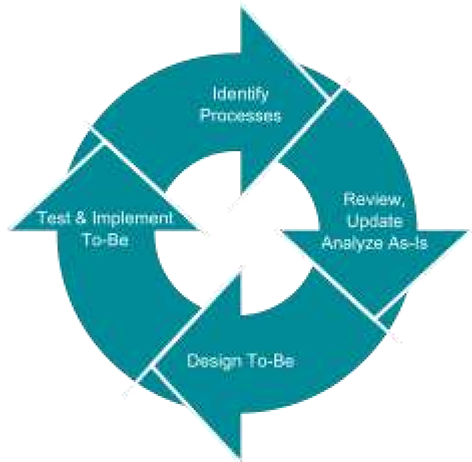
15. What is Business Process Reengineering (BPR)?

- Business process re-engineering: Business process re-engineering (BPR) is the analysis and redesign of workflows within and between enterprises in order to optimize end-to-end processes and automate non-value-added tasks.
- Business Process Reengineering (BPR) is the practice of rethinking and redesigning the way work is done to better support an organization's mission and reduce costs.
- Reengineering starts with a high-level assessment of the organization's mission, strategic goals, and customer needs.

16. Define Management Information System.

A management information system (MIS) is a computerized database of financial information organized and programmed in such a way that it produces regular reports on operations for every level of management in a company.

17. Depict the neat diagram of Business Process Reengineering cycle.



Business Process Reengineering Cycle

18. List out the advantages of BPR.

Advantages of BPR are

- Satisfactions
- Growth of knowledge
- Solidarity of the company
- Demanding jobs
- Authority

19. Write a note on Decision Support System (DSS).

- Decision support system (DSS) is a computer-based information system that supports business or organizational decision-making activities.
- DSSs serve the management, operations, and planning levels of an organization (usually mid and higher management) and help people make decisions about problems that may be rapidly changing and not easily specified in advance.
- Unstructured and Semi-Structured decision problems.
- Decision support systems can be either fully computerized, human-powered or a combination of both.

20. What are the various phases are available in DSS.

Decision Support System consists of four phases:

- **Intelligence**: Searching for conditions that call for decision.
- **Design**: Developing and analyzing possible alternative actions of solution.
- **Choice**: Selecting a course of action among those.
- **Implementation**: Adopting the selected course of action in decision situation.

21. Define Executive Support System.

- An executive information system (EIS), also known as an executive support system (ESS), is a type of management information system that facilitates and supports senior executive information and decision-making needs.
- It provides easy access to internal and external information relevant to organizational goals.
- It is commonly considered a specialized form of decision support system (DSS).
- EIS emphasizes graphical displays and easy-to-use user interfaces.

22. What is On-Line Analytical Processing (OLAP)?

- OLAP (Online Analytical Processing) is the technology behind many Business Intelligence (BI) applications.
- OLAP is a powerful technology for data discovery, including capabilities for limitless report viewing, complex analytical calculations, and predictive "what if" scenario (budget, forecast) planning.

23. Define Multidimensional OLAP.

- MOLAP (multi-dimensional online analytical processing) is the classic form of OLAP and is sometimes referred to as just OLAP.
- MOLAP stores this data in an optimized multi-dimensional array storage, rather than in a relational database.

24. Mention the advantages of Multidimensional OLAP.

- Fast query performance due to optimized storage, multidimensional indexing and caching.
- Smaller on-disk size of data compared to data stored in relational database due to compression techniques.
- Automated computation of higher level aggregates of the data.
- It is very compact for low dimension data sets.
- Array models provide natural indexing.
- Effective data extraction achieved through the pre-structuring of aggregated data.

25. Draft a note on Supply Chain Management (SCM).

- Supply chain management (SCM) is the management of the flow of goods and services.
- It includes the movement and storage of raw materials, work-in-process inventory, and finished goods from point of origin to point of consumption.

26. Define Customer Relationship Management (CRM).

Customer relationship management (CRM) is an approach to managing a company's interaction with current and future customers. It often involves using technology to organize, automate, and synchronize sales, marketing, customer service, and technical support.

27. What are the characteristic of CRM?

- Sales force automation, which implements sales promotion analysis, automates the tracking of a client's account history for repeated sales or future sales, and coordinates sales, marketing, call centers, and retail outlets.
- Data warehouse technology, used to aggregate transaction information, to merge the information with CRM products, and to provide key performance indicators.
- Opportunity management which helps the company to manage unpredictable growth and demand, and implement a good forecasting model to integrate sales history with sales.

28. Differentiate B2C and B2B.

- B2B companies have smaller contact databases than B2C.
- The volume of sales in B2B is relatively small.
- In B2B there are less figure propositions, but in some cases they cost a lot more than B2C items.
- Relationships in B2B environment are built over a longer period of time.

29. Explain ROLAP.

- ROLAP – Relational OLAP works directly with relational databases.
- The base data and the dimension tables are stored as relational tables and new tables are created to hold the aggregated information.
- It depends on a specialized schema design.
- This methodology relies on manipulating the data stored in the relational database to give the appearance of traditional OLAP's slicing and dicing functionality.

30. Name some of the applications of EIS.

Applications of EIS are,

- Manufacturing
- Marketing
- Financial

PART – B (11 MARKS)

1. Discuss about ERP and its Functional Components.

DEFINITION OF ERP:

- Enterprise Resource Planning (ERP) is business management software.
- It typically a suite of integrated applications that a company can use to collect, store, manage and interpret data from many business activities, including product planning, cost , Manufacturing or service delivery, Marketing and sales, Inventory .
- An important goal of ERP is to facilitate the flow of information so business decisions can be data-driven.
- ERP software suites are built to collect and organize data from various levels of an organization to provide management with insight into key performance indicators (KPIs) in real time.
- ERP is the backbone of e-business.
- ERP integrates application suites.
- ERP is not a single system, but a framework that includes administrative apps (finance, accounting), human resource apps (payroll, benefits), and manufacturing resource planning (MRP) apps (procurement, production planning). ERP unites major business processes – order processing, general ledger, payroll, and production within a single family of software modules.

Fundamental Components of ERP:

The following are the fundamental components of the ERP.

1. Financial Management
2. Business Intelligence
3. Supply Chain Management
4. Human Resource Management
5. Manufacturing Operations
6. Integration

Financial Management:

- At the core of ERP are the financial modules, including general ledger, accounts receivable, accounts payable, billing and fixed asset management.
- If organization is considering the move to an ERP system to support expansion into global markets, make sure that multiple currencies and languages are supported, as well as regulatory compliance in the U.S. and in foreign countries.

- Other functionality in the financial management modules will include budgets, cash-flow, expense and tax reporting.
- The evaluation team should focus on areas that are most important to support the strategic plans for your organization.

Business Intelligence:

- Business Intelligence (BI) has become a standard component of most ERP packages.
- In general, BI tools allow users to share and analyze the data collected across the enterprise and centralized in the ERP database.
- BI can come in the form of dashboards, automated reporting and analysis tools used to monitor the organization business performance.
- BI supports informed decision making by everyone, from executives to line managers and accountants.

Supply Chain Management:

- Supply Chain Management (SCM), sometimes referred to as logistics, improves the flow of materials through an organization by managing planning, scheduling, procurement, and fulfillment, to maximize customer satisfaction and profitability.
- Sub modules in SCM often include production scheduling, demand management, distribution management, inventory management, warehouse management, and procurement and order management.
- Any company dealing with products, from manufacturers to distributors, needs to clearly define their SCM requirements to properly evaluate an ERP solution.
- It easy for a vendor to focus on their applications strengths and not address the full needs of the company.

Human Resource Management:

- Human resource management ERP modules should enhance the employee experience – from initial recruitment to time tracking.
- A Sub module can include payroll, performance management, time tracking, benefits, compensation and workforce planning.
- Self-service tools that allow managers and employees to enter time and attendance, choose benefits and manage PTO are available in many ERP solutions.

Manufacturing Operations:

- Manufacturing modules make manufacturing operations more efficient through product configuration, job costing and bill of materials management.
- ERP manufacturing modules often include Capacity Requirements Planning, Materials Requirements Planning, forecasting, Master Production Scheduling, work-order management and shop-floor control.

Integration:

- Key to the value of an ERP package is the integration between modules, so that all of the core business functions are connected.
- Information should flow across the organization so that BI reports on organization-wide results.
- ERP can be easier than you imagine – Microsoft Dynamics ERP is cost effective and familiar to your users. If you are thinking about upgrading your systems to a fully integrated ERP system, give us a call.

NEED FOR ERP:

1. Real-time information for decisions
2. Best practice procedures
3. Improved visibility
4. Faster month-end close
5. Increased customer satisfaction
6. Managed and controlled costs
7. Better operational efficiency
8. Accurate records
9. Balance of supply and demand
10. Reduced lead times and increased throughput.

Real-time information for decisions:

- Without an ERP system, team is flying blind.
- They make decisions based on guesswork and rules of thumb because they don't have the data they need.
- Sometimes they are the right decisions, but more often, they are sub-optimum decisions that can cost you money and customer goodwill.

Best practice procedures:

- Software companies often design their ERP systems to support specific industries or verticals.
- As they add customers, they learn industry best practices and incorporate them into the software.
- By implementing an ERP system designed for your industry, you automatically make your business processes more efficient.

Improved visibility:

- If customers want to know when their order will ship or if you need to know whether you have enough of a critical component to accept a rush order, an ERP system gives you instant visibility into your operations and your supply chain.

Faster month-end close:

- ERP systems automatically process transactions and generate audit trails and financial reports that can simplify period-end closings.
- They flag anomalies so you can investigate quickly, and they simplify repetitive journal entries and other activities that make closing so complex and time consuming. Faster closes mean you know the health of your business sooner.

Increased customer satisfaction:

- Customers like accurate delivery dates, and ERP can help you provide them with improved inventory and shop floor visibility.
- In addition, the increased visibility and accuracy will help you improve your DIFOT rate (delivery in full on time), which helps keep customers happy.

Managed and controlled costs:

- ERP systems calculate and collect costs so you always have an accurate picture of your product cost and margins.

Better operational efficiency:

- By helping you to plan production more effectively, your operational efficiency will improve as you reduce setups and teardowns or unnecessary downtime.

Accurate records:

- The uniformity of record data that an ERP system instills will help ensure that your records are more accurate, which will increase process accuracy across the board.

Balance of supply and demand:

- MRP, a component of ERP systems, will help you balance supply and demand so you can reduce inventory while keeping customers happy.

Reduced lead times and increased throughput:

- Better scheduling and accurate records ensure that your schedules focus on priorities, leading to shorter lead times. Since you won't have as many orders waiting for tooling or parts, your throughput will increase.

2. **Explain in detail about Implementation of ERP, key issues and its characteristics.**

Implementation of ERP

- Implementation of ERP System, is a complex exercise, involving many process alterations and several legacy issues.
- Organizations need an implementation strategy encompassing both pre implementation and implementation stages. The fallout of a poor strategy is unpreparedness of employees, implementation not in conformity with wider business strategy, poor business process redesign and time and cost overrun.
- Following issues must be carefully thought out and formulated, as a part of implementation strategy, before embarking on actual implementation:

Business Process:

- Hypothetically, company insiders should know best about the processes of their organization. But employees often constrained to work in departmental silos and overlook wood for the tree.
- Under most circumstances, prevailing business practices are not properly defined and no "as is" flow charts, documenting existing processes, are available.
- An ERP implementation could be a great occasion to assess and optimize existing business processes, control points, breaking points between departments, and interfaces with trading partners.
- But, often, due to resistance to changes and departmental clouts, ERP implementation is comprehended as an exercise to automate legacy processes.
- This may lead to little improvement in underlying business processes, resulting no appreciable return on investment.
- Automating existing manual processes peculiar to a company necessitates, significant source code customization, as even a best fit ERP product match to a maximum of 85% to 90% of legacy processes.

- Source code customization will not only require changing of software objects but also need changing data models. The efforts needed to make such changes are significant in terms of development, testing and documentation.
- The future cost of maintenance and upgrades will be substantial, affecting entire life cycle of the system.
- Unless a considered view favoring process changes is taken as a part of implementation strategy, pressure will mount subsequently for more and more customization, when the exercise of Business Process Mapping and Gap Analysis is taken up during implementation.
- ERP systems are highly configurable and contain series of design trade off to meet various nuances of the same business cycles / processes.
- This should, normally, be sufficed to cover needed processes, probably with a little bit of swapping whenever needed.
- At occasions, it may be imperative to change source code to account for some unique core processes of the organizations.
- Procedure for authorization of such changes, normally requiring attention from sponsor, should also form part of the strategy document.

Implementation Methodology:

- Selection of implementation methodology constitutes an important component of implementation strategy. Most popular implementation methodology is "big bang" approach where on a scheduled cut-off date; entire system is installed throughout the organization.
- All users move to the new system and manual / legacy systems are discontinued. The implementation is swift and price tag is lesser than a phased implementation.
- On the flip side, risk element is much higher and resources for training, testing and hand holding are needed at a much higher level, albeit for a shorter period of time
- Another major implementation strategy is "phased implementation", where roll out is done over a period. This method is less focused, prolonged and necessitates maintenance of legacy System over a period of time.
- But, phased implementation is less risky, provides time for user's acquaintance and fall back scenarios are less complicated.

  There are various choice of phasing such as
  - Phased roll out by locations for a multi-location company
  - Phased roll out by business unit e.g. human resources

- Phased roll out by module e.g. general ledger.

Implementation methodology is depends on several factors

- Size
- Industry
- Sales volume

The Common basic factors of ERP is

- Physical Scope, BPR, Resource Allocation

The 3 Broad implementation strategies are:

1. Big Bang
2. Middle-road
3. Vanilla

1. BIG BANG APPROACH

- What is the “Big Bang” approach?

  A straightforward ERP Implementation approach. It means all ERP modules, such as financials, manufacturing, and human resources, etc., are implemented in all business units’ at all geographic locations at the same time.
- It will push the entire organization to use the new system at the same time.
- The old system will be entirely shut down

IMPLEMENTATION PROCESS

- Converting the system.

  Planning, convert data from old system, load data in new system, test data in new system, execute off-line trials, and check to verify validity.
- Releasing parts of the system.

  Release converted database, release produced application, release infrastructure.
- Training the future users.

  Create main buffer of experienced staff, training all users.
- Releasing the whole system.
- Turn down the old system and load the new system.

Middle Road Approach

- Not all ERP modules are implemented
- Industry specific modules and sub modules can be chosen
- Some legacy systems to remain functional.

Vanilla Approach

- Only essential modules are chosen
- Industry specific modules are discouraged
- Minimized risk

Challenges in implementing ERP solutions are quite normal. Though it is not completely a technical job, a lot of planning and proper communication is very much essential to implement ERP across the organization.

Below are the 7 common challenges we have noticed companies experience, when ERP is implemented.

- It is very important, that **implementation is done in stages**. Trying to implement everything at once will lead to a lot of confusion and chaos.
- **Appropriate training is very essential** during and after the implementation. The staff should be comfortable in using the application or else, it will backfire, with redundant work and functional inefficiencies.
- **Lack of proper analysis of requirements** will lead to non-availability of certain essential functionalities. This might affect the operations in the long run and reduce the productivity and profitability.
- **Lack of Support from Senior Management** will lead to unnecessary frustrations in work place. Also, it will cause delay in operations and ineffective decisions. So, it is essential to ensure that the Senior Management supports the transformation.
- **Compatibility Issues with ERP Modules** lead to issues in integration of modules. Companies associate different vendors to implement different ERP modules, based on their competency. It is very essential that there is a way to handle compatibility issues.
- **Cost Overheads** will result, if requirements are not properly discussed and decided during the planning phase. So, before execution, a detailed plan with a complete breakdown of requirements should be worked out.

- **Investment in Infrastructure** is very essential. ERP applications modules will require good processing speed and adequate storage. Not allocating suitable budget for infrastructure will result in reduced application speed and other software issues. Hardware and Software Security is also equally important.

Characteristics of ERP

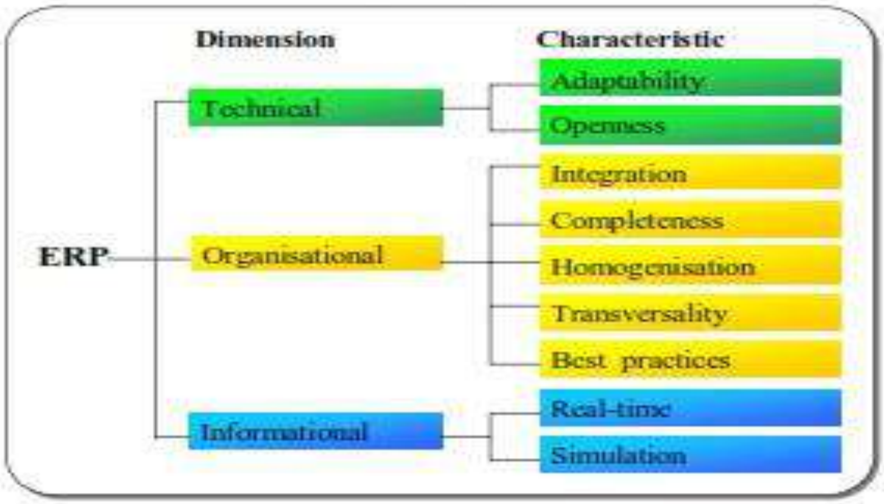
- ERP (Enterprise Resource Planning) systems typically include the following characteristics:
- An integrated system that operates in (or near) real time without relying on periodic updates
- A common database that supports all applications
- A consistent look and feel across modules
- Installation of the system with elaborate application/data integration by the Information Technology department provided the implementation is not done in small steps.
- Flexible
- Modular and Open
- Comprehensive
- Online-Connection with Other ERP System
- Best Business Practices
- Multi-Facilities
- Strategic planning
- Optimize the data
- Project Management
- Automatic Functions

Key issues

Legacy data: Gathering, cleaning and removing of duplicate data.

Hardware and software: Addition and updating of existing resources. Compatibility with existing Operating system and Database

Project structure: Project champions and competency center



Adaptability

- ERP system was not adaptable, it would limit the organization's development potential in a changing environment.
- Also, given that the ERP system integrates various functions of the organization (Production, finance, HRM, etc.), there seems to be both an opposing and a complementary.

Openness

- Openness is a characteristic that appears to be redundant in regard to flexibility.
- The modularity and portability (openness) that allow an ERP system to evolve with the organization can be considered as factors of flexibility.

Integration

- Integration is without a doubt the most important ERP characteristics.
- It distinguishes ERP systems from traditional IS whose informational fragmentation has been criticized in that it reflects a vision of the organization as a set of islands or functional silos that cannot communicate with each other, or that communicate little or with great difficulty.

Completeness

- Completeness (the generic functionality) is a characteristic that, pushed to the extreme, becomes unrealistic.
- A generic system that would work for all types of firms and industries is in fact very difficult if not impossible to design.

Homogenization

- Homogenization refers to the existence of a unique data referential, the uniformity of human-machine interfaces, and to the unicity of the application system's administration.

Transversality

- Transversality refers to the process-oriented view of an ERP system.
- Its importance comes from the fact that ERP systems are composed of functional modules, and are generally implemented module by module, that is, in a vertical manner.

Best Practices

- Imbedding best practices would not be considered as an essential characteristic of ERP systems.
- The reason is that this notion is based on adopting generic processes that, despite being exemplary, offer few possibilities of gaining a competitive advantage.

Real Time

- The capacity for real-time operation and simulation in an ERP system are consequences of its successful integration.
- It is this integration that allows the same information to be communicated in real time to all parts of the organization, and allows simulating the effect of an input on all activities of the organization

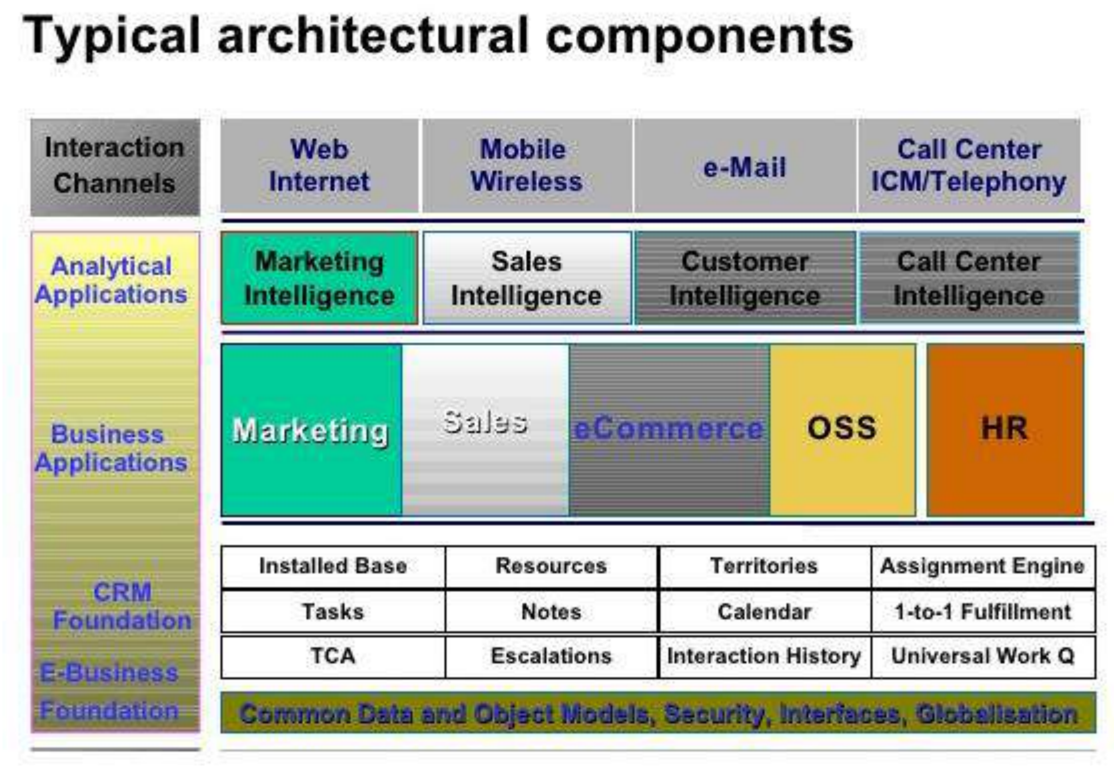
CHARACTERISTICS OF TYPICAL ARCHITECTURE COMPONENTS OF ERP

Integration

- Seamless integration of all the information flowing through a company – financial and accounting, human resource information, supply chain information, and customer information.

Packages

- Enterprise systems are not developed in-house
- IS life cycle is different
    1. Mapping organizational requirements to the processes and terminology employed by the vendor and
    2. Making informed choices about the parameter setting.
- Organizations that purchase enterprise systems enter into long-term relationships with vendors. Organizations no longer control their own destiny.



Best Practices

- ERP vendors talk to many different businesses within a given industry as well as academics to determine the best and most efficient way of accounting for various transactions and managing different processes. The result is claimed to be "industry best practices".
- The general consensus is that business process change adds considerably to the expense and risk of an enterprise systems implementation. Some organizations rebel against the inflexibility of these imposed business practices.

Evolving

- Enterprise Systems are changing rapidly
- Architecturally: Mainframe, Client/Server, Web-enabled, Object-oriented, Componentization
- Functionally: front-office (i.e. sales management), supply chain (advanced planning and scheduling), data warehousing, specialized vertical industry solutions, etc.

3. Sketch and explain about ERP System Architecture.

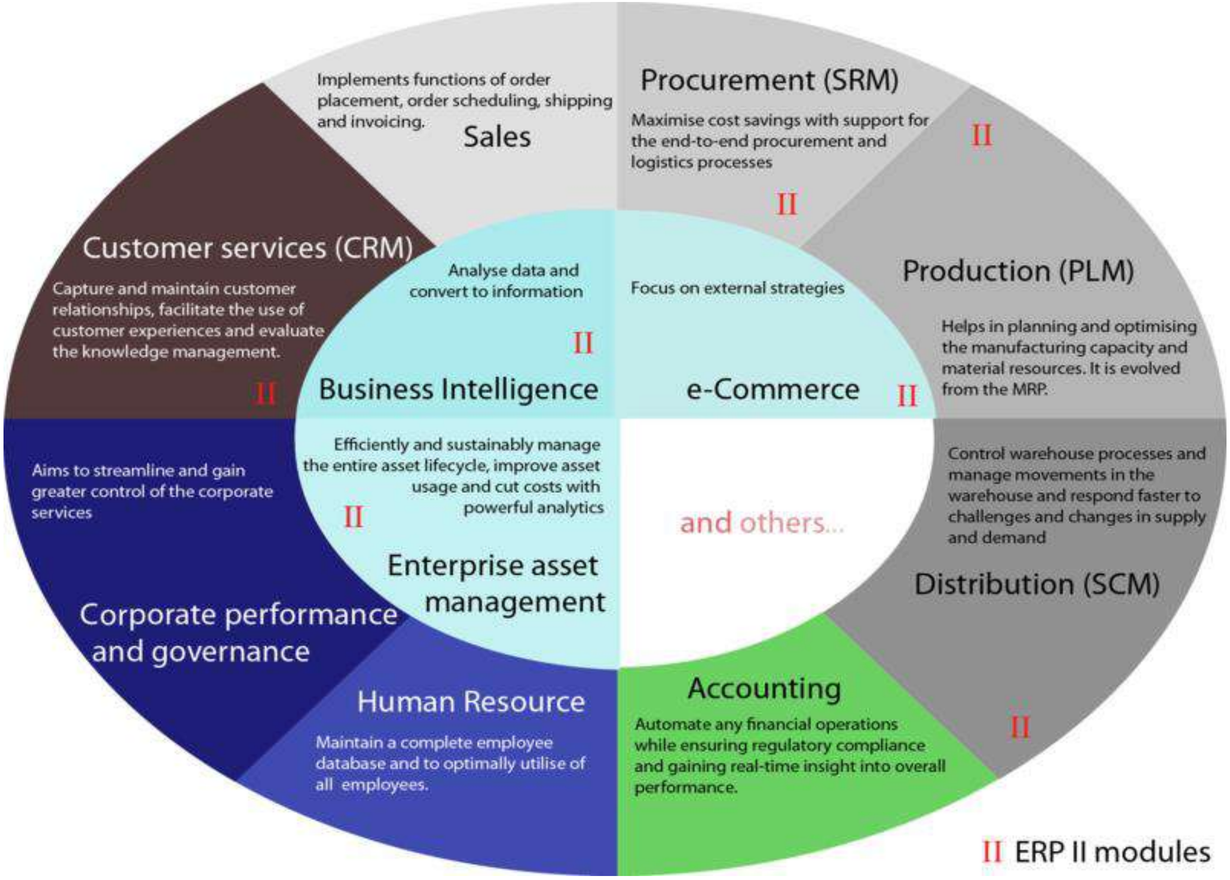
ERP SYSTEM ARCHITECTURE:

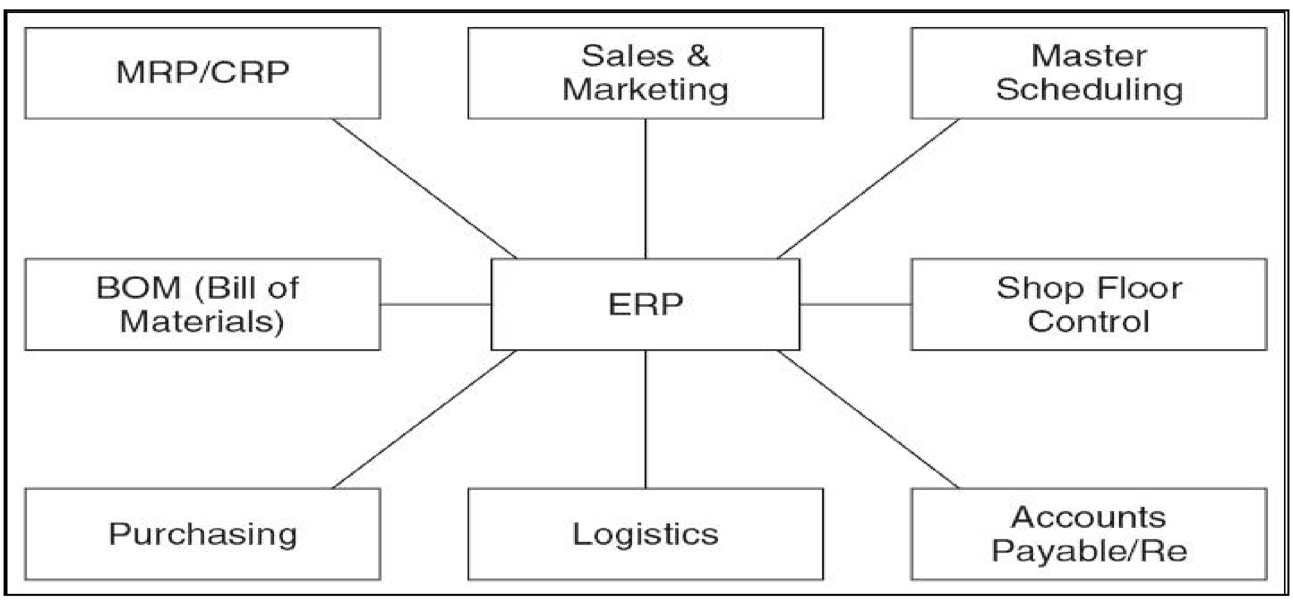
- Enterprise resource planning (ERP) is business-management software typically a suite of integrated applications that an organization can use to collect, store, manage and interpret data from many business activities, including:
    - Product planning, cost
    - Manufacturing or service delivery
    - Marketing and sales
    - Inventory management
    - Shipping and payment
- ERP provides an integrated view of core business processes, often in real-time, using common databases maintained by a database management system.
- ERP systems track business resources such as cash, raw materials, production capacity and the status of business commitments: orders, purchase orders, and payroll.
- The applications that make up the system share data across various departments (manufacturing, purchasing, sales, accounting, etc.) that provide the data. ERP facilitates information flow between all business functions, and manages connections to outside stakeholders.
- The ERP system is considered a vital organizational tool because it integrates varied organizational systems and facilitates error-free transactions and production. However, developing an ERP system differs from traditional system development.
- ERP systems run on a variety of computer hardware and network configurations, typically using a database as an information repository

ERP modules:

- The key role of an ERP system is to provide support for such business functions as accounting, sales, inventory control, and production.
- ERP vendors, including SAP, Oracle, and Microsoft, etc. provide modules that support the major functional areas of a business.
- The ERP software embeds best business practices that implement the organization's policy and procedure via business rules

Typical ERP Modules:





- ERP applications are most commonly deployed in a distributed and often widely dispersed manner. While the servers may be centralized, the clients are usually spread to multiple locations throughout the enterprise.

- Generally there are three functional areas of responsibility that is distributed among the servers and the clients.
- First, there is the database component - the central repository for all of the data that is transferred to and from the clients.
- Then, of course, the clients - here raw data gets inputted, requests for information are submitted, and the data satisfying these requests is presented.
- Lastly, we have the application component that acts as the intermediary between the client and the database. Where these components physically reside and how the processes get distributed will vary somewhat from one implementation to the next.

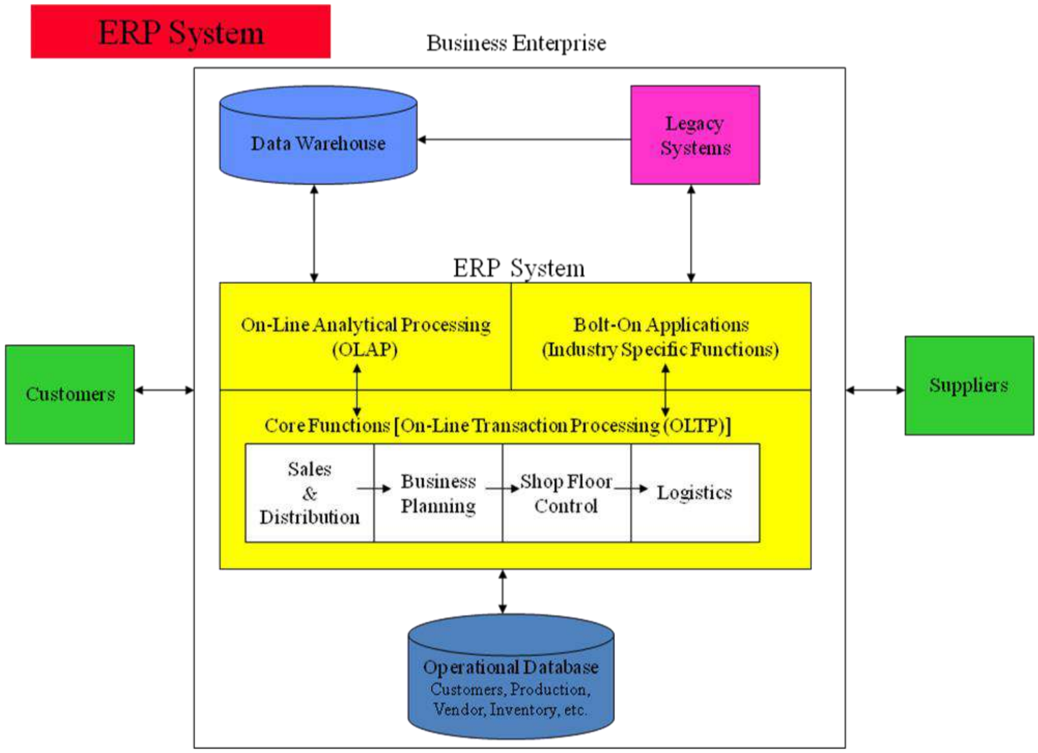
The two most commonly implemented architectures are outlined below.

Two-tier Implementations

- In typical two-tier architecture, the server handles both application and database duties. The clients are responsible for presenting the data and passing user input back to the server.
- While there may be multiple servers and the clients may be distributed across several types of local and wide area links, this distribution of processing responsibilities remains the same.

Three-tier Client/Server Implementations

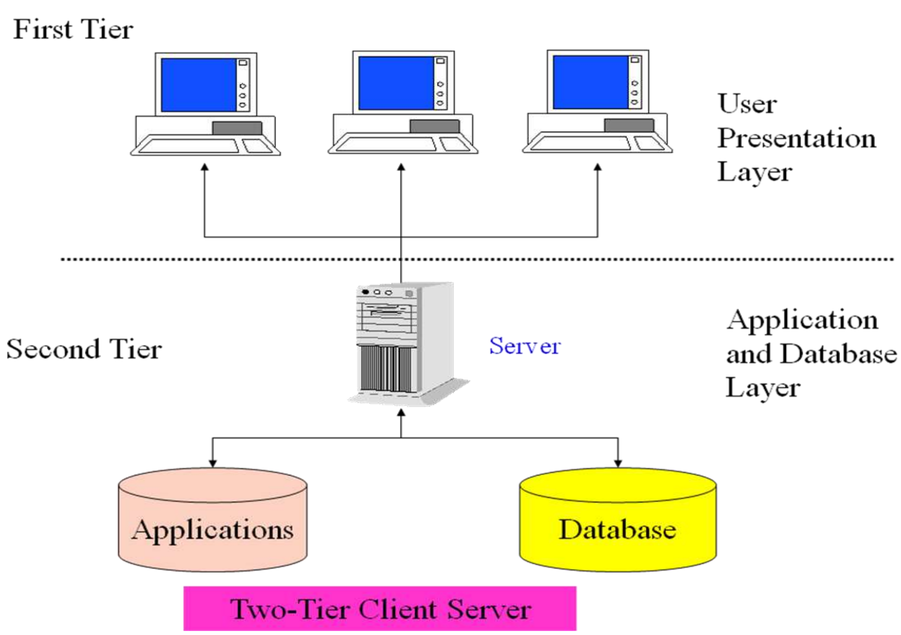
- In three-tier architectures, the database and application functions are separated. This is very typical of large production ERP deployments.
- In this scenario, satisfying client requests requires two or more network connections. Initially, the client establishes communications with the application server. The application server then creates a second connection to the database server.



ERP System Configurations: Client-Server Network Topology

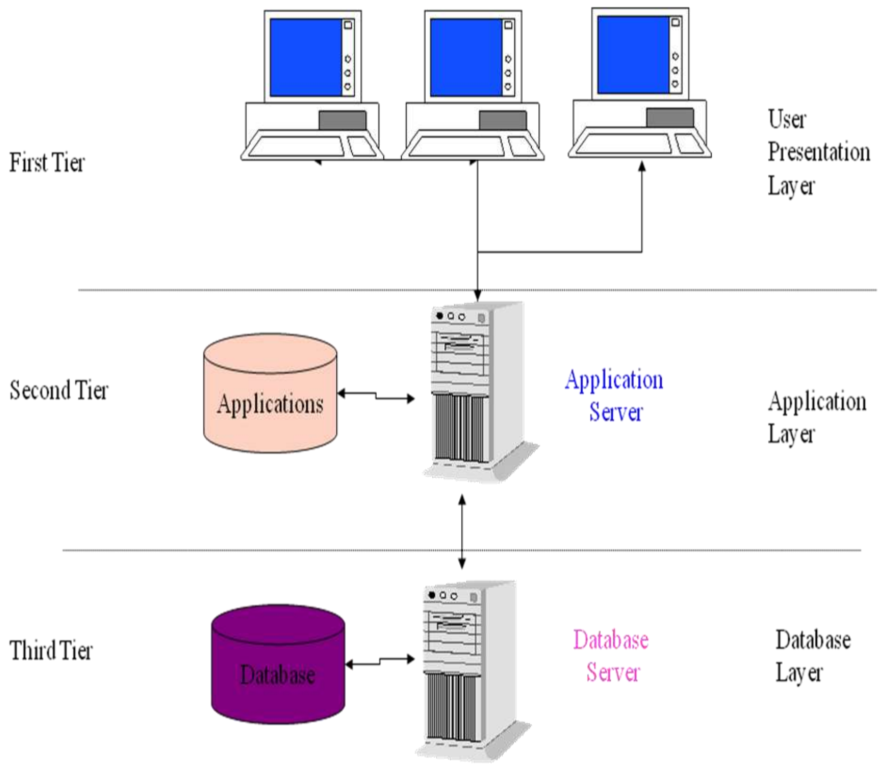
<u>Two-tier:</u>

- It common server handles both application and database duties.
- It is used especially in LANs.



Three-tier:

- The client links to the application server which then initiates a second connection to the database server.
- It is used especially in WANs



Three-Tier Client Server

4. Discuss the advantages and disadvantages of ERP.

<u>ADVANTAGES</u>

- Complete **visibility** into all the important processes, across various departments of an organization (especially for senior management personnel).
- Automatic and coherent **workflow** from one department/function to another, to ensure a smooth transition and quicker completion of processes. This also ensures that all the inter-departmental activities are properly tracked and none of them is 'missed out'.
- A unified and single **reporting** system to analyze the statistics/status etc. in real-time, across all functions/departments.
- Since **same (ERP) software** is now used across all departments, individual departments having to buy and maintain their own software systems is no longer necessary.

- Certain ERP vendors can extend their ERP systems to provide Business **Intelligence** functionalities that can give overall insights on business processes and identify potential areas of problems/improvements.
- Advanced **e-commerce integration** is possible with ERP systems – most of them can handle web-based order tracking/ processing.
- There are **various modules** in an ERP system like Finance/Accounts, Human Resource Management, Manufacturing, Marketing/Sales, Supply Chain/Warehouse Management, CRM, Project Management, etc.
- Since ERP is a **modular software** system, it's possible to implement either a few modules (or) many modules based on the requirements of an organization. If more modules implemented, the integration between various departments may be better.
- Since a Database system is implemented on the backend to store all the information required by the ERP system, it enables **centralized storage/back-up** of all enterprise data.
- ERP systems are **more secure** as centralized security policies can be applied to them. All the transactions happening via the ERP systems can be tracked.
- ERP systems provide better company-wide visibility and hence enable **better/faster collaboration** across all the departments. 12. It is possible to integrate other systems (like bar-code reader, for example) to the ERP system through an API (Application Programing Interface). 13. ERP systems make it easier for order tracking, inventory tracking, revenue tracking, sales forecasting and related activities.
- ERP systems are especially helpful for managing globally dispersed enterprise companies, better.

DISADVANTAGES

- The cost of ERP Software, planning, customization, configuration, testing, implementation, etc. is too high.
- ERP deployments are highly time-consuming – projects may take 1-3 years (or more) to get completed and fully functional.
- Too little **customization** may not integrate the ERP system with the business process & too much customization may slow down the project and make it difficult to upgrade.
- The cost savings/payback may not be realized immediately after the ERP implementation & it is quite difficult to measure the same.

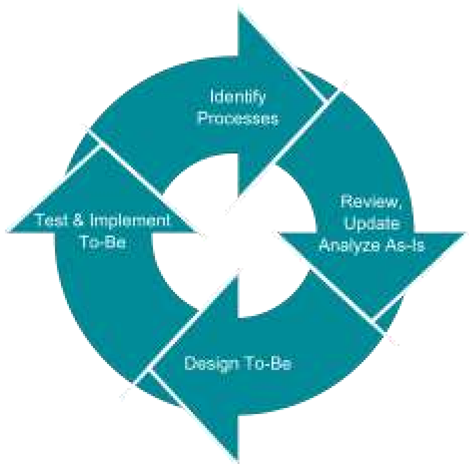
- The **participation** of users is very important for successful implementation of ERP projects – hence, exhaustive user training and simple user interface might be critical. But ERP systems are generally difficult to learn (and use).
- There may be **additional indirect** costs due to ERP implementation – like new IT infrastructure, upgrading the WAN links, etc.
- **Migration** of existing data to the new ERP systems is difficult (or impossible) to achieve. Integrating ERP systems with other standalone software systems is equally difficult (if possible). These activities may consume a lot of time, money & resources, if attempted.
- ERP implementations are difficult to achieve in **decentralized organizations** with disparate business processes and systems.
- Once an ERP systems is implemented it becomes a **single vendor lock-in** for further upgrades, customizations etc. Companies are at the discretion of a single vendor and may not be able to negotiate effectively for their services.
- **Evaluation** prior to implementation of ERP system is critical. If this step is not done properly and experienced technical/business resources are not available while evaluating, ERP implementations can (and have) become a failure.

5. Explain in detail about Business Process Re-engineering (BPR).

Business Process Re-engineering (BPR).

- Business process re-engineering (BPR) is the analysis and redesign of workflows within and between enterprises in order to optimize end-to-end processes and automate non-value-added tasks.
- BPR is the practice of rethinking and redesigning the way work is done to better support an organization's mission and reduce costs.
- Reengineering starts with a high-level assessment of the organization's mission, strategic goals, and customer needs.
- Within the framework of this basic assessment of mission and goals, re-engineering focuses on the organization's business processes—the steps and procedures that govern how resources are used to create products and services that meet the needs of particular customers or markets.
- As a structured ordering of work steps across time and place, a business process can be decomposed into specific activities, measured, modeled, and improved.
- It can also be completely redesigned or eliminated altogether. Re-engineering identifies, analyzes, and re-designs an organization's core business processes with the aim of achieving dramatic improvements in critical performance measures, such as cost, quality, service, and speed.

- Re-engineering recognizes that an organization's business processes are usually fragmented into sub processes and tasks that are carried out by several specialized functional areas within the organization.
- Re-engineering maintains that optimizing the performance of sub processes can result in some benefits, but cannot yield dramatic improvements if the process itself is fundamentally inefficient and outmoded.
- For that reason, re-engineering focuses on re-designing the process as a whole in order to achieve the greatest possible benefits to the organization and their customers.



Business Process Reengineering Cycle

The most notable definitions of reengineering are:

- The fundamental rethinking and radical redesign of business processes to achieve dramatic improvements in critical contemporary modern measures of performance, such as cost, quality, service, and speed.
- Encompasses the envisioning of new work strategies, the actual process design activity, and the implementation of the change in all its complex technological, human, and organizational dimensions."

Model based on PRLC approach:

- Although the labels and steps differ slightly, the early methodologies that were rooted in IT-centric BPR solutions share many of the same basic principles and elements.
- The following outline is one such model, based on the PRLC (Process Reengineering Life Cycle). Simplified schematic outline of using a business process approach, exemplified for pharmaceutical R&D.

- Structural organization with functional units
- Introduction of New Product Development as cross-functional process
- Re-structuring and streamlining activities, removal of non-value adding tasks

BPR Success & Failure Factors:

- Factors that are important to BPR success include:
- BPR team composition.
- Business needs analysis.
- Adequate IT infrastructure.
- Effective change management.
- Ongoing continuous improvement

There are many reasons for sub-optimal business processes which include:

- One department may be optimized at the expense of another
- Lack of time to focus on improving business process
- Lack of recognition of the extent of the problem
- Lack of training
- People involved use the best tool they have at their disposal which is usually Excel to fix problems
- Inadequate infrastructure
- Overly bureaucratic processes
- Lack of motivation

BPR team composition:

Once organization-wide commitment has been secured from all departments involved in the reengineering effort and at different levels, the critical step of selecting a BPR team must be taken. The determinants of an effective BPR team may be summarized as follows:

- Competency of the members of the team, their motivation,
- Their credibility within the organization and their creativity,
- Team empowerment, training of members in process mapping and brainstorming techniques,
- Effective team leadership,
- Proper organization of the team,

- Complementary skills among team members, adequate size, interchangeable accountability, clarity of work approach, and
- Specificity of goals.
- The most effective BPR teams include active representatives from the following work groups: top management, business area responsible for the process being addressed, technology groups, finance, and members of all ultimate process users' groups.

The BPR team should be mixed in depth and knowledge. For example, it may include members with the following characteristics:

- Members who do not know the process at all.
- Members who know the process inside-out.
- Customers, if possible.
- Members representing impacted departments.
- One or two members of the best, brightest, passionate, and committed technology experts.
- Members from outside of the organization

Business needs analysis:

This plan includes the following:

- Identifying specific problem areas.
- Solidifying particular goals and
- Defining business objectives.
- The business needs analysis contributes tremendously to the re-engineering effort by helping the BPR team to prioritize and determine where it should focus its improvements efforts.

6. Explicate about Management Information System (MIS).

Management Information Systems (MIS), are information systems, typically computer based, that are used within an organization. WordNet described an information system as "a system consisting of the network of all communication channels used within an organization".

Definition:

MIS is a computer – based system that optimizes the collection, collation, transfer and presentation of information throughout an organization, through an integrated structure of databases and information flow.

Characteristics of MIS

- MIS supports data processing functions of transaction handling and record keeping.
- MIS uses an integrated database and supports a variety of functional areas.
- MIS provides operational, tactical and strategic levels of organization with timely, structured information.
- MIS is flexible and can adapt to the changing needs of the organization.

Importance of MIS

- As an area of study it is commonly referred to as information technology management.
- The study of information systems is usually a commerce and business administration discipline, and frequently involves software engineering, but also distinguishes itself by concentrating on the integration of computer systems with the aims of the organization.
- The area of study should not be confused with Computer Science which is more theoretical and mathematical in nature or with Computer Engineering which is more engineering.
- In business, information systems support business processes and operations, support decision making, and support competitive strategies.

The major differences between a management information system and a Data Processing system are:

- The integrated database of the MIS enables greater flexibility in meeting the information needs of the management.
- The MIS integrates the information flow between functional areas (accounting, marketing, manufacturing, etc.) whereas data processing systems tend to support a single functional area.
- MIS caters to the information needs of all levels of management whereas data processing systems focus on departmental-level support.

- Management’s information needs are supported on a timelier basis with the MIS (with its on-line query capability) than with a data processing system.

	MIS		DPS(DATA PROCESSING SYSTEM)
1	It uses an integrated database.	1	It does not use an integrated database.
2	It provides greater flexibility to the management	2	It provides no such flexibility
3	Integrates the information flow between functional areas.	3	DPS tends to support a single functional area.
4	Focus on information needs of all levels of management	4	DPS focus on departmental-level support

Advantages:

The following are some of the benefits that can be attained using MISs.

- Companies are able to identify their strengths and weaknesses due to the presence of revenue reports, employees' performance record etc. Identifying these aspects can help a company improve its business processes and operations.
- Giving an overall picture of the company.
- Acting as a communication and planning tool.
- The availability of customer data and feedback can help the company to align its business processes according to the needs of its customers. The effective management of customer data can help the company to perform direct marketing and promotion activities.
- MISs can help a company gain a competitive advantage. Competitive advantage is a firm’s ability to do something better, faster, cheaper, or uniquely, when compared with rival firms in the market.

7. **Describe about Decision support system (DSS).**

Decision Support Systems (DSS)

- Decision support systems (DSS) are interactive software-based systems intended to help managers in decision-making by accessing large volumes of information generated from various related information systems involved in organizational business processes such as office automation system, transaction processing system, etc.
- DSS uses the summary information, exceptions, patterns, and trends using the analytical models. A decision support system helps in decision-making but does not necessarily give a decision itself.
- The decision makers compile useful information from raw data, documents, personal knowledge, and/or business models to identify and solve problems and make decisions.

Programmed and Non-programmed Decisions

- There are two types of decisions - programmed and non-programmed decisions. Programmed decisions are basically automated processes, general routine work, where:
  - These decisions have been taken several times.
  - These decisions follow some guidelines or rules.
- Non-programmed decisions occur in unusual and non-addressed situations, so:
  - It would be a new decision
  - There will not be any rules to follow.
  - These decisions are made based on the available information.
  - These decisions are based on the manger's discretion, instinct, perception and judgment.

Attributes of a DSS

- Adaptability and flexibility
- High level of Interactivity
- Ease of use
- Efficiency and effectiveness
- Complete control by decision-makers
- Ease of development
- Extendibility
- Support for modeling and analysis
- Support for data access
- Standalone, integrated, and Web-based

Characteristics of a DSS

- Support for decision-makers in semi-structured and unstructured problems.

- Support for managers at various managerial levels, ranging from top executive to line managers.
- Support for individuals and groups. Less structured problems often requires the involvement of several individuals from different departments and organization level.
- Support for interdependent or sequential decisions.
- Support for intelligence, design, choice, and implementation.
- Support for variety of decision processes and styles.
- DSSs are adaptive over time

Classification of DSS

- There are several ways to classify DSS. Hoi Apple and Whinstone classifies DSS as follows:
- **Text Oriented DSS:** It contains textually represented information that could have a bearing on decision. It allows documents to be electronically created, revised and viewed as needed.
- Database Oriented DSS: Database plays a major role here; it contains organized and highly structured data.
- **Spreadsheet Oriented DSS:** It contains information in spread sheets that allows create, view, modify procedural knowledge and also instructs the system to execute self-contained instructions. The most popular tool is Excel and Lotus 1-2-3.
- **Solver Oriented DSS:** It is based on a solver, which is an algorithm or procedure written for performing certain calculations and particular program type.
- Rules Oriented DSS: It follows certain procedures adopted as rules.
- Procedures are adopted in rules oriented DSS. Export system is the example.
- **Compound DSS:** It is built by using two or more of the five structures explained above

Components of a DSS

- **Database Management System (DBMS):** To solve a problem the necessary data may come from internal or external database. In an organization, internal data are generated by a system such as TPS and MIS. External data come from a variety of sources such as newspapers, online data services, databases (financial, marketing, human resources).
- **Model Management System:** It stores and accesses models that managers use to make decisions. Such models are used for designing manufacturing facility, analyzing the financial health of an organization, forecasting demand of a product or service, etc.
- **Support Tools**: Support tools like online help; pulls down menus, user interfaces, graphical analysis, error correction mechanism, facilitates the user interactions with the system.

8. **Detail about Executive Information System/Executive Support Systems.**

- Executive support systems are intended to be used by the senior managers directly to provide support to non-programmed decisions in strategic management.
- These information are often external, unstructured and even uncertain. Exact scope and context of such information is often not known beforehand. This information is intelligence based:
    1. Market intelligence
    2. Investment intelligence
    3. Technology intelligence
- A computerized system intended to provide current and appropriate information to support executive decision making for managers using a networked workstation.
- The emphasis is on graphical displays and an easy to use interface that present information from the corporate database.
- They are tools to provide canned reports or briefing books to top-level executives. They offer strong reporting and drill-down capabilities. An early term for a sophisticated data-driven DSS targeted to senior executives.
- Executive information systems (EIS) provide a variety of internal and external information to top managers in a highly summarized and convenient form. EIS are becoming an important tool of top-level control in many organizations.
- They help an executive spot a problem, an opportunity, or a trend

Characteristics of Executive Information Systems

- EIS provide immediate and easy access to information reflecting the key success factors the company and of its units.
- A User-seductive@ interfaces, presenting information through color graphics or video, allow an EIS user to grasp trends at a glance. 3. EIS provide access to a variety of databases, both internal and external, through a uniform interface.
- Both current status and projections should be available from EIS.
- An EIS should allow easy tailoring to the preferences of the particular users or group of users.
- EIS should offer the capability to Adrill down@ into the data.

DSS are primarily used by middle and lower level managers to project the future, EIS's primarily serve the control needs of higher level management.

- EISs primarily assist top management in uncovering a problem or an opportunity. Analysts and middle managers can subsequently use a DSS to suggest a solution to the problem.
- At the heart of an EIS lies access to the data. EISs may work on the data extraction principal, as DSSs do, or they may be given access to the actual corporate databases or data warehouses.
- EISs can reside on personal workstations or servers

Developing EIS

- EIS's should make it easy to track the critical success factors (CSF) of the enterprise, that is, the few vital indicators of the firm's performance.
- With the use of this methodology, executives may define just the few indicators of corporate performance they need. With the drill-down capability, they can obtain more detailed data behind the indicators.
- Strategic business objectives methodology of EIS development takes a company-wide perspective of the strategic business objectives of the firm where the critical businesses are identified and prioritized.
- Then the information needed to support these processes is defined, to be obtained with the EIS that is being planned. Finally, an EIS is developed to report on the CSFs. This methodology avoids the frequent pitfall of aligning an EIS too closely to a particular sponsor.

Advantages of ESS

- Easy for upper level executive to use
- Ability to analyze trends
- Augmentation of managers' leadership capabilities
- Enhance personal thinking and decision-making
- Contribution to strategic control flexibility
- Enhance organizational competitiveness in the market place
- Instruments of change
- Increased executive time horizons
- Better reporting system
- Improved mental model of business executive
- Help improve consensus building and communication

Disadvantage of ESS

- Functions are limited
- Hard to quantify benefits
- Executive may encounter information overload
- System may become slow
- Difficult to keep current data
- May lead to less reliable and insecure data
- Excessive cost for small company

9. Explain in Detail about OLAP (Online Analytical Processing).

- OLAP is an acronym for On Line Analytical Processing. It is an approach to quickly provide the answer to analytical queries that are dimensional in nature.
- It is part of the broader category business intelligence, which also includes Extract transform load (ETL), relational reporting and data mining.
- The typical applications of OLAP are in business reporting for sales, marketing, management reporting, business performance management (BPM), budgeting and forecasting, financial reporting and similar areas.
- The term OLAP was created as a slight modification of the traditional database term OLTP (On Line Transaction Processing).
- Databases configured for OLAP employ a multidimensional data model, allowing for complex analytical and ad-hoc queries with a rapid execution time.
- Nigel Pendse has suggested that an alternative and perhaps more descriptive term to describe the concept of OLAP is Fast Analysis of Shared Multidimensional Information (FASMI). They borrow aspects of navigational databases and hierarchical databases that are speedier than their relational

Functionality

- OLAP takes a snapshot of a set of source data and restructures it into an OLAP cube. The queries can then be run against this. It has been claimed that for complex queries OLAP can produce an answer in around 0.1% of the time for the same query on OLTP relational data.
- The cube is created from a star schema or snowflake schema of tables. At the center is the fact table which lists the core facts which make up the query. Numerous dimension tables are linked to the fact tables. These tables indicate how the aggregations of relational data can be analyzed. The

number of possible aggregations is determined by every possible manner in which the original data can be hierarchically linked.

- For example a set of customers can be grouped by city, by district or by country; so with 50 cities, 8 districts and two countries there are three hierarchical levels with 60 members.
- These customers can be considered in relation to products; if there are 250 products with 20 categories, three families and three departments then there are 276 product members.
- With just these two dimensions there are 16,560 (276 * 60) possible aggregations. As the data considered increases the number of aggregations can quickly total tens of millions or more.
- The calculation of the aggregations AND the base data combined make up an OLAP cube, which can potentially contain all the answers to every query which can be answered from the data (as in Gray, Bosworth, Layman, and Pirahesh, 1997). Due to the potentially large number of aggregations to be calculated, often only a predetermined number are fully calculated while the remainder are solved on demand.

Types of OLAP

- **Multidimensional or MOLAP:** MOLAP is the classic form of OLAP and is sometimes referred to as just OLAP. MOLAP uses database structures that are generally optimal for attributes such as time period, location, product or account code. The way that each dimension will be aggregated is defined in advance by one or more hierarchies.
- **Relational or ROLAP:** ROLAP works directly with relational databases. The base data and the dimension tables are stored as relational tables and new tables are created to hold the aggregated information. Depends on a specialized schema design.
- **Hybrid or HOLAP:** There is no clear agreement across the industry as to what constitutes "Hybrid OLAP", except that a database will divide data between relational
- and specialized storage.• For example, for some vendors, a HOLAP database will use relational tables to hold the larger quantities of detailed data, and use specialized storage for at least some aspects of the smaller quantities of more- aggregate or less-detailed data

Comparison:

- Each type has certain benefits, although there is disagreement about the specifics of the benefits between providers.

- Some MOLAP implementations are prone to database explosion. Database explosion is a phenomenon causing vast amounts of storage space to be used by MOLAP databases when certain common conditions are met: high number of dimensions, pre-calculated results and sparse multidimensional data.
- The typical mitigation technique for database explosion is not to materialize all the possible aggregation, but only the optimal subset of aggregations based on the desired performance vs. storage trade off.
- MOLAP generally delivers better performance due to specialized indexing and storage optimizations. MOLAP also needs less storage space compared to ROLAP because the specialized storage typically includes compression techniques.
- ROLAP is generally more scalable. However, large volume pre-processing is difficult to implement efficiently so it is frequently skipped. ROLAP query performance can therefore suffer.
- Since ROLAP relies more on the database to perform calculations, it has more limitations in the specialized functions it can use.
- HOLAP encompasses a range of solutions that attempt to mix the best of ROLAP and MOLAP. It can generally pre-process quickly, scale well, and offer good function support.

Advantages of MOLAP

- Fast query performance due to optimized storage, multidimensional indexing and caching.
- Smaller on-disk size of data compared to data stored in relational database due to compression techniques.
- Automated computation of higher level aggregates of the data.
- It is very compact for low dimension data sets.
- Array models provide natural indexing.
- Effective data extraction achieved through the pre-structuring of aggregated data.

Disadvantages of MOLAP

- Within some MOLAP Solutions the processing step (data load) can be quite lengthy, especially on large data volumes. This is usually remedied by doing only incremental processing, i.e., processing only the data which have changed (usually new data) instead of reprocessing the entire data set.
- Some MOLAP methodologies introduce data redundancy

10. Discuss in detail about Supply chain management (SCM).

Supply chain management (SCM)

- Supply chain management is the systemic, strategic coordination of the traditional business functions and tactics across these business functions - both within a particular company and across businesses within the supply chain- all coordinated to improve the long-term performance of the individual companies and the supply chain as a whole.
- In a traditional manufacturing environment, supply chain management meant managing movement and storage of raw materials, work-in-progress inventory, and finished goods from point of origin to point of consumption.
- It involves managing the network of interconnected smaller business units, networks of channels that take part in producing a merchandise of a service package required by the end users or customers. With businesses crossing the barriers of local markets and reaching out to a global scenario, SCM is now defined as:
- Design, planning, execution, control, and monitoring of supply chain activities with the objective of creating net value, building a competitive infrastructure, leveraging worldwide logistics, synchronizing supply with demand and measuring performance globally.
- SCM consists of:
    - Operations management
    - Logistics
    - Procurement
    - information technology
    - integrated business operations

Objectives of SCM

- To decrease inventory cost by more accurately predicting demand and scheduling production to match it.
- To reduce overall production cost by streamlining production and by improving information flow.
- To improve customer satisfaction.

Features of SCM

- Integrated behavior
- Mutually sharing information
- Mutually sharing channel and risk rewards

- Focus on serving customers
- Co-operation
- Partnership to build maintain long term relationship
- Integration of process

SCM Processes

- Customer Relationship Management
- Customer Service Management
- Demand Management
- Customer Order Fulfillment
- Manufacturing Flow Management
- Procurement Management
- Product Development and Commercialization
- Returns Management

Advantages of SCM

SCM have multi-dimensional advantages:

- To the suppliers:
    - Help in giving clear-cut instruction
    - Online data transfer reduce paper work
- Inventory Economy:
    - Low cost of handling inventory
    - Low cost of stock outage by deciding optimum size of replenishment orders
    - Achieve excellent logistical performance such as just in time
- Distribution Point:
    - Satisfied distributor and whole seller ensure that the right products reach the right place at right time
    - Clear business processes subject to fewer errors
    - Easy accounting of stock and cost of stock
- Channel Management:
    - Reduce total number of transactions required to provide product assortment
    - Organization is logically capable of performing customization requirements
- Financial management:

    - Low cost
    - Realistic analysis
- Operational performance:
    - It involves delivery speed and consistency.
- External customer:
    - Conformance of product and services to their requirements
    - Competitive prices
    - Quality and reliability
- To employees and internal customers:
    - Teamwork and cooperation
    - Efficient structure and system

11. Illustrate about Customer relationship management (CRM).

- CRM is an enterprise application module that manages a company's interactions with current and future customers by organizing and coordinating, sales and marketing, and providing better customer services along with technical support.
- Atul Parvatiyar and Jagdish N. Sheth provide an excellent definition for customer relationship management in their work titled.
- 'Customer Relationship Management: Emerging Practice, Process, and Discipline': Customer Relationship Management is a comprehensive strategy and process of acquiring, retaining, and partnering with selective customers to create superior value for the company and the customer.
- It involves the integration of marketing, sales, customer service, and the supply-chain functions of the organization to achieve greater efficiencies and effectiveness in delivering customer value.

Why CRM?

- To keep track of all present and future customers.
- To identify and target the best customers.
- To let the customers know about the existing as well as the new products and services.
- To provide real-time and personalized services based on the needs and habits of the existing customers.
- To provide superior service and consistent customer experience.
- To implement a feedback system.

CHARACTERISTICS OF CRM

- Builds a database that describes the customers and the relationship they hold with the company.
- Database: a collection of information that is organized in a way that allows it to be easily accessed, managed and updated.
- Provides enough detail so that the company can offer the client the product/service that matches their need the best.
- May contain information about their past purchases, who is involved with the account and a summary of all conversations.

GOALS OF CRM

- Make sure the primary goal (ie: customer service) is standard in the software. Customization is very expensive.
- Be able to integrate smoothly and flawlessly.
- Make sure the program doesn't offer more then you need.
- Compare with other companies of equivalently the same size who have purchased the same CRM software and examine their results.

BENEFITS OF CRM

- Help marketing departments identify and target their best customers, manage campaigns as well as discover qualified leads.
- Qualified Leads: prospects who seem most likely to buy because of some information known about them.
- Improve sales and streamline existing processes.
- Form individualized relationships with customers.
- Give employees information needed to improve customer service and also to better understand customer needs.
- Tracks all points of contact between a customer and the company.
- CRM quickly manages the scheduling of follow-up sales calls to assess the satisfaction of customers and their repurchase probabilities (when and how much).
- Identifying prospects and helps them become customers.
- Closing sales more effectively and efficiently.

- Allowing customers to perform business transactions quickly and easily.
- Providing better service and support following a sale.

CUSTOMER SERVICE

- Helps make call centers more efficient.
- Aids in cross and up selling products.
- Cross Selling: Provide additional products/services.
- Up Selling: Upgrade existing products/services.
- Helps sales staff close deals faster.
- Simplifies marketing and sales processes.
- Allows companies to discover new customers.

COMPARISON OF CRM AND BI

- CRM is closely related to business intelligence because both methods involve using technology to gather, analyze and organize data in order to develop relevant information.
- CRM is just a more specific form of BI that concentrates strictly on customer's behaviors and actions, for both past and future information.

Advantages of CRM

- Provides better customer service and increases customer revenues.
- Discovers new customers.
- Cross-sells and up-sells products more effectively.
- Helps sales staff to close deals faster.
- Makes call centers more efficient.
- Simplifies marketing and sales processes.

Disadvantages of CRM

- Sometimes record loss is a major problem.
- Overhead costs.
- Giving training to employees is an issue in small organizations.

DEPARTMENT OF CSE

CS T61 ENTERPRISE SOLUTIONS

Unit-II – SAP AND ABAP/4

QUESTION BANK WITH ANSWER

SAP: History – SAP R/2 – SAP R/3 – Characteristics of SAP R/3 – Architecture of SAP R/3 - SAP Modules, NetWeaver, Customer Relationship Management, Business Warehouse, Advanced Planner and Optimizer. **ABAP/4**: Workbench - Workbench Tools - ABAP/4 Data Dictionary - ABAP/4 Repository Information – Structure of ABAP/4 program - ABAP/4 syntax – Data types – Constants and Variables – Statements : DATA, PARAMETERS, TABLE, MOVE, MOVE-CORRESPONDING, CLEAR, WRITE,CHECK, FORMAT. LOOP STRUCTURES. Sample programs.

PART A (2 MARKS)

1. **Define SAP.**
    - Systems Applications and Products in Data Processing (SAP) is business software that integrates all applications running in an organization.
    - These applications represent various modules on the basis of business areas, such as finance, production and planning, and sales and distribution, and are jointly executed to accomplish the overall business logic.
    - SAP integrates these modules by creating a centralized database for all the applications running in an organization. We can customize SAP according to your requirements by using the Advanced Business Application Programming (ABAP) language.

2. **Differentiate SAP R/2 and SAP R/3.**

SAP R/2	SAP R/3
SAP R/2: is 2-tier architecture. In which all 3 layers [Presentation + Application +Database] are installed in two separate systems/server. (Server One – Presentation , Server Two – Application +Database)	SAP R/3: is 3-tier architecture. In which all 3 layers [Presentation + Application +Database] are installed in three separate systems/server. (server One -Presentation, server Two – Application, Server Three – Database)

3. **Define SAP R/3.**
    - It is based on a client-server model, was officially launched on July 6, 1992. This version is compatible with multiple platforms and operating systems, such as UNIX and Microsoft Windows.
    - SAP R/3 introduced a new era of business software—from mainframe computing architecture to a three-tier architecture consisting of the Database layer, the Application layer (business logic), and the Presentation layer.
    - The SAP R/3 system contains various standard tables to execute various types of processes, such as reading data from the tables or processing the entries stored in a table.
    - The SAP R/3 system uses these databases to handle the queries of the users

4. **What are the types of views in SAP R/3 system?**

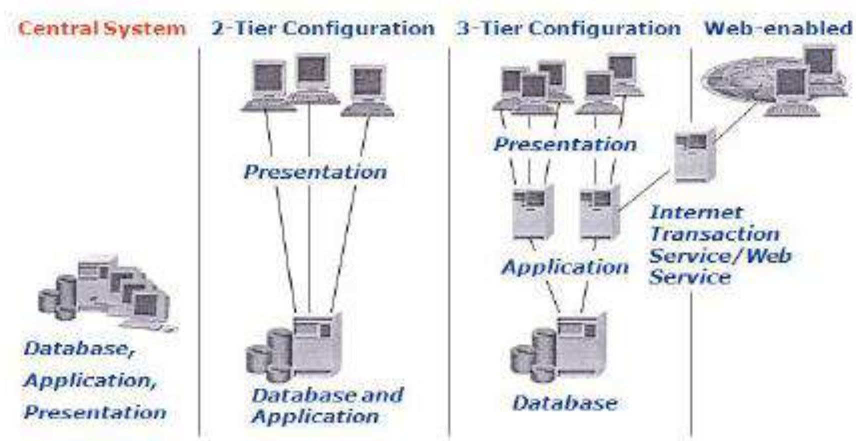
    The SAP R/3 system consists of the following three types of views:
    - Logical view
    - Software-oriented view
    - User-oriented view

5. **Name any five modules in SAP.**

    The SAP R/3 system provides the following set of applications, also known as functional modules, functional areas, or application areas:
    - Financial Accounting (FI)
    - Production Planning (PP)
    - Material Management (MM)
    - Sales and Distribution (SD)
    - Controlling (CO)
    - Asset Management (AM)
    - Human Resources (HR)
    - Project System (PS)
    - Industry Solutions (IS)
    - Plant Maintenance (PM)
    - Quality Management (QM)
    - Workflow (WF)

6. **List the characteristics and issues of SAP R/2 with diagram.**
    - The SAP R/2 system was introduced in 1980.
    - SAP R/2 was a packaged software application on a mainframe computer, which used the time-sharing feature to integrate the functions or business areas of an enterprise, such as accounting, manufacturing processes, supply chain logistics, and human resources.

- SAP R/2 system was based on a two-tier client-server architecture, where an SAP client connects to an SAP server to access the data stored in the SAP database.
- SAP R/2 was implemented on the mainframe databases, such as DB/2, IMS, and Adabas.
- The SAP R/2 system was designed to handle different languages and currencies.
- SAP R/2 system delivered a higher level of stability compared to the earlier version.



7. What is the purpose of NetWeaver?

- NetWeaver is an application builder from SAP for integrating business processes and databases from a number of sources while exploiting the leading Web services technologies.
- NetWeaver is getting a lot of industry attention as the first fully interoperable Web-based cross-application platform that can be used to develop not only SAP applications but others as well.
- NetWeaver allows a developer to integrate information and processes from geographically dispersed locations using diverse technologies, including Microsoft's .NET, IBM's WebSphere and Sun's Java technologies.

8. Write a note on VM Container.

- The VM Container technology enables a Java VM to be integrated into the ABAP work process.
- The execution of applications on the Java VM integrated in the ABAP work process is only intended for components developed by SAP.
- The VM Container technology in connection with a Java VM integrated in the ABAP work process

9. Define SAP CRM.

- SAP CRM is a part of SAP business suite. It can implement customized business processes, integrate to other SAP and non-SAP systems, and help achieve CRM strategies.
- SAP CRM can help an organization to stay connected to customers. This way organization can achieve customer expectations with the types of services and products that he or she actually needs.

10. What are the goals of CRM?

The goals of CRM are.

- Providing better and best customer service
- Selling products more effectively
- Discovering new customers and maintaining relation with existing customers.

11. List out the advantages of CRM.

- Increases customer satisfaction and attract new customers.
- Growth in number of customers in a high competition business world. It reduces cost and maximize profits.
- Boosts employee productivity and morale.

12. Draft a note on SAP business warehouse.

- SAP Business Warehouse (BW) is one of the important technical module of SAP and it stands for business information warehouse. The most important thing in the running of a business is data.
- Every business activity generates data. This data is used by a company is all areas of its business; It is used by the employees of the company in their work, to help them make a better decision.

13. What is meant by SAP APO?

- SAP APO stands for advanced planning and optimization and it is one of the most important component of SAP SCM (supply chain management).
- SAP is one the biggest vendor of business ERP application software in the world. One of the products that SAP has worked very hard on is the Supply Chain Management business suite, which helps a business to manage its supply chain better, while adjusting to constant changes in market conditions and holding on to their business advantage simultaneously.

14. Mention the important components of SAP APO.

The important components of SAP APO (advanced planning and optimization) are as follows.

- Demand Planning
- Supply Network Planning
- Production Planning and Detailed Scheduling (PP/DS)
- Global Availability and Available to Promise (ATP)

15. Define ABAP workbench.

- ABAP Workbench is a graphical programming environment in the SAP R/3 system to develop different applications by using the ABAP language.
- It provides different tools, such as ABAP Dictionary, ABAP Editor, and Screen Painter, to create ABAP applications.
- Using these tools, you can perform different tasks, such as defining data structures in ABAP Dictionary, developing data programs in ABAP Editor, and designing interfaces in Screen Painter and Menu Painter. Besides these tasks, you can also create user-defined reports, transactions, and enhancements within ABAP.

16. List out the different tools provided by ABAP workbench.

Different tools provided by ABAP Workbench:

- ABAP Dictionary
- ABAP Editor
- Class Builder
- Function Builder
- Screen Painter
- Menu Painter
- Object Navigator
- Maintain Message
- ABAP Text Elements
- Maintain Transaction

17. What is of ABAP dictionary?

- ABAP Dictionary is one of the important tools of ABAP Workbench. It is used to create and manage data definitions without redundancies.
- ABAP Dictionary always provides the updated information of an object to all the system components.

- The presence and role of ABAP Dictionary ensures that data stored in an SAP system is consistent and secure.

18. Draft a note on ABAP Front-End New and Old editor.

Front-End Editor (New)

- Front-End Editor (New) is the latest programming tool to create applications in an SAP system.
- It provides various new features, such as code hints, syntax highlighting, and auto-completion of language structures and elements

Front-End Editor (Old)

- As stated earlier, Front-End Editor (New) is used to edit the source code of the development objects.
- The classic mode of Front-End Editor (Old) is also used to edit the development objects, such as ABAP programs, function modules, and type groups. The source code of a program in Front- End Editor (Old) appears in plain text.

19. Write the ABAP syntax.

- ABAP programs are made up of individual statements.
- The first word in the statement is called an ABAP keyword.
- Each statement must end with a period.
- Statements can be intended.
- Can have multiple statements in a single line.

20. List the different types of ABAP data types.

The following are the different types of data type. They are

- Data elements
- Structure
- Table Types

21. Write a note on ABAP/4 repository information.

- The description of the ABAP Repository Information System given here is restricted to aspects of relevance for work with the Data Modeler. Using the ABAP Repository Information System, you can search for all modeling objects of the Data Modeler that you wish to display or edit.
- The ABAP Repository Information System provides you with the two basic functions Find and Where-used list.
- The Find function allows you to find objects of a specific object class.
- The Where-used list function allows you to determine the other objects in which a particular object is used.

22. Define MOVE statement and write the syntax for MOVE statement.

- The MOVE statement is used to assign the value of a data object to another data object; that is, to transfer the content of one field to another.

  Syntax:

  MOVE <f1> TO <f2>.

  Where <f1> is the data object whose data has to be transferred to another data object, <f2>. The equivalent statement for the MOVE statement is <f2> = <f1>.
- MOVE statement to assign values to multiple data objects, such as <f4> = <f3> =<f2> = <f1>.

  Syntax of MOVE statement for multiple assignments:

  Move <f1> TO <f2>

  Move <f2> TO <f3>

  Move <f3> TO <f4>

23. What is the purpose of CLEAR statement?

- It is used to reset the value of a data object. Resetting the values of the data objects depends on the data type of the data objects.

  CLEAR <f>.

  In this syntax, <f> is either a field or a field string.

24. Write a sample program for CLEAR statement.

The sample program:

```
REPORT ZCLEAR.
*-----------------------------------------*
*/ Data Declarations
*/
DATA number TYPE I VALUE '80'.
*-----------------------------------------* WRITE number.
*----------------------------------------**
/Using CLEAR statement
CLEAR number. WRITE / number.
*-------------------------------------*
```

25. Describe WHILE statement and show the sample program using WHILE statement.

- Conditional loops execute several statements only after certain conditions are fulfilled. This can be accomplished with the WHILE statement.
- The WHILE statement allows you to execute a block of statements so long as the condition mentioned in the WHILE statement evaluates to true.
- The syntax of the WHILE statement is:

```
WHILE <condition> [VARY <f> FROM <f1> NEXT <f2>]
        [statement_block]
ENDWHILE.
```

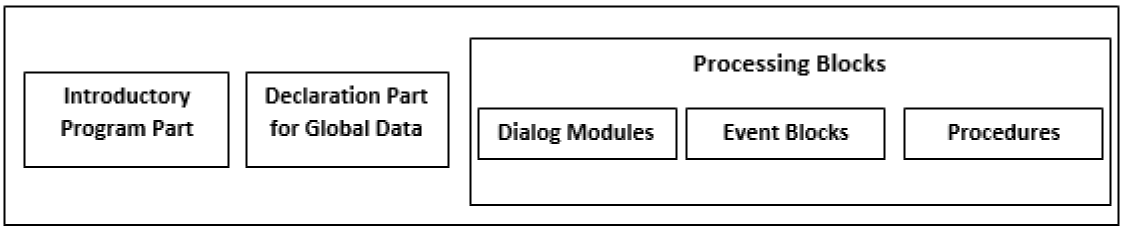
Sample program

```
REPORT ZWHILE_DEMO.
*---------------------------------*
*/ Data Declarations
*/
DATA: LENGTH TYPE I VALUE 0, VAR1 TYPE I VALUE 0,
TEXT1 (50) TYPE C VALUE 'Calculating the Length of the Text
String'.
*----------------------------------* VAR1 = STRLEN (TEXT1).
*----------------------------------*
*/Using WHILE statement WHILE TEXT1 NE SPACE. WRITE TEXT1 (1).
LENGTH = SY-INDEX.
SHIFT TEXT1. ENDWHILE.
*----------------------------------* WRITE: / 'STRLEN: ',
VAR1.
```

WRITE: / 'Length of string:' LENGTH.

26. Show the structure of ABAP program.

- The structure of an ABAP program includes the following:
- Introductory program part
- Global declaration part
- Processing blocks (consisting of different functions, such as procedures, dialog box, and event blocks) below figure shows the structure of an ABAP program:



27. Mention the different types of ABAP programs.

- The program types that can be selected are:
    1. Executable programs
    2. Module pools
    3. Subroutine pools
    4. Include programs
- Besides the preceding program types, some other types of ABAP programs exist that are not created by ABAP Editor. Instead, ABAP Workbench provides specific tools to create and maintain these programs. These program types are:
    1. Function pools
    2. Class pools
    3. Interface pools

28. What are the statements are used to declare variable statically in ABAP?

The statements that declared variable statically are

- DATA
- STATICS
- CLASS-DATA

- PARAMETERS
- SELECT- OPTIONS
- RANGES

29. Write the syntax for CONSTANT statement.

- Constants are named data objects created statically by using declarative statements

  The **syntax** of the CONSTANTS statement is:

  CONSTANTS <f> ... [TYPE <type>|LIKE <obj>]... [VALUE <val>].

- Notice that the syntax of the CONSTANTS statement is similar to the DATA statement.

30. List out the different types of FORMAT statements available in ABAP and write its syntax.

- Different types of syntaxes are available for specific actions performed with the FORMAT statement

FORMAT Statement	**Description**
FORMAT <option1> [ON|OFF] <option2> [ON|OFF]........	Specifies that the formatting options are set statically in a program. The option mentioned in the <option> expression is applicable to all output until the formatting option is turned off by using the OFF clause, as indicated in the statement. Note that the ON or OFF clause is optional.
FORMAT <option1> = <var1> <option2> = <var2>....	Specifies that the formatting options are set dynamically at runtime. The variables var1, var2, ... are interpreted as numbers and should be declared with the data type as I. If the content in the var1 and var2 variables is zero, the variable has the same effect as the OFF clause. If the numbers in the var1 and var2 variables are not equal to zero, either the color is shown according to the numbers stored in the variables or the variables have the same effect as that of the ON clause. Note that the formatting options are all set to their default values whenever they are used for the new event.
FORMAT RESET	Sets all the formatting options to off.

PART – B (11 MARKS)

1. Explain the history of SAP.

History of SAP Systems

- SAP is a translation of the German term Systeme, Anwendungen, und Produkte in der Datenverarbeitung.
- It was developed by SAP AG, Germany.
- The basic idea behind developing SAP was the need for standard application software that helps in real-time business processing.
- The development process began in 1972 with five IBM employees: Dietmar Hopp, Hans-Werner Hector, Hasso Plattner, Klaus Tschira, and Claus Wellenreuther in Mannheim, Germany.

SAP R/1 Systems

- A year later, the first financial and accounting software was developed; it formed the basis for continuous development of other software components, which later came to be known as the SAP R/1 system.
- Here, R stands for real-time data processing and 1 indicates single-tier architecture, which means that the three networking layers, Presentation, Application, and Database, on which the architecture of SAP depends, are implemented on a single system.
- SAP ensures efficient and synchronous communication among different business modules, such as sales and distribution, production planning, and material management, within an organization.
- These modules communicate with each other so that any change made in one module is communicated instantly to the other modules, thereby ensuring effective transfer of information

The SAP R/2 system

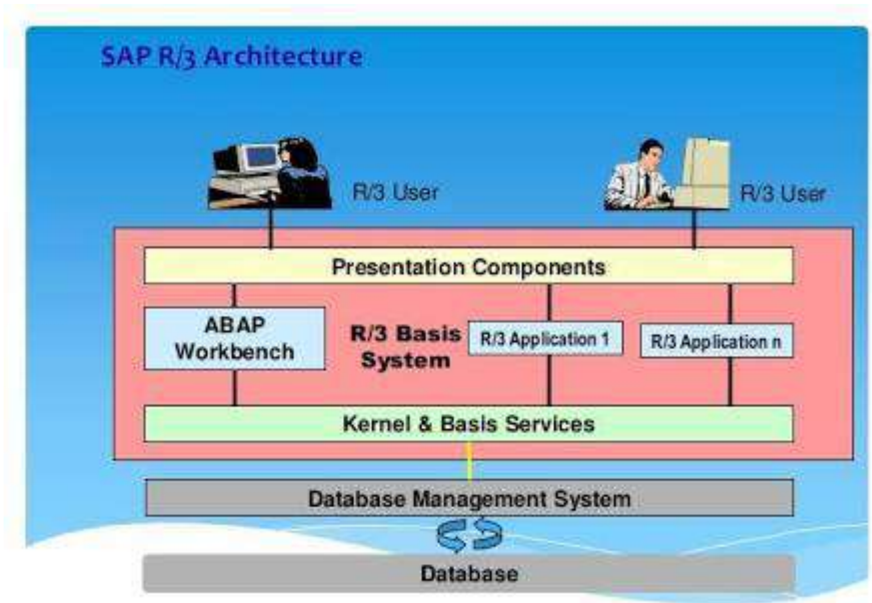
- It was introduced in 1980.
- SAP R/2 was a packaged software application on a mainframe computer, which used the time-sharing feature to integrate the functions or business areas of an enterprise, such as accounting, manufacturing processes, supply chain logistics, and human resources.
- The SAP R/2 system was based on a two-tier client-server architecture, where an SAP client connects to an SAP server to access the data stored in the SAP database.
- Keeping in mind that SAP customers belong to different nations and regions, the SAP R/2 system was designed to handle different languages and currencies.

SAP R/3 Systems

- It is based on a client-server model, was officially launched on July 6, 1992.
- This version is compatible with multiple platforms and operating systems, such as UNIX and Microsoft Windows.
- SAP R/3 introduced a new era of business software—from mainframe computing architecture to a three-tier architecture consisting of the Database layer, the Application layer (business logic), and the Presentation layer.
- The three-tier architecture of the client-server model is preferred to the mainframe computing architecture as the standard in business software because a user can make changes or scale a particular layer without making changes in the entire system.
- The SAP R/3 system is a customized software with predefined features that you can turn on or off according to your requirements.
- The SAP R/3 system contains various standard tables to execute various types of processes, such as reading data from the tables or processing the entries stored in a table.
- The SAP R/3 system uses these databases to handle the queries of the users.
- With the passage of time, a business suite that would run on a single database was required. This led to the introduction of the mySAP ERP application as a follow-up product to the SAP R/3 system.

2. **Discuss in detail about architecture of SAP R/3.**

- As stated earlier, the SAP R/3 system evolved from the SAP R/2 system, which was a mainframe.
- The SAP R/3 system is based on the three-tier architecture of the client-server model.
- SAP R/3 architecture shows how the R/3 Basis system forms a central platform within the R/3 system. The architecture of the SAP R/3 system distributes the workload to multiple R/3 systems.



SAP R/3 Architecture

- The link between these systems is established with the help of a network.
- The SAP R/3 system is implemented in such a way that the Presentation, Application, and Database layers are distributed among individual computers in the SAP R/3 architecture.
- The SAP R/3 system consists of the following three types of views:
    1. Logical view
    2. Software-oriented view
    3. User-oriented view

The Logical View

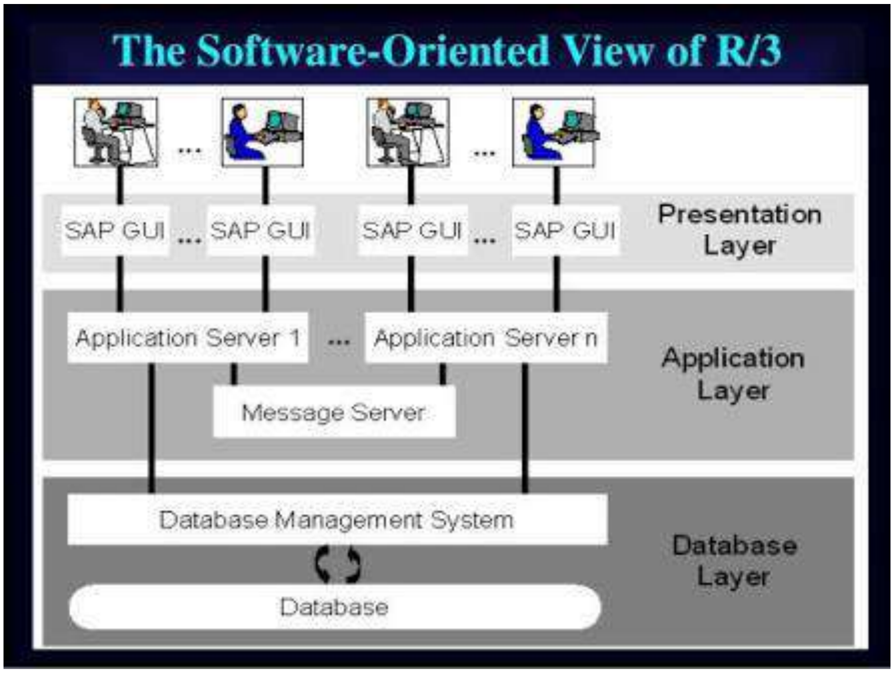
- The logical view represents the functionality of the SAP system. In this context, the R/3 Basis component controls the functionality and proper functioning of the SAP system.
- Therefore, in the logical view of the SAP R/3 system, we describe the services provided by the R/3 Basis component that help to execute SAP applications.

The following is a description of the various services provided by the R/3 Basis component:

- **Kernel and Basis services**—provide a runtime environment for all R/3 applications.
  The tasks of the Kernel and Basis services are as follows:
  - Executing all R/3 applications on software processors (virtual machines).
  - Handling multiple users and administrative tasks in the SAP R/3 system
  - Accessing the database in the SAP R/3 system.
  - Facilitating communication of SAP R/3 applications with other SAP R/3 systems and with non- SAP systems.
  - Monitoring and controlling the SAP R/3 system when the system is running.
- **ABAP Workbench service**—provides a programming environment to create ABAP programs by using various tools, such as the ABAP Dictionary, ABAP Editor, and Screen Painter.
- **Presentation Components service**—Helps users to interact with SAP R/3 applications by using the presentation components (interfaces) of these applications.

The Software-Oriented View

- The software-oriented view displays various types of software components that collectively constitute the SAP R/3 system.
- It consists of SAP Graphical User Interface (GUI) components and Application servers, as well as a Message server, which make up the SAP R/3 system.
- Since the SAP R/3 system is a multitier client-server system, the individual software components are arranged in tiers.
- These components act as either clients or servers, based on their position and role in a network.



- Software-oriented view of the SAP R/3 system consists of the following three layers:
    1. Presentation layer
    2. Application layer
    3. Database layer

Presentation Layer

- The Presentation layer consists of one or more servers that act as an interface between the SAP R/3 system and its users, who interact with the system with the help of well-defined SAP GUI components.
- For example, using these components, users can enter a request, to display the contents of a database table.
- The Presentation layer then passes the request to the Application server, which processes the request and returns a result, which is then displayed to the user in the Presentation layer.

Application Layer

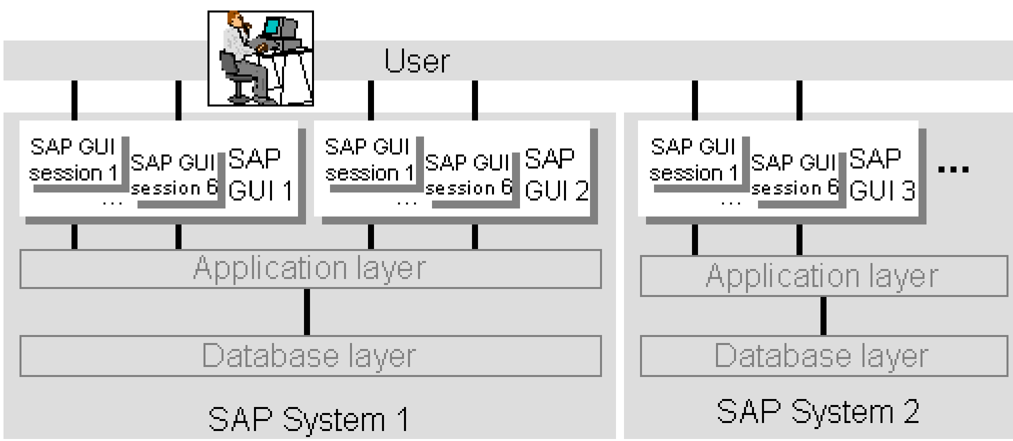
- The Application layer executes the application logic in the SAP R/3 architecture.
- This layer consists of one or more Application servers and Message servers.
- Application servers are used to send user requests from the Presentation server to the Database server and retrieve information from the Database server as a response to these requests.
- Application servers are connected to Database servers with the help of the local area network

Database Layer

- The Database layer of the SAP R/3 architecture comprises the central database system.
- The central database system has two components, DBMS and the database itself. The SAP R/3 system supports various databases, such as Adabas D, DB2/400 (on AS/400), DB2/Common Server, DB2/MVS, Informix, Microsoft SQL Server, Oracle, and Oracle Parallel Server.
- The database in the SAP R/3 system stores the entire information of the system, except the master and transaction data.

The User-Oriented View

- The user-oriented view displays the GUI of the R/3 system in the form of windows on the screen.
- These windows are created by the Presentation layer.
- To view these windows, the user has to start the SAP GUI utility, called the SAP Logon program, or simply SAP Logon.
- After starting the SAP Logon program, the user selects an SAP R/3 system from the SAP Logon screen.



User Oriented View

- The SAP Logon program then connects to the Message server of the R/3 Basis system in the selected SAP R/3 system and retrieves the address of a suitable Application server; i.e., the Application server with the lightest load.
- The SAP Logon program then starts the SAP GUI connected to the Application server.

- The SAP GUI starts the logon screen. After the user successfully logs on, the initial screen of the R/3 system appears. This initial screen starts the first session of the SAP R/3 system. Figure 1.5 shows the user-oriented view of the SAP R/3 system:
- A user can open a maximum of six sessions within a single SAP GUI. Each session acts as an independent SAP GUI. You can simultaneously run different applications on multiple open R/3 sessions. The processing in an opened R/3 session is independent of the other opened R/3 sessions.

3. Detail about Customer Relationship Management, Business Warehouse and Advanced Planner and Optimizer.

<u>Customer Relationship Management:</u>

What is SAP CRM?

- SAP CRM stands for customer relationship management. SAP CRM module is one of the best tool provided by SAP and <u>supports customer related process end to end</u>.
- The main concept of SAP CRM is managing and maintaining relationship with customers by firms. It deals with acquisition of customers and to its final termination.
- SAP CRM enable a firm to obtain a <u>360 degree over there customers and various functions such as accounts receivable (A/R), invoicing, fulfilment, delivery, decision making and various functions</u>.
- SAP CRM module is integrated with other SAP modules such as SCM (Supply chain management), PLM (Product life cycle management), and SRM (Supplier relationship management) and with various modules.

Why CRM

- To have good relationship with customers and attract new customers.
- Effective Communication with customers.
- To increase the sales and maximize the profits.
- Analyzing customers, vendors and partners.

Goals of CRM:

- Providing better and best customer service
- Selling products more effectively
- Discovering new customers and maintaining relation with existing customers.

Versions:

- SAP CRM 7.0 released in 2009
- SAP CRM 6.0 released in 2007
- SAP CRM 5.0 released in 2006

ERP with SAP CRM

- SAP CRM is one of the import tool in SAP ERP functionality.
- CRM enables the firms to know about their customers and maximize profit and expand the revenue.
- It enable trade promotional management to improve visibility and control in to the trade promotion process

CRM key Features

- SAP CRM consists of three sub modules – Marketing, Sales & Services

Advantages of CRM

- Increases customer satisfaction and attract new customers.
- Growth in number of customers in a high competition business world. It reduces cost and maximize profits.
- Boosts employee productivity and morale.

Business Warehouse:

- SAP BW is one of the important technical module of SAP and it stands for business information warehouse. The most important thing in the running of a business is data. Every business activity generates data. This data is used by a company is all areas of its business; It is used by the employees of the company in their work, to help them make a better decision.
- SAP business information warehouse shortened to business warehouse are BW is a package comprehensive business intelligence product center on a data warehousing platform.

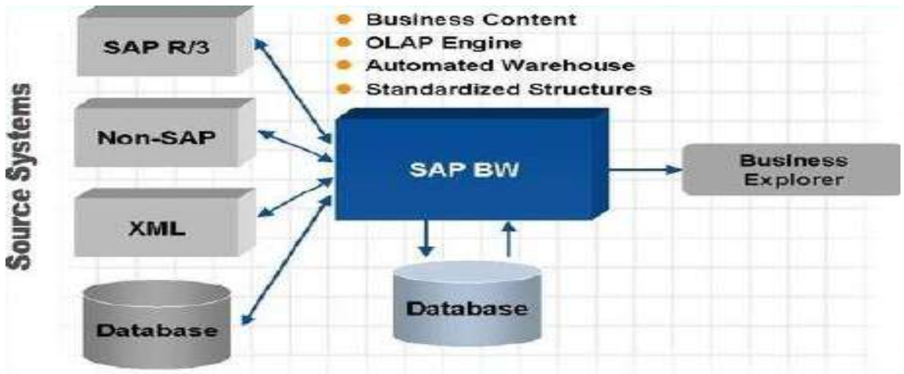
USES:

- Optionally designed for quick response time.
- Improves overall processing performance.
- No possibility of corrupting data.

SAP BW Provides the Following:

- Data warehousing system with optimized data structures for reporting and analysis

- Separate system
- OLAP engine and tools
- Comprehensive data-warehousing architectural base
- Automated data warehouse management
- Pre configuration using SAP global business know-how.

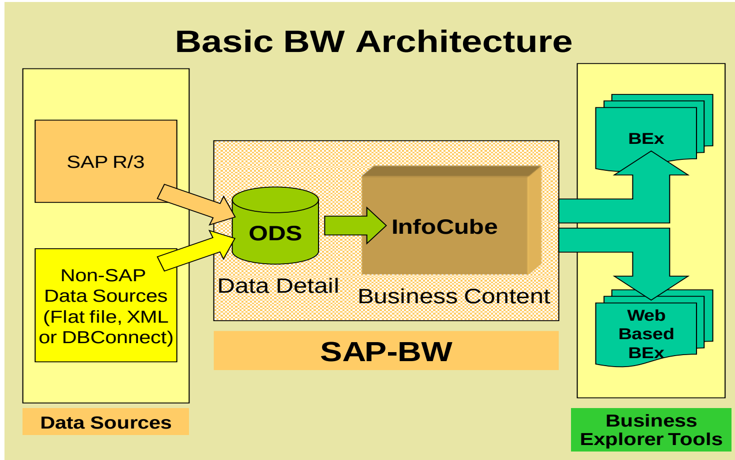


SAP BW Tools

- SAP BW has most comprehensive tools, business processes and functions for access and visualization.
- These tools enable the executive working with SAP BW to display the detailed information gathered by him along with extensive – planning data and business data analyses. He can do so in exhaustive detail or as a brief overview depending on how he wishes to present the data. Plus, he can present the data in any work environment of his choice – this could be in the form of Web based data or Microsoft Excel.
- The SAP BW tools enable the flexible distribution of the most vital data to all individuals involved in the process of decision making. All individuals can be notified of the latest information by email through a corporate eportal.

BW Reporting Tools

- MS-Excel-Based Analyzer (Web or GUI)
- Formatted Reporting
- Web Application Designer



Terminology and Object in SAP BW InfoObjects

- Business analysis-objects(customer, sales volume, and so on)are called infoObject in SAP BW
- Characteristics can be further divided into units, time characteristics, and technical characteristics.
- Key figures are all data fields that are used to store values or quantities.

Terminology and Objects in SAP BW InfoCube

- The central data containers that form the basis for reports and analyses in SAP BW are called infoCubes.
- They contain key figures (sales volumes, incoming order, actual characteristics (that is, to the master data of the SAP BW system [cost center, customers and so on).

Terminology and Objects in SAP BW InfoProvider

- InfoProvider is the super-ordinate term for an object that you can use to create reports in Business Explorer (BEx). InfoProviders deliver data from physical data stores..
    - InfoCubes
    - InfoObjects
    - ODS Objects

or logical views of physical data stores…

- InfoSets
- Remote Cubes
- Virtual InfoCubes
- MultiProviders

Terminology and Objects in SAP BW Operational Data Store (ODS)

- ODS is a data store in which data is stored at a basic lever (document lever).
- This data is often from various data sources and/or source systems.
- ODS objects should be reported very selectively because ODS objects are not built for reporting purposes other than in drill-down paths to retrieve a few detailed records.

Advantages of SAP BW

- Every business seeks a competitive advantage, an "XFactor" to give it a leg-up over the competition. SAP BW gives a business that advantage. This is because SAP BW does critical business functions, such as reporting, analysis and interpretation of business data.
- It helps a business to <u>optimize its business processes and access the most important and most relevant data</u> pertaining to the most crucial aspects of the business within seconds.
- The several tools available with SAP BW allows a business to integrate data from several SAP modules, , transform it into a more legible and easily understandable form, and consolidate data from various sources into one solid, concrete information that throws light on the intricacies of a business process.
- The SAP BW provides a well thought out, high performance, well-structured infrastructure that allows the end user to evaluate and interpret complex data, simplify it and to decipher its meaning.

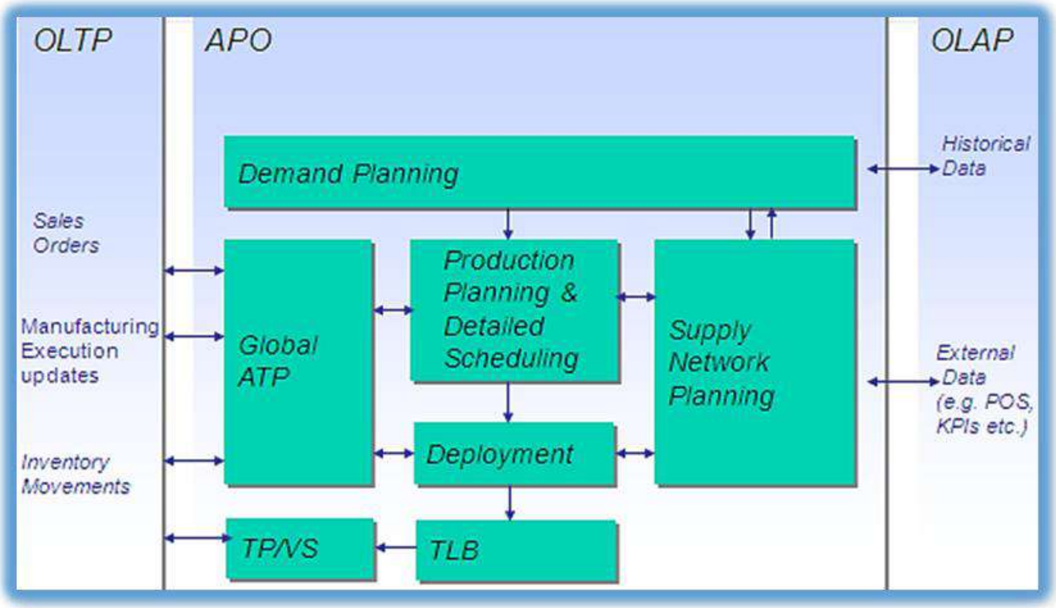
<u>SAP Advanced Planner and Optimizer:</u>

- Advanced Planning and Optimization (APO) is one of the most important module of SAP.
- SAP APO offers a <u>fully integrated pallet of functions that you use to plan and execute your supply chain processes.</u>
- SAP is one the biggest vendor of business ERP application software in the world.
- One of the products that SAP has worked very hard on is the Supply Chain Management business suite, which helps a business to manage its supply chain better, while adjusting to constant changes in market conditions and holding on to their business advantage simultaneously.

SAP APO supports the following:

- Business collaboration on a strategic, tactical, and operational planning level.
- Co-operation between partners at all stages of the supply chain process; from order receipt, stock monitoring, through final shipping of the product.

- Cultivation of customer and business partner relationships.
- Constant optimization and evaluation of the supply chain network's efficiency.



APO Architecture

Modules of APO

The sub modules of Advanced Planning and Optimization are:

1. Demand Planning.
2. Supply Network Planning / CIF.
3. Production Planning / Detailed Scheduling.
4. Global Available to promise / Deployment.
5. Transportation Management.

Demand Planning

- This is a subset of Demand Management.
- It is a business planning process that enables sales teams (and customers) to develop demand forecasts and inputs to feed various planning processes, production, inventory planning, procurement planning and revenue planning.
- Based on estimated product demand, a firm can plan for deployment of its resources to meet this demand.

- It is a bottom-up process as different from any top-down management process.
- It is also seen as a multistep operational SCM process to create reliable demand forecasts.
- Effective demand planning helps to improve accuracy of revenue forecasts, align inventory levels in line with demand changes and enhance product-wise/channel-wise profitability.
- Its purpose can be seen as to drive the supply chain to meet customer demands thro effective management of company resources.

Product Planning/Detailed Scheduling

The Production Planning and Detailed Scheduling (PP/DS) component:

- To create procurement proposals for in-house production or external procurement to cover product requirements.
- To optimize and plan the resource schedule and the order dates/times in detail.
- You can take the resource and component availability into account here.
- Reduce lead times
- Increase on-time delivery performance
- Increase the throughput of products and reduce the stock costs, through better coordination of resources, production, and procurement

Supply Network Planning (CIF)

- Supply planning, interactive supply planning, integration with other SCM components.
- Planning methods: Heuristics, optimization and CTM.
- Deployment.
- Transport load builder.
- Releasing supply plans to DP, PPDS.
- CIF core interface
- Monitoring and handling CIF errors.
- Comparison and reconciliation.
- Initial and change transfers, background jobs.

Global Available To Promise

- Global ATP is a type of availability check and business function that provides a <u>response to customer order inquiries, based on resource availability.</u>

- Execution of an ATP run generates available quantities of the requested product and delivery due dates.
- This helps companies support order promising and fulfillment, aiming to manage demand and match it to production and procurement plans while maximizing on customer service.

Transportation Management

- Purpose
    - Moving goods from one plant to another
    - Transportation planning
- Benefits
    - Transportation execution monitoring
- Key Process Steps
    - Create Stock Transfer Order
    - Creating an Outbound Delivery for Purchase Orders
    - Checking the Generated Outbound Delivery
    - Generating a Shipment Document
    - Creating a New Transfer Order Order with the Same Plant
    - Creating an Outbound Delivery for the New Stock Transfer Order
    - Adding a New Delivery to the Existing Shipment
    - Maintaining Delivery Picking and Posting Goods Issue
    - Goods Receipt with reference to Purchase Order

Benefits of Advanced Planner and Optimizer

- Reduction in costs due to simulation, 'what-if' capabilities.
- Integrating companies in the supply chain.
- Simultaneous planning of the problem to reach a synchronized sales, procurement, distribution and transport planning across the supply chain.
- Robust Supply Chain modeling capabilities.
- Decision making capabilities for the optimal location of facilities for the supply chain network
- All modules (DP/SNP/GATP/PPDS) are tightly integrated so that the supply chain entities can change their plans according to the needs of the market and this change can be seamlessly integrated across all the module.

- Tightly Integrated with ECC using Core Interface – enabling easy sync between planning and execution.

4. Describe about ABAP/4 workbench tools.

ABAP WORKBENCH

- ABAP Workbench is a graphical programming environment in the SAP R/3 system to develop different applications by using the ABAP language.
- It provides different tools, such as ABAP Dictionary, ABAP Editor, and Screen Painter, to create ABAP applications.
- Using these tools, you can perform different tasks, such as defining data structures in ABAP Dictionary, developing data programs in ABAP Editor, and designing interfaces in Screen Painter and Menu Painter. Besides these tasks, you can also create user-defined reports, transactions, and enhancements within ABAP

ABAP WORKBENCH TOOLS

- The ABAP Workbench tools are used to write ABAP code, design the screens, and create user interfaces. The following are different tools provided by ABAP Workbench.
    1. ABAP Dictionary
    2. ABAP Editor
    3. Class Builder
    4. Function Builder
    5. Screen Painter

6. Menu Painter
7. Object Navigator
8. Maintain Message
9. ABAP Text Elements
10. Maintain Transaction

1. ABAP Dictionary

- ABAP Dictionary is one of the important tools of ABAP Workbench.
- It is used to create and manage data definitions without redundancies. ABAP Dictionary always provides the updated information of an object to all the system components.
- The presence and role of ABAP Dictionary ensures that data stored in an SAP system is consistent and secure.
- The data definitions stored in ABAP Dictionary allow you to create various objects, such as tables and views.
- The objects stored in ABAP Dictionary are logically related to each other in the context of an application

The initial screen of ABAP Dictionary provides the following object types.

- **Database table**— Creates table definitions in ABAP Dictionary, independent of any database. The created table definition can then be used to create a table (with the same table structure) in the database.
- **View**— Creates view definitions in ABAP Dictionary. A view is a virtual table that does not store the data physically. It has a logical structure that represents or points to the data from one or more database tables. From the created view definition, you can create a view (with the same view structure) in the database.
- **Data type**— creates a definition of a user-defined type in ABAP Dictionary. The created definition of a user-defined type can then be used in a program to define data types and data objects.
- **Type group**— Creates groups of data types in ABAP Dictionary.
- **Domain**— Creates domains in ABAP Dictionary. A domain is used to describe the technical attributes of a field, such as a range of values and the type of data that are acceptable in a field.
- **Search help**— creates a help document (F4 help) or called input help for fields. A help document related to a field provides you help to enter a set of valid values in the field.

- **Lock object**— creates a local or lock object that helps synchronize the access of multiple users simultaneously. This is because an SAP system does not allow multiple users to access an object simultaneously. In such a situation, ABAP Dictionary allows you to create a lock or local object.

2. ABAP Editor

- ABAP Editor is used to create ABAP programs by writing code. You can use the ABAP Editor to define the class methods, function modules, screen flow logic, type groups, and logical databases for the ABAP Programs.
- ABAP Editor is available in three different modes for the front-end and back-end purposes.
    1. Front-End Editor (New)
    2. Front-End Editor (Old)
    3. Back-End Editor

Front-End Editor (New)

- Front-End Editor (New) is the latest programming tool to create applications in an SAP system.
- It provides various new features, such as code hints, syntax highlighting, and auto-completion of language structures and elements

Front-End Editor (Old)

- As stated earlier, Front-End Editor (New) is used to edit the source code of the development objects.
- The classic mode of Front-End Editor (Old) is also used to edit the development objects, such as ABAP programs, function modules, and type groups.
- The source code of a program in Front- End Editor (Old) appears in plain text.

Back-End Editor

- Back-End Editor, another classic editor, allows users to edit ABAP code.
- This editor is line-based and is used to perform normal editor functions, such as cut, copy, and paste.

3. CLASS BUILDER

Class Builder is a tool of ABAP Workbench used to create, define, change, and test the global ABAP classes and interfaces. ABAP classes and interfaces are object types defined in Repository Browser.

The following actions by using the Class Builder tool:

- Displaying an overview of global object types and their relationships in Class Browser
- Maintaining the already existing global classes or interfaces easily
- Creating new global classes and interfaces
- Implementing inheritance between two or more global classes
- Creating compound interfaces
- Creating and specifying the attributes, methods, and events of global classes and interfaces

4. FUNCTION BUILDER

- Function Builder plays an important role in defining and maintaining the ABAP functional modules. Function modules are procedures or routines defined in an ABAP program.
- They are stored in function groups, which act as containers for function modules that are logically related to each other.
- The Function Builder tool is used to create a function group and function module.

5. SCREEN PAINTER

- Screen Painter is used to design and manage screens and their elements. It facilitates users to create GUI screens for transactions.

Layout Editor	Manages a screen's layout. The screen layout describes both the screen elements and their layout.
Element List	Manages the ABAP Dictionary or program fields for a screen, and assigns a program field to the OK_CODE (a predefined field name) field in Screen Painter. Screen elements include I/O fields, text fields, check boxes, radio
Attributes	Manage screen's attributes. Screen attributes determine the type of a screen and the program to which the screen belongs.
Flow Logic	Manage screen's flow logic. The flow logic controls the flow of execution of a program.

6. MENU PAINTER

- Menu Painter is used to design user interfaces for ABAP programs.
- The main components of Menu Painter are GUI status, menu bars, menu lists, F-key settings, functions, and titles.

- In other words, using Menu Painter, you can create custom menus in SAP.

7. OBJECT NAVIGATOR

- Object Navigator is another important tool of ABAP Workbench.
- It helps create objects in an SAP system and navigate from one object to another.
- It acts as the central area of ABAP Workbench, where you can access any object of an SAP system.
- Object Navigator perform the following tasks:
    - Select a browser and navigate in the object list
    - Use the tools for development objects
    - Navigate from one window to another window
    - Perform syntax checks in the integrated window
    - Open an object in a new session using an Additional dialog box

8. MESSAGE MAINTENANCE

- Messages help an SAP system communicate with the users.
- For example, error messages display if the users make invalid entries on a screen, making the users aware that a wrong entry has been made. These messages are displayed by using a message class.
- Each message class has an ID and usually contains a set of messages. A message contains a single text line and may contain placeholders for variables.

9. ABAP TEXT ELEMENTS

- The text elements of program texts in different languages can be maintained easily with the help of a tool known as Text Element.
- Creation, maintenance, and translation of the text elements can be done with the help of the Text.
- The following are the various types of text elements:
    - **Text symbols—** Used by the Write statement.
    - **List and column headers**— Appear in an ABAP list.
    - **Selection texts**— Appear on selection screens.

10. MAINTAIN TRANSACTION

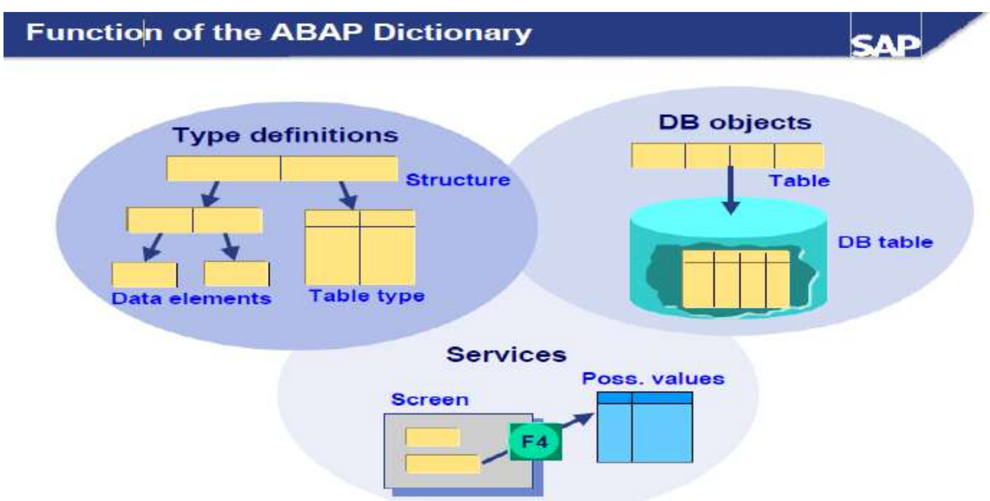
- Maintain Transaction is an ABAP Workbench tool used to create and manage user-defined transactions.

- Transaction screen contains a transaction code name, Ztransaction, in the Transaction Code field.

5. Explain in detail about ABAP/4 data dictionary and repository information.

ABAP/4 Data Dictionary

- The ABAP Dictionary is the central repository for all data definitions in the R/3 system.
- It permits common, non-redundant description of the data.
- It is also used to describe the logical structure of the objects in the application development environment, and shows how they are mapped to structure in the underlying physical database.
- FUNCTIONS OF ABAP DICTIONARY
- In the ABAP Dictionary you can create user-defined types (data elements, structures and table types) for use in ABAP programs or in interfaces of function modules.
- Database objects such as tables and database views can also be defined in the ABAP
- Dictionary and created with this definition in the database.
- The ABAP Dictionary also provides a number of services that support program development.



- Example, setting and releasing locks, defining an input help (F4 help) and attaching a field help (F1 help) to a screen field are supported.

DATABASE OBJECT IN ABAP DICTIONARY

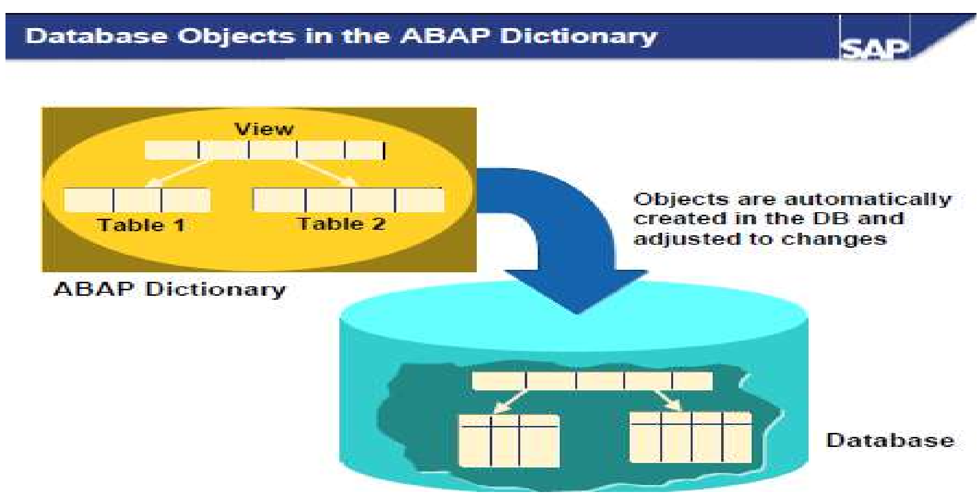
- Tables and database views can be defined in the ABAP Dictionary.
- These objects are created in the underlying database with this definition.
- Changes in the definition of a table or database view are also automatically made in the database.
- Indexes can be defined in the ABAP Dictionary to speed up access to data in a table.

There are three different type categories in the ABAP Dictionary:

- **Data elements:** Describe an elementary type by defining the data type, length
- **Structures:** Consist of components that can have any type.
- **Table types:** Describe the structure of an internal table.
- Any complex user-defined type can be built from these basic types

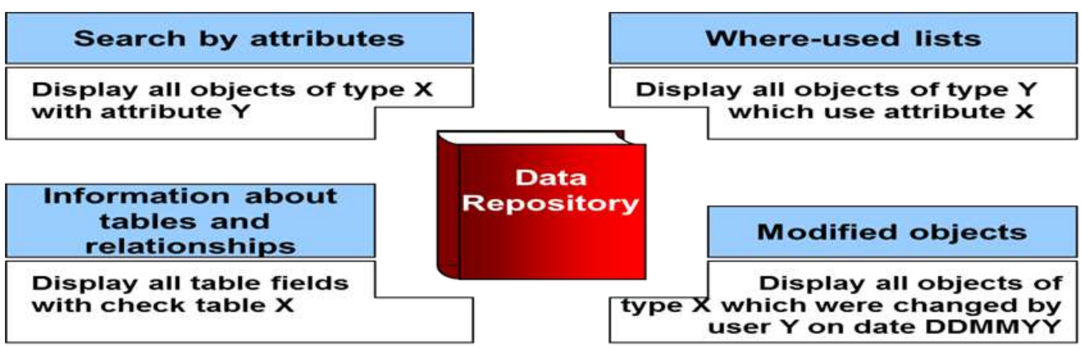
SERVICES OF ABAP DICTIONARY

- The ABAP Dictionary supports program development with a number of services:
- Input helps (F4 helps) for screen fields can be defined with search helps.
- Screen fields can easily be assigned a field help (F1 help) by creating documentation for the data element.
- An input check that ensures that the values entered are consistent can easily be defined for screen fields using foreign keys.
- The ABAP Dictionary provides support when you set and release locks.



ABAP/4 Repository Information

- Data Repository is a Central catalog that contains the descriptions of an organization's data and provides information about the relationships between the data and its use in programs and screens.
- The Repository, the central store for all ABAP workbench development objects, is also cross-client. It contains all Dictionary objects (tables, data elements, domains), and also all ABAP programs, menus and screens. Repository objects are grouped together to form packages.
- A repository is a place where data are stored and maintained.
- The ABAP Repository Information System is a tool which allows you to perform quick searches for information on all ABAP development objects
- The two basic functions available in the Repository Info System are the Find and Where-used list functions.



- The Find function allows you to search for objects from a specific object class meeting selection criteria.
- The Where-used list function allows you to determine the use of an object in other objects. Both functions produce a list of results which met the specified search criteria.
- The ABAP/4 Repository Information System allows you to obtain information about objects (tables, fields, domains, etc.) in the ABAP/4 Repository.

6. Detail about ABAP syntax and datatypes.

ABAP Syntax

- ABAP programs are made up of individual statements.
- The first word in the statement is called an ABAP keyword.
- Each statement must end with a period.
- Statements can be intended.
- Can have multiple statements in a single line
- Every statement provided in the ABAP programming language has a predined syntax associated with it.
- Apart from this syntax, certain conventions need to be followed while writing the syntax for the statements.

<table>
<tr><th>Term</th><th>Description</th></tr>
<tr><td>Statement</td><td>Specifies that ABAP statements and clauses are given in uppercase. It is not mandatory to use the ABAP keywords in uppercase; for instance, the WRITE keyword also can be Write or write.</td></tr>
<tr><td>Operand</td><td>Specifies that the operands used in the statement are in lowercase.</td></tr>
<tr><td>[ ]</td><td>Specifies that the element enclosed in the square brackets is optional.</td></tr>
<tr><td>|

( )</td><td>Specifies that there are elements on both sides of this symbol, and you can use either element in a program.
Specifies that parentheses is a part of the syntax of a statement and must be used while using the statement.</td></tr>
<tr><td>,</td><td>Separates one or more variables that you need to specify in a statement.</td></tr>
<tr><td>f1, f2</td><td>Represents variables indicated with indices. You can list as many variables in a program as you want.</td></tr>
<tr><td>............</td><td>Represents the rest of the code in a program besides the syntax of ABAP statements.</td></tr>
</table>

Data Types

- Data types describe data objects. ABAP provides predefined data types that can be used in an ABAP program.
- Apart from the predefined data types, data types can be customized either locally in a program or globally in the SAP R/3 Repository.
- The following are three kinds of data types:
    1. Elementary types
    2. Complex types

3. Reference types

Describing Elementary Types

- Elementary types are the smallest undivided unit of predefined data types. These elementary types are already defined in the SAP system and are visible in all ABAP programs.
- ABAP programs use these predefined elementary types to define local data types (structures) and data objects (fields) used in a program.
- The following are two types of predefined elementary data types, which can be used in an ABAP program:
  i. Fixed-length
  ii. Variable-length
- These types are not derived from other types. The main difference between the fixed-and variable-length types is that the memory space required by the data objects of variable-length data types can change dynamically at runtime.

The Fixed Length Data Type

- As the name suggests, the length of the fixed length data types are fixed at runtime.
- The fixed-length data types are:
  1. Character types
  2. Hexadecimal types
  3. Numeric types

Data Type	Initial Field Length (in bytes)	Valid Field Length (in bytes)	Initial Value	Description
Numeric Types				
I	4	4	0	Specifies an integer (whole number)
F	8	8	0	Represents a floating point number
P	8	1-16	0	Specifies a packed number
Character Type				
C	1	1-65535	'……..'	Denotes a text field (alphanumeric characters)
D	8	8	'00000000	Specifies a date field

				(Format:YYYYMMDD)
N	1	1-65535	‘0….0’	Specifies a numeric text field (numeric characters)

Variable-length data type

- A variable-length data type is used for data objects whose length cannot be fixed, as the length varies according to the specified data.
- The variable-length data type is of two types, STRING and XSTRING described in below table.

Variable Length Data Type	**Description**
STRING	Specifies a sequence of characters with variable lengths. A string can contain any number of alphanumeric characters. The length of the string can be calculated by multiplying the number of characters with the length required for the internal representation of a single character.
XSTRING	Represents a string used for byte strings of a hexadecimal type. A byte string has a variable string and can contain any number of bytes. The length of a byte string is same as the number of bytes.

7. Explicate about constants and variables.

Variables

- Variables are named data objects used to store values within the allotted memory area of a program.
- As the name suggests, users can change the content of variables with the help of ABAP statements.

- Below table shows the statements used to declare a variable statically:

Statement	Description
DATA	Declares the variable whose lifetime is linked to the context of the declaration.
STATICS	Declares the variables that can be used in subroutines, function modules, and static methods.
CLASS-DATA	Declares variables within the classes.
PARAMETERS	Declares the elementary data objects that are linked to input fields on a selection screen.
SELECT-OPTIONS	Declares the internal tables that are linked to input fields on a selection screen
RANGES	Declares internal tables with the same structure as defined in the SELECT- OPTIONS statement, provided the declared internal table is not linked to a selection screen.

- Apart from declaring the variables statically, you can also declare the variables dynamically; that is, whenever you call procedures, such as subroutines.
- For example, the variables declared in the FORM statement (also known as formal parameters) inherit the technical characteristics, such as the data type and length of the parameters defined with the PERFORM statement (also known as actual parameters).
- In addition to this, the values of the actual parameters are assigned to the formal parameters.
- Now, let's discuss the DATA statements in detail.

Data statement

- The DATA statement is used to declare variables in an ABAP program. Variables defined by the DATA statement have a predefined or user-defined data type.

  Syntax

  DATA <f>... [TYPE <type>|LIKE <obj>]... [VALUE <val>].

Description of the clauses of the DATA statement

- **<f>** - Specifies the name of a variable. The name of the variable can be up to 30 characters long.

- **TYPE <type>-** Specifies that the type of <f> variable. Any data type with fully specified technical attributes is known as <type>.
- The possible <type> allotted to a variable can be a non-generic predefined ABAP type (D, F, I, T, STRING, XSTRING) or any existing local data type in a program (the TYPES statement is used to define local data types in a program) or an ABAP Dictionary data type.
- **LIKE <obj>** - Shows that the declared variable name <f> inherits the same technical attributes as an existing data object, <obj>. Predefined data objects that do not need to be declared separately are represented by <obj>.
- **VALUE <val>-** Specifies the initial value of the <f> variable. In case you define an elementary fixed- length variable, the DATA statement automatically populates the value of the variable with the type-specific initial value, as listed in and Other possible values for <val> can be a literal, constant, or an explicit clause, such as IS INITIAL

The following conventions are used while naming a variable in an ABAP program:

- You cannot use special characters such as "+" and "," to name variables.
- The name of the predefined data objects cannot be changed.
- The name of the variable cannot be the same as any ABAP keyword or clause.
- The name of the variables must convey the meaning of the variable without the need for further comments.
- Hyphens are reserved to represent the components of structures.
- Therefore, avoid using hyphens in variable names.
- The underscore character can be used to separate compound words

The following code snippet shows how to define variables by using the DATA statement:

```
DATA      d1(2)  TYPE C.
DATA      d2     LIKE d1.
DATA min_value TYPE I VALUE 10.
```

- In the previous code snippet, d1 is a variable of C type,d2 is a variable of d1 type, and min_value is a variable of I type.

The following code snippet demonstrates another way to declare variables

```
*---------------------------------------------*
*/ Structure Declarations
```

```
*/
TYPES:          BEGIN    OF    number,
                Number_1 TYPE I,
                Number_2 TYPE p DECIMALS 2,
                End of number.
*---------------------------------------------*
*/Data Declarations
*/
DATA:           n_number TYPE number, Num
        LIKE n_number-Number_1, Date LIKE SY-DATUM,
        Year TYPE i.
```

CONSTANT

- Constants are named data objects created statically by using declarative statements.
- In a program, a constant is declared by assigning a value to it, which is stored in the program's memory area.
- The value assigned to a constant cannot be changed during the execution of the program. When the value of the constant is changed, a syntax or runtime error may occur.
- These named data objects are declared with the help of the CONSTANTS statement.

  The syntax of the CONSTANTS statement is:

  CONSTANTS <f> ... [TYPE <type>|LIKE <obj>]... [VALUE <val>].

- Notice that the syntax of the CONSTANTS statement is similar to the DATA statement.

The following code snippet shows how to define constants by using the CONSTANTS statement:

```
CONSTANTS: abc TYPE P DECIMALS 5 VALUE '1.23456', def TYPE C
VALUE IS INITAIL.
```

The following code snippet shows how to define complex constants:

```
*---------------------------------------------*
*/ Defining a complex constant
*/
CONSTANTS: BEGIN OF EMPLOYEE,
           Name (20) TYPE c VALUE 'SHIVAM SRIVASTAVA', Company(50)
           TYPE c VALUE 'Software Solutions Incorporation',
```

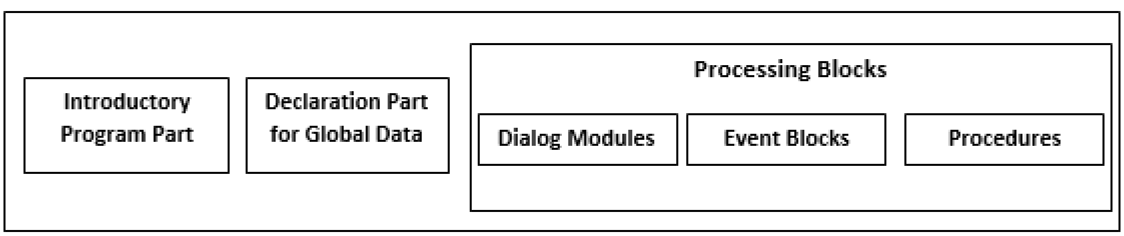
City (10) TYPE c VALUE 'New Delhi', Pincode(8)
TYPE i VALUE '110002',
END OF EMPLOYEE.

- In the preceding code snippet, EMPLOYEE is a complex constant that is composed of the Name, Company, City, and Pin code fields.

8. Sketch and explain in detail about structure of ABAP/4 program and its types.

Structure of ABAP/4 Program

- ABAP programs are responsible for data processing within the individual dialog steps of an application program. This means that the program cannot be constructed as a single sequential unit, but must be divided into sections that can be assigned to the individual dialog steps.
- To meet this requirement, ABAP programs have a modular structure. Each module is called a processing block.
- A processing block consists of a set of ABAP statements. When you run a program, you effectively call a series of processing blocks. They cannot be nested.
- The following diagram shows the structure of an ABAP program:



Structure of an ABAP program

The structure of an ABAP program includes the following:

1. Introductory program part
2. Global declaration part
3. Processing blocks (consisting of different functions, such as procedures, dialog box, and event blocks)

Introductory Program Part

- Every ABAP program begins with an introductory statement, such as REPORT, PROGRAM, or FUNCTION- POOL.
- Each program has a type, such as executable program, module pool program, or function group program, associated to it.
- The introductory statement of an ABAP program depends on the type of the ABAP program.
- Below table shows the introductory statements for different types of programs.

Introductory Statement	**Description of the Program Type**
REPORT	Represents an executable program.
PROGRAM	Represents a module pool program and subroutines
FUNCTION-POOL	Represents a function group.
CLASS-POOL	Represents a class pool program.
INTERFACE-POOL	Represents an interface pool program.
TYPE-POOL	Defines the type group.

Declaration of Global data, Class Definitions, and Selection Screens

You always declare global data, classes, and selection screens after the introductory program part of an ABAP program. The declaration of these objects includes:

- **Global data**— Defines global data in an ABAP program. Global data is specified by declaration statements, such as TYPES, TABLES, and DATA. The global data defined in a program is visible in all the internal processing blocks of the program.
- **Selection screens**— Represent special screens that are generated by using ABAP statements instead of Screen Painter. These screens allow you to enter either a single value for one or more fields or a selection criterion.
- **Local class definitions**— Contains the definitions of local classes in an ABAP program. Local class definitions are created with the help of the CLASS DEFINITION statement. The local classes are a part of the ABAP Objects, which are object-oriented extensions of ABAP.

Processing Blocks

- Another part of an ABAP program structure is the processing block.
- A processing block is a set of ABAP statements that represents a module of an ABAP program.
- As we know, ABAP programs are created to process data within the dialog steps of an application
- The second part of an ABAP program contains all of the processing blocks for the program. The following types of processing blocks are allowed:
    1. Dialog modules (no local data area)
    2. Event blocks (no local data area)
    3. Procedures (methods, subroutines and function modules with their own local data area).
- Whereas dialog modules and procedures are enclosed in the ABAP keywords which define them, event blocks are introduced with event keywords and concluded implicitly by the beginning of the next processing block.

Types of ABAP Program

- Each ABAP program has a program type that determines whether a program can be executed.
- An ABAP program can be executed by either entering the program name in the initial screen of ABAP Editor or using a transaction code in the Command field.
- The Type field of ABAP Editor is used to specify the type of a program. The program types that can be selected are:
    1. Executable programs
    2. Module pools
    3. Subroutine pools
    4. Include programs
- Besides the preceding program types, some other types of ABAP programs exist that are not created by ABAP Editor.
- Instead, ABAP Workbench provides specific tools to create and maintain these programs. These program types are:
    1. Function pools
    2. Class pools

3. Interface pools

Defining Executable Programs

- Executable programs are often called report programs because the source code of an executable program starts with the REPORT statement.
- These programs are created by processing the data stored in an SAP database.
- Executable programs cannot only retrieve the data from the database but also modify the data stored in the database.
- Executable programs can contain almost every kind of processing block, such as dialog modules, methods, and event blocks, except function modules and local classes.

Defining Module Pools

- A module pool is a program used to display data in and add functionality to a screen, such as a button, radio button, group box, or menu bar. You can write the screen flow logic using a module pool program; a subroutine is not used for this purpose

Defining Subroutine Pools

- A subroutine pool is a program that contains a collection of subroutines and local or global classes and interfaces that must be called externally in other ABAP programs.

Defining Include Programs

- Include programs are not complete programs because they do not have a memory and cannot be executed, unlike other stand-alone programs

Defining Function Pools

- Function pools, called function groups, are programs that contain function modules. A function pool program is created by using the FUNCTION-POOL statement.
- Each function pool contains global data, such as data objects, subroutines, or screens, which are shared by all the function modules of the function pool.

Defining Class Pools

- A class pool is a special ABAP program used to store global classes and interfaces. All the ABAP programs can access these global classes and interfaces.
- A class pool is created by using the CLASS-POOL statement or by using.

Defining Interface Pools

- Interface pools are programs that store global interfaces. An interface pool acts as a container that can store exactly one global interface. You can use an interface pool to implement the methods, which are defined in an interface of a class.

9. List out and explain the ABAP statements.

There are ten ABAP statements are available. They are

1. DATA
2. PARAMETERS
3. TABLE
4. MOVE
5. MOVE-CORRESPONDING
6. CLEAR
7. WRITE
8. CHECK
9. FORMAT
10. LOOP STRUCTURES

<u>1. DATA</u>

- The DATA statement is <u>used to declare variables</u> in an ABAP program. Variables defined by the DATA statement have a predefined or user-defined data type.

  Syntax

  DATA <f>... [TYPE <type>|LIKE <obj>]... [VALUE <val>].

Description of the clauses of the DATA statement

- **<f>** - Specifies the name of a variable. The name of the variable can be up to 30 characters long.
- **TYPE <type>-** Specifies that the type of <f> variable. Any data type with fully specified technical attributes is known as <type>.
- The possible <type> allotted to a variable can be a non-generic predefined ABAP type (D, F, I, T, STRING, XSTRING) or any existing local data type in a program (the TYPES statement is used to define local data types in a program) or an ABAP Dictionary data type.
- **LIKE <obj>** - Shows that the declared variable name <f> inherits the same technical attributes as an existing data object, <obj>. Predefined data objects that do not need to be declared separately are represented by <obj>.
- **VALUE <val>-** Specifies the initial value of the <f> variable. In case you define an elementary fixed- length variable, the DATA statement automatically populates the value of the variable with the type-specific initial value, as listed in and Other possible values for <val> can be a literal, constant, or an explicit clause, such as IS INITIAL

The following conventions are used while naming a variable in an ABAP program:

- You cannot use special characters such as "+" and "," to name variables.
- The name of the predefined data objects cannot be changed.
- The name of the variable cannot be the same as any ABAP keyword or clause.
- The name of the variables must convey the meaning of the variable without the need for further comments.
- Hyphens are reserved to represent the components of structures.
- Therefore, avoid using hyphens in variable names.
- The underscore character can be used to separate compound words

The following code snippet shows how to define variables by using the DATA statement:

```
DATA      d1(2)  TYPE C.
DATA      d2     LIKE d1.
DATA min_value TYPE I VALUE 10.
```

- In the previous code snippet, d1 is a variable of C type,d2 is a variable of d1 type, and min_value is a variable of I type.

The following code snippet demonstrates another way to declare variables

```
*----------------------------------------------*
```

```
*/ Structure Declarations
*/
TYPES:          BEGIN    OF    number,
                Number_1 TYPE I,
                Number_2 TYPE p DECIMALS 2,
                End of number.
*---------------------------------------------*
*/Data Declarations
*/
DATA:           n_number TYPE number, Num
         LIKE n_number-Number_1, Date LIKE SY-DATUM,
         Year TYPE i.
```

2. PARAMETER

- The PARAMETERS statement is used to enter the values of the variables on both the standard selection screen as well as user-defined selection screens.
- These variables (also known as parameters) are defined with the help of the PARAMETERS statement.
- Each parameter defined with the PARAMETERS statement appears as an input field on the relevant selection screen.
- The PARAMETERS statement is used to enter single values in the input fields.
- The flow of the program can be controlled with the help of parameters.

Syntax

```
PARAMETERS <p> [(<length>)] [TYPE <type>|LIKE <obj>] [DECIMALS <d>]
```

- In the preceding syntax, the use of the TYPE or LIKE clause is optional, but the data must be of the C type with length 1.

Describes the clauses used in the PARAMETERS statement:

- **<p>** - Represents the name of a parameter. The maximum length for naming parameters is 8 instead of 30, as in the DATA statement.
- **<length>** - Specifies the length of a variable.
- **TYPE <length>** - Specifies a data type with fully specified technical attributes. The possible <type> allotted to a variable can be a predefined ABAP type (D, I, T, STRING, XSTRING) or an ABAP Dictionary data type. The TYPE keyword shows that the declared parameter name <p> inherits the same technical attributes as that of an existing data type <type>.-

- **LIKE <obj> -** Specifies a predefined data object that is already present and need not be declared separately, such as system variables. Fields used to provide information about the current state of the SAP system are known as system variables. The LIKE keyword specifies that the declared parameter name <p> inherits the same technical attributes as that of the existing data object <obj>.
- **DECIMALS <d> -** Specifies the number of decimal places for numeric values.

Sample program using Using the PARAMETERS statement

```
REPORT ZPARAMETERDISPLAY. *---------------------------------------------*
*/Creating Parameters
*/
PARAMETERS: NAME (10) TYPE C, CLASS TYPE I,
SCORE TYPE P DECIMALS 2,
CONNECT TYPE MARA-MATNR.
*---------------------------------------------*
```

Four parameters are created and displayed on the standard selection screen. The parameters are:

- NAME—Represents a parameter of 10 characters
- CLASS—Specifies a parameter of integer type with the default size in bytes
- SCORE—Represents a packed type parameter, with values up to two decimal places
- CONNECT—Refers to the MARA-MATNR type of ABAP Dictionary

3. TABLE

- The TABLES statement is used to create a structure having the same name as a database table, view, or structure defined in ABAP Dictionary.
- This statement actually defines a table work area, which is a kind of interface work area, used to create in the shared area of a program.

  Syntax

  TABLES <dbtab>.
- In the preceding syntax, <dbtab> represents a structure with the same data type and name of a database table, a view, or a structure from ABAP Dictionary.

4. MOVE

- The MOVE statement is used to assign the value of a data object to another data object; that is, to transfer the content of one field to another.

  Syntax

  MOVE <f1> TO <f2>.

- In this syntax, <f1> is the data object whose data has to be transferred to another data object, <f2>. The equivalent statement for the MOVE statement is <f2> = <f1>. The MOVE statement assigns the value of one data object to another, but the value of the original data object remains unchanged.
- You can use the MOVE statement to assign values to multiple data objects, such as

  <f4> = <f3> = <f2> = <f1>.

- The following syntax shows how to use the MOVE statement for multiple assignments:

  MOVE <f1> TO <f2>.

  MOVE <f2> TO <f3>.

  MOVE <f3> TO <f4>.

Sample program using MOVE statement

```
REPORT ZMOVE_DEMO.
*-----------------------------------------------*
*/ Data Declarations
*/
DATA: Number_1(10) TYPE c,
Number_2 TYPE p DECIMALS 2, Number_3 TYPE i.
*-----------------------------------------------* Number_1 = 100.
*----------------------------------------------**
/MOVE statement */
MOVE '5.75' TO Number_2.
*-----------------------------------------------* Number_3 = Number_1.
WRITE: 'Number_1 =',Number_1.
WRITE: / 'Number_2 =',Number_2. WRITE: / 'Number_3 =',Number_3.
```

5. MOVE CORRESPONDENCE

- The MOVE-CORRESPONDING statement is used to assign values between the components of two or more structures.

Syntax

MOVE-CORRESPONDING <struct1> TO <struct2>.

6. CLEAR

- The CLEAR statement is used to reset the value of a data object. Resetting the values of the data objects depends on the data type of the data objects.
- For example, data objects of type I are set to zero and data objects of type C are set to null.

  Syntax

  CLEAR <f>.

Sample program using CLEAR statement

```
REPORT ZCLEAR.
*-----------------------------------------*
*/ Data Declarations
*/
DATA number TYPE I VALUE '80'.
*-----------------------------------------* WRITE number.
*----------------------------------------**
/Using CLEAR statement
CLEAR number. WRITE / number.
*-------------------------------------*
```

7. WRITE

WRITE TO

- The WRITE TO statement is an assignment statement that converts the content of the source field into a field of type C.

  Syntax

  WRITE <f1> TO <f2> [<option>].

- In the preceding syntax, the WRITE TO statement converts the content of the data object <f1> to type C and places the string into the <f2> variable. Listing 6.7 shows the use of the WRITE TO statement

Sample program using WRITE TO statement

```
REPORT ZWRITE_DEMO.
*--------------------------*
```

***/ Data Declarations**
***/ DATA: Name(10) TYPE C VALUE 'SHIVAM', TEXT(10).**

***/Using WRITE TO statement to show source and target names**
***/**
WRITE :'Source name is :', Name.WRITE: Name TO TEXT. WRITE: /'Target name is :', TEXT.

WRITE

- The WRITE statement is a <u>formatting statement used to display data on a screen</u>. There are different formatting options for the WRITE statement.

  Syntax

  WRITE <format> <f> <options>.

8. FORMAT

- Different types of syntaxes are available for specific actions performed with the FORMAT statement.
- **FORMAT <option1> [ON|OFF] <option2> [ON|OFF]........** -Specifies that the formatting options are set statically in a program. The option mentioned in the <option> expression is applicable to all output untill the formatting option is turned off by using the OFF clause, as indicated in the statement. Note that the ON or OFF clause is optional.
- **FORMAT <option1> = <var1> <option2> =<var2>....-**Specifies that the formatting options are set dynamically at runtime. The variables var1, var2, ... are interpreted as numbers and should be declared with the data type as I. If the content in the var1 and var2 variables is zero, the variable has the same effect as the OFF clause. If the numbers in the var1 and var2 variables are not equal to zero, either the color is shown according to the numbers stored in the variables or the variables have the same effect as that of the ON clause. Note that the formatting options are all set to their default values whenever they are used for the new event.
- **FORMAT RESET** -Sets all the formatting options to off.

Using the FORMAT Statement Clauses

- The FORMAT statement is also used to display the output in a colored format.

  Syntax

  FORMAT COLOR <num> [ON] INTENSIFIED [ON|OFF] INVERSE [ON|OFF].

The following syntax is used to set colors at runtime:

FORMAT COLOR = <const> INTENSIFIED = <int> INVERSE = <inv>.

- In both syntaxes, the COLOR clause is used to set the background color of a line

9. CHECK

- Conditionally leaves a loop or processing block.

  Syntax

  CHECK <logexp>.

- If the logical expression <logexp> is true, the system continues with the next statement. If it is false, processing within the loop is interrupted at the current loop pass, and the next loop pass is performed. Otherwise the system leaves the current processing block.
- In conjunction with selection tables, and inside GET events, you can use an extra variant of the CHECK statement.

10. LOOPING

- A statement block is executed repeatedly in a program by using loops in the program. The loops are categorized into two types:
  i. Unconditional loops by using the DO statement.
  ii. Conditional loops by using the WHILE statement

Unconditional Loops Using the DO Statement

- Unconditional loops repeatedly execute several statements without specifying any condition.
- The DO statement implements unconditional loops by executing a set of statements (statement block) several times unconditionally.

  Syntax

```
DO [n TIMES] [VARYING <f> FROM <fi> NEXT <f2>].
<statement block>
ENDDO.
```

Sample program using DO statement

```
REPORT demo_do.
*---------------------------------------------*
*/Using DO Statement
*/ DO.
WRITE / SY-INDEX. IF SY-INDEX = 3. EXIT.
ENDIF.
ENDDO.
*---------------------------------------------*
```

Conditional Loops Using the WHILE Statement

- Conditional loops execute several statements only after certain conditions are fulfilled. This can be accomplished with the WHILE statement.
- The WHILE statement allows you to execute a block of statements so long as the condition mentioned in the WHILE statement evaluates to true.

  Syntax

  ```
  WHILE <condition> [VARY <f> FROM <f1> NEXT <f2>]
  [statement_block]
  ENDWHILE.
  ```
- You can nest the WHILE loop any number of times and combine the nested WHILE loop with other loop forms.

TERMINATING LOOP

- A loop can be terminated prematurely with the help of two types of termination statements provided by
- ABAP. The two types of termination statements are:
  1. Statements that apply to a loop, such as CONTINUE, CHECK, and EXIT.
  2. Statements that apply to an entire processing block, such as STOP and REJECT.

- The CONTINUE statement can be used only in a loop statement. On the other hand, the CHECK and EXIT statements are context-sensitive. When used in the loop, the CHECK and EXIT statements are responsible for the execution of the loop itself. However, outside the loop, the statements terminate the entire processing block, such as a subroutine, dialog module, or event block, in which they occur.
- CONTINUE, CHECK, and EXIT can be used in looping statements, such as DO, WHILE, LOOP, and SELECT.

DEPARTMENT OF CSE

CS T61 ENTERPRISE SOLUTIONS

Unit-III – ORACLE SUITE

QUESTION BANK WITH ANSWER

Oracle Suite : Oracle Apps 11i - Application Framework - File System - Workflow Analysis - SQL / PLSQL fundamentals - Creating Forms - Oracle Reports. Oracle Electronic Data Interchange – functions of EDI – Data File Structure - Oracle Data, Oracle Database - Oracle Database - DW vs OLTP - DW Connectors.

PART A (2 MARKS)

1. What is Oracle Application Architecture?

The Oracle Applications Architecture is a framework for multi-tiered, distributed computing that supports Oracle Applications products. In this model, various servers are distributed among multiple levels, or tiers.

2. List the three tiers in an Oracle E-Business Suite installation.

- **Database tier**, which supports and manages the Oracle database
- **Application tier,** which supports and manages the various Applications components, and is sometimes known as the middle tier
- **Desktop tier,** which provides the user interface via an add-on component to a standard web browser

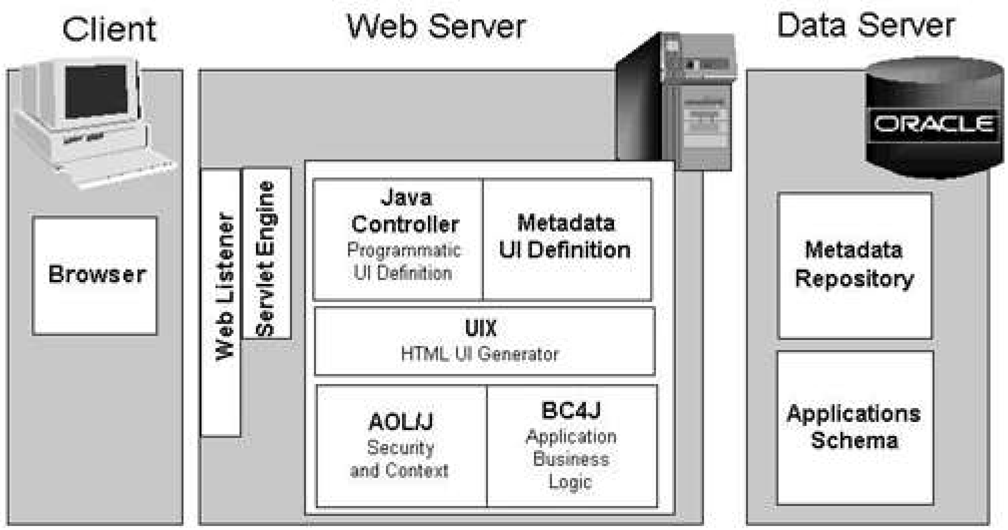
3. Define Web server.

- The Oracle HTTP server (powered by Apache) acts as the Web server.
- It processes the requests received over the network from the desktop clients, and includes additional components such as:
    - Web Listener
    - Java Servlet Engine
    - Java Server Pages (JSP)

4. What is Oracle Application Framework?

- The Oracle Applications Framework is the development platform for HTML-based applications.
- It consists of a Java-based application tier framework and associated services, designed to facilitate the rapid deployment of HTML-based applications.

5. Draw Oracle Applications Framework Architecture.



6. Write a note on Discoverer Server.

Discoverer is an ad hoc query, reporting, analysis, and publishing tool that allows business users at all levels of an organization to gain immediate access to information from data marts, data warehouses, and online transaction processing (OLTP) systems.

7. What are operations from Admin server?

- Upgrading Oracle Applications
- Applying database patches to Oracle Applications
- Maintaining Oracle Applications data

8. Write a note on Daily Business Intelligence (DBI).

- Daily Business Intelligence (DBI) is a reporting framework that is integrated with Oracle E-Business Suite.
- It replaces the Business Intelligence System (BIS), and includes a new set of materialized views that pre-summarize transaction data. Using Daily Business Intelligence overview pages, managers can view summarized information across multiple organizations, drilling down to specific transaction details on a daily basis.

9. Define oracle overflow.

Oracle Workflow provides an infrastructure for the enterprise-wide communication of data related to defined business events, providing the capabilities needed to:

- Manage enterprise business processes that may span trading partners
- Support standard and personalized business rules
- Streamline and automate transaction flows
- Manage exceptions without manual intervention

10. What is Oracle Workflow Business Event System and oracle workflow builder?

Oracle Workflow Business Event System

- It provides a workflow-enabled solution for your enterprise application integration requirements.
- The Business Event System supports the following types of integration:
  - Message-based point-to-point system integration
  - System integration messaging hubs
  - Distributed applications messaging

Oracle workflow builder

- Oracle Workflow Builder provides a graphical drag and drop process designer.
- We can create and evolve business processes to incorporate existing business practices between your organization and customers or suppliers, without modifying existing business processes and without changing applications code.

11. Define Oracle Workflow Event Manager.

- It enables registration of significant business events for selected applications, including functions that generate the XML event messages.

12. List the features in Oracle E-Business Suite Release 11i.

Release 11i utilizes various Oracle database features. They are

- Optimize performance
- Scalability, and
- Business intelligence capacity

13. Mention the use of cost-based optimization (CBO).

- The Oracle optimizer evaluates many factors to calculate the most efficient way to execute a SQL statement.
- The optimizer compares the costs of the execution plans and <u>chooses the one with the smallest cost</u>, i.e. optimum execution characteristics.

14. Draft a note on Real Application Clusters.

- Real Application Clusters (RAC) harness the <u>processing power of multiple interconnected computers.</u>
- RAC software and a collection of computers (known as a cluster) harness the processing power of each component to create a robust and powerful computing environment.
- A large task divided into subtasks and distributed among multiple nodes is completed more quickly and efficiently than if the entire task was processed on one node.
- Cluster processing also facilitates deployment of additional hardware resources for larger workloads and rapidly growing user populations.

15. Define Structured Query Language (SQL).

- An internationally standardized language that is used to access data in a relational database.
- SQL statements enable you to perform the following tasks:
  - Query data
  - Insert, update, and delete rows in a table
  - Create, replace, alter, and drop objects
  - Control access to the database and its objects
  - Guarantee database consistency and integrity

16. Write short notes on Data Definition Language (DDL).

Data definition language (DDL) statements define, structurally change, and drop schema objects. For example, DDL statements enable you to:

- Create, alter, and drop schema objects and other database structures, including the database itself and database users. Most DDL statements start with the keywords CREATE, ALTER, or DROP.
- Delete all the data in schema objects without removing the structure of these objects (TRUNCATE)
- Grant and revoke privileges and roles (GRANT, REVOKE).
- Turn auditing options on and off (AUDIT, NOAUDIT).
- Add a comment to the data dictionary (COMMENT).

17. Define Data Manipulation Language (DML).

DML statements are the most frequently used SQL statements and enable you to:

- Retrieve or fetch data from one or more tables or views (SELECT).
- Add new rows of data into a table or view (INSERT) by specifying a list of column
- Values or using a subquery to select and manipulate existing data.
- Change column values in existing rows of a table or view (UPDATE).
- Update or insert rows conditionally into a table or view (MERGE).
- Remove rows from tables or views (DELETE)

18. What is PL/SQL?

- PL/SQL provides a server-side, stored procedural language that is easy-to-use, seamless with SQL, robust, portable, and secure.
- We can access and manipulate database data using procedural schema objects called PL/SQL program units.

19. List the advantages of PL/SQL subprograms.

- Improved performance
- Memory allocation
- Improved productivity
- Security with definer's rights procedures

20. What is PL/SQL Packages?

- A PL/SQL package is a group of related subprograms, along with the cursors and use, stored together in the database for continued use as a unit.

21. List out the Advantages of PL/SQL Packages.

- Encapsulation
- Data security
- Better Performance

22. Define Cursor.

- A cursor is a pointer to this context area. PL/SQL controls the context area through a cursor. A cursor holds the rows (one or more) returned by a SQL statement.
- The set of rows the cursor holds is referred to as the active set.

23. What is Electronic Data Interchange (EDI)?

- EDI is an electronic exchange of information between trading partners.
- Data files are exchanged in a standard format to minimize manual effort, speed data processing, and ensure accuracy.

24. List out the functions that can be performed by EDI.

- Define trading partner groups and trading partner locations.
- Enable transactions for trading partners.
- Provide location code conversion between trading partner location codes and codes used in Oracle Applications.
- Provide general code conversion between trading partner codes or standard codes.
- Define interface data files so that application data can interface with EDI translators.

25. Write a note on Data File Structure.

- Oracle Database creates a data file for a table space by allocating the specified amount of disk space plus the overhead for the data file header.
- The operating system under which Oracle Database runs is responsible for clearing old information and authorizations from a file before allocating it to the database

- The data file header contains metadata about the data file such as its size and checkpoint SCN. Each header contains an absolute file number and a relative file number.
- The absolute file number uniquely identifies the data file within the database. The relative file number uniquely identifies a data file within a table space.

26. List some unstructured data.

- Spatial data
- Multimedia data
- Text data

27. What is meant by LOB?

- The large object (LOB) data types enable you to store and manipulate large blocks of unstructured data in binary or character format.
- LOBs provide efficient, random, piece-wise access to the data.

28. Write a short note on external LOBs.

- A BFILE (binary file LOB) is an external LOB because the database stores a pointer to a file in the operating system. The external data is read-only.
- Overview of XML in Oracle Database
- Oracle XML DB is a set of Oracle Database technologies related to high-performance XML storage and retrieval. XML DB provides native XML support by encompassing both SQL and XML data models in an interoperable manner.

29. What is Oracle Spatial and Oracle multimedia?

Oracle Spatial

- Oracle Spatial makes spatial data management easier to users of location-enabled applications and geographic information system (GIS) applications.

Oracle Multimedia

- Oracle Multimedia enables Oracle Database to store, manage, and retrieve images, medical images that follow the Digital Imaging and Communications in Medicine (DICOM) standard, audio, and video data in an integrated fashion with other enterprise information.

30. Mention the advantages of Oracle Text.

- Oracle Text allows text searches to be combined with regular database searches in a single SQL statement. The Text index is in the database, and Text queries are run in the Oracle Database process.
- The optimizer can choose the best execution plan for any query, giving the best performance for ad hoc queries involving Text and structured criteria.

31. Define DBMS and RDBMS.

DBMS

- A database management system (DBMS) is software that controls the storage, organization, and retrieval of data

RDBMS

- An RDBMS moves data into a database, stores the data, and retrieves it so that it can be manipulated by applications.
- Oracle Database is an RDBMS. RDBMS that implements object-oriented features such as user-defined types, inheritance, and polymorphism is called an object-relational database management system (ORDBMS).

32. What is transaction management?

- Oracle Database is designed as a multiuser database.
- The database must ensure that multiple users can work concurrently without corrupting one another's data.

33. What is meant by Data warehousing?

- A data warehouse is a relational database designed for query and analysis rather than for transaction processing.
- For example, a data warehouse could track historical stock prices or income tax records.

- A warehouse usually contains data derived from historical transaction data, but it can include data from other sources.

34. List the Characteristics of Data warehouse?

- Subject-Oriented
- Nonvolatile
- Time-Variant

35. Write Data warehouse Vs OLTP?

Characteristics	Data warehouse	OLTP
Workload	Designed to accommodate adhoc queries. We may not know the workload of our data warehouse in advance, so it should be optimized to perform well for a wide variety of possible queries	Supports only predefined operations. Our applications might be specifically tuned or designed to support only these operations.
Data Modifications	Updated on a regular basis by the ETL process using bulk data modification techniques. End user of a data warehouse do not directly update the database	Subjected to individual DML statements routinely issued by end users. The OLTP database is always up to date and reflects the current state of each business transcation.
Scheme design	Uses denormalized or partially denormalized schemes (such as star scheme) to optimize query performance.	Uses fully normalized schemes to optimize DML performance and to guarantee data consistency

PART – B (11 MARKS)

1. Explain in detail about Oracle Apps 11i and Oracle Application Framework.

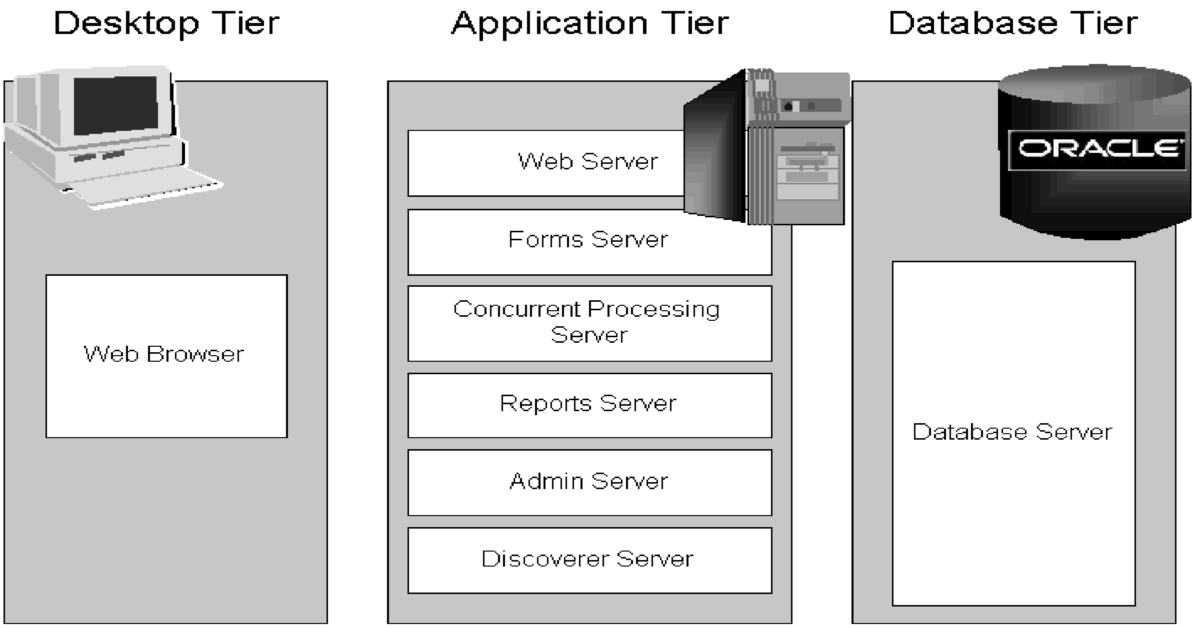
<u>Oracle Suite: Oracle Apps 11i</u>

- Oracle Applications <u>comprise the applications software or business software of the Oracle Corporation.</u>
- Oracle's E-Business Suite (also known as Applications/Apps or EB-Suite/EBS) consists of a collection of enterprise resource planning (ERP), customer relationship management (CRM), and supply-chain management (SCM) computer applications either developed or acquired by Oracle. Enterprise applications available as part of Oracle e-Business Suite.

- New 11i end-to-end Implementations or module add-ons in existing implementations or up gradations to new versions.
- Project management
- Requirement Analysis
- Solution Function design
- Gap Fitment Analysis
- Customization Development
- Comprehensive User Training
- Post implementation Support & Maintenance

ORACLE APPLICATIONS ARCHITECTURE

- Oracle Applications 11i eBusiness Suite (Oracle ERP) use the n-tier architecture (the architecture made for the web).
- This architecture can be seen on the following graphic.



Oracle Applications Architecture

<u>The Desktop Tier:</u>

- In Oracle Applications Release 11i, each user logs in to Oracle Applications through the E-Business Suite Home Page on a desktop client web browser.
- The E-Business Suite Home Page provides a single point of access to HTML-based applications, Forms-based applications, and Business Intelligence applications .Once logged in via the E-Business Suite Home Page, you need not sign on again to access other parts of the system.
- Oracle Applications does not prompt again for user name and password, even when you navigate to other tools and products

Forms Client Applet:

- The Forms client applet is a general-purpose presentation applet that supports all Oracle Applications Forms-based products, including those with customizations and extensions

Oracle JInitiator:

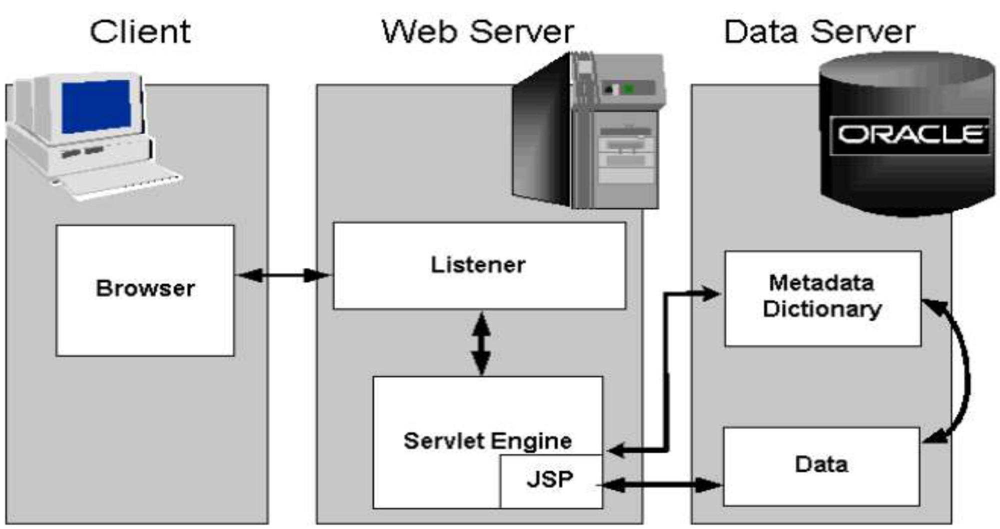
- The Forms client applet must run within a Java Virtual Machine (JVM) on the desktop client. The Oracle JInitiator component allows use of the Oracle JVM on web clients, instead of the browser's own JVM. JInitiator is implemented as a plug-in (Netscape) or ActiveX component (Microsoft Internet Explorer).

<u>Application Tier</u>

- The application tier has a dual role: hosting the various servers that process the business logic, and managing communication between the desktop tier and the database tier.
- This tier is sometimes referred to as the middle tier.
- Oracle9i Application Server comprise the application tier for Oracle Applications
    - Web server
    - Forms server
    - Reports server
    - Discoverer server (optional)
    - Concurrent Processing server
    - Admin server

Web server

- The Oracle HTTP server (powered by Apache) acts as the Web server. It processes the requests received over the network from the desktop clients.



Web server

1. The user clicks the hyperlink of a function from a browser.
2. The browser makes a URL request to the Web listener.
3. The Web listener contacts the Servlet engine (JServ) where it runs a JSP.
4. The JSP obtains the content from the Applications tables and uses information from the metadata dictionary to construct the HTML page.
5. The resulting HTML page is passed back to the browser, via the Web server.

Forms server

- The Forms server hosts the Oracle Applications forms and associated runtime engine that support the professional interface and mediates the communication between the desktop client and the Oracle database server,
- The Forms server communicates with the desktop client using these protocols:
  - Standard HTTP network connection
  - Secure HTTPS network connection
  - TCP/IP connection

Reports server:

- The Reports server dynamically selects the report language at runtime, so users see their reports in their preferred language.

Discoverer Server:

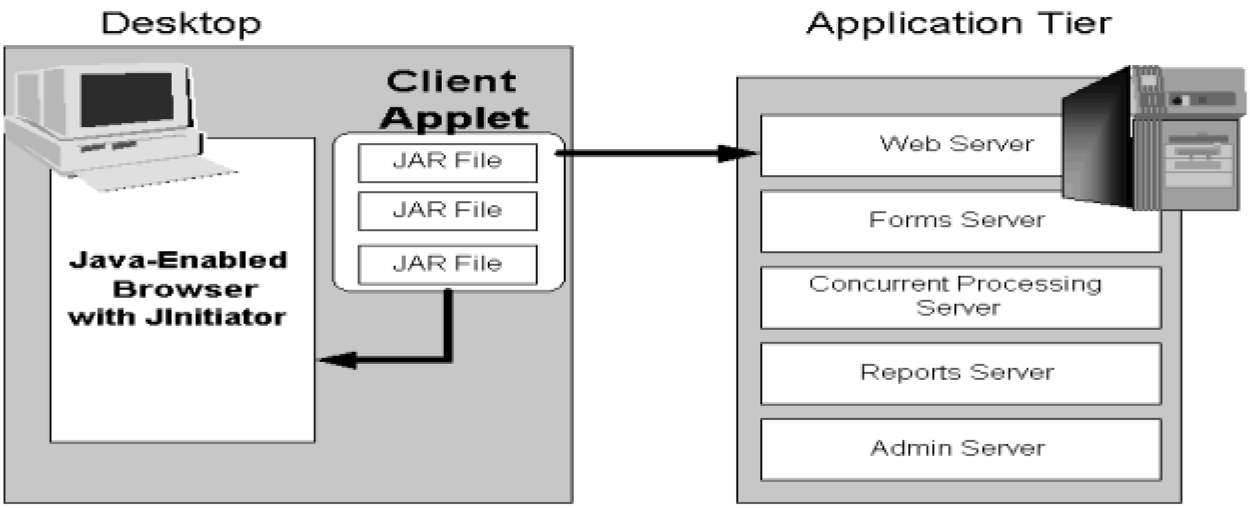
- Discoverer is an ad hoc query, reporting, analysis, and publishing tool that allows business users at all levels of an organization to gain immediate access to information from data marts, data warehouses, and online transaction processing (OLTP) systems.

Admin Server:

- The Admin (Administration) server is located on the node where you maintain the data model and data in your Oracle Applications database.

Database Tier

- The database tier contains the Oracle database server, which stores all the data maintained by Oracle Applications.
- The database also stores the Oracle Applications online help information.
- The database server does not communicate directly with the desktop clients, but rather with the servers on the application tier, which mediate the communications between the database server and the clients.



Database Tier

Oracle Applications DBA (AD):

- The Applications DBA product provides a set of tools for administration of the Oracle Applications file system and database.
- AD tools are used for installing, upgrading, maintaining, and patching the Oracle Applications system.

Oracle Common Modules (AK):

- The Oracle Common Modules can be used to develop inquiry applications for the HTML-based Applications, without the need for any programming.

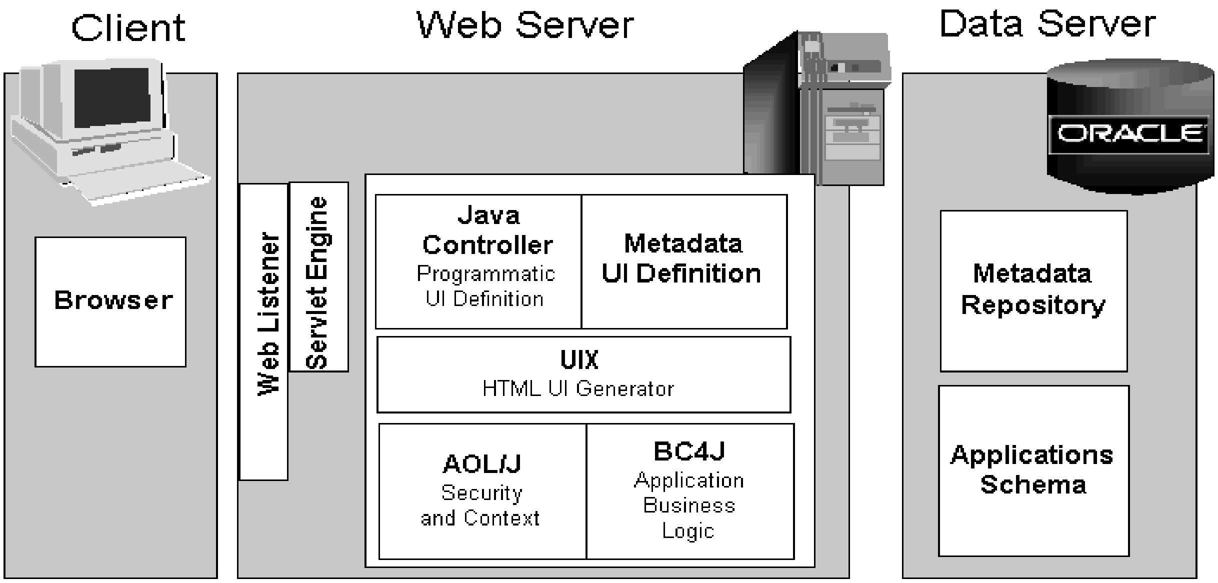
Oracle Applications Utilities (AU):

- The Applications Utilities (AU) component is used to maintain the Oracle Applications system. AU hosts a collection of files copied from other products.

ORACLE APPLICATIONS FRAMEWORK ARCHITECTURE

Following is a more detailed description of the processing that occurs in Step 4 of the previous section. Oracle Applications. Framework uses additional components, including:

- BC4J (Business Components for Java), included in Oracle JDeveloper, is used to create Java business components for representing business logic. It also provides a mechanism for mapping relational tables to Java objects. It allows separation of the application business logic from the UI.
- AOL/J provides the Oracle Applications Framework with underlying security and applications Java services. It provides the Oracle Applications Framework with its connection to the database and with application-specific functionality, such as attachments.



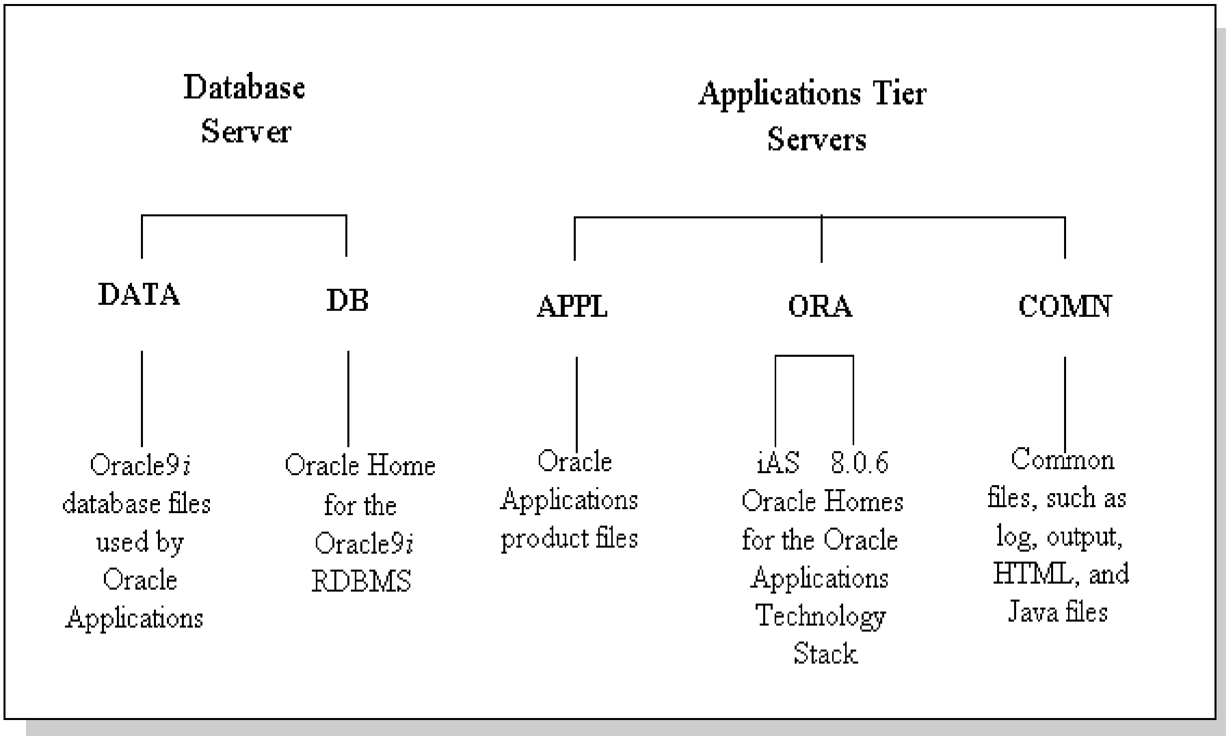
Oracle Applications Framework Architecture

- AOL/J validates user access to the page.
- The page definition is loaded from Metadata Repository on the database tier into the applications tier (Metadata UI Definition).
- The BC4J objects that contain the applications logic and access the database are
- Instantiated.
- The Java Controller programmatically manipulates the page definition as necessary based on dynamic UI rules.
- UIX (HTML UI Generator) interprets the page definition and creates corresponding HTML in accordance with UI standards and sends the page to the browser.

2. Discuss in detail about Oracle File Systems.

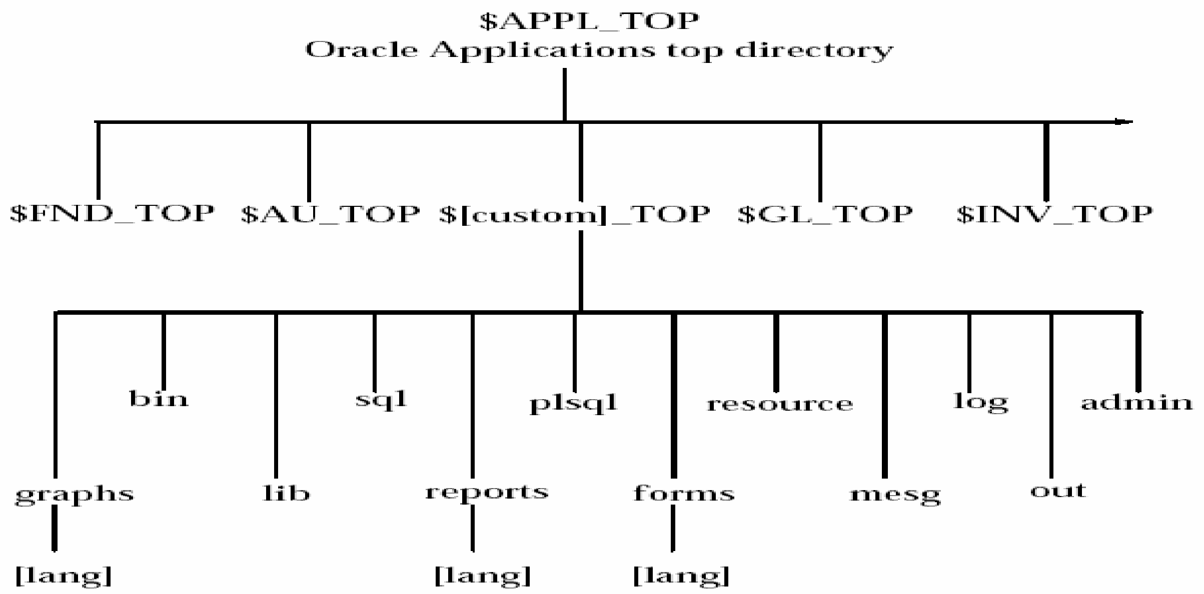
- An Oracle Applications Release 11i system utilizes components from many Oracle products.
- These product files are stored below a number of key top-level directories on the database and application server machines.
- File systems in detail, including the following:
  - Top-Level Directories
  - Product Directories
  - Environment Settings

Top-Level Directories:



- The **<dbname>DATA or DATA_TOP** directory is located on the database server and it contains the system tablespaces, redo log files, data tablespaces, index tablespaces, and database files.
- The **<dbname>DB directory** is located on the database server and it contains Oracle Home for the Oracle9i RDBMS.
- The **<dbname>APPL or APPL_TOP** directory contains the product directories and files for Oracle Applications.
- The **<dbname>ORA directory** contains the Oracle Homes for the Applications technology stack components.
- The **<dbname>COMN or COMN_TOP (or COMMON_TOP**) directory contains directories and files used across products.

Application Directory Structure



Data Directory

- This file system contains .dbf files of oracle db.
- Rapid installation installs system, data and index files in directories below several mount points (Can be specified during installation).

DB & ORA Directory

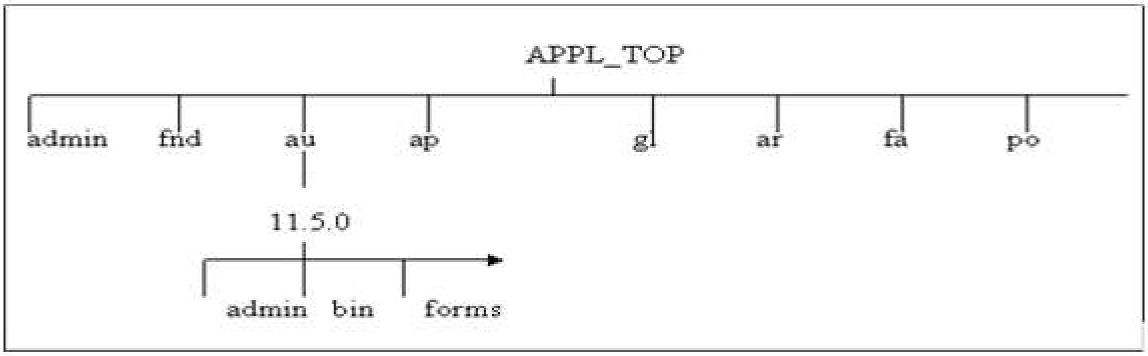
- Oracle application supports linking program using tool from 2nd / 3rd version of DB ORACLE_HOME is used for backward compatibility.

Application Directory

- All application files are stored here. It is also known as APPL_TOP. It contains:
- Core technology files.
- Product files & directories.
- .env application environment file (UNIX) & .cmdon (Windows).

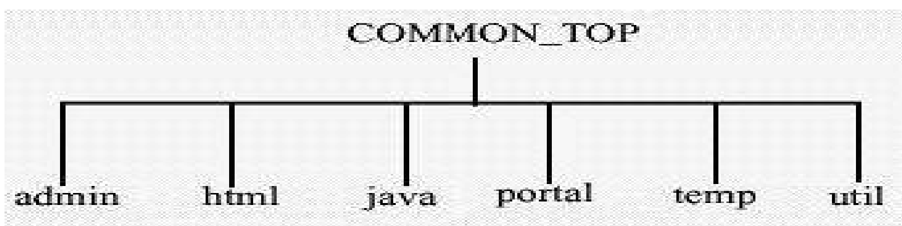
Product Directory

- Each product has its own directory under APPL_TOP.
- Sub Directories are named based on the product standard abbreviation. E.g. GL for general ledger.



COMN Directory

- COMN or COMMON_TOP directory contains files used by different oracle application products and used with third party products.



Admin

- It is default location for concurrent manager log and output directories.

Html

- OA_HTML environmental settings points to HTML Directory. Oracle application based sign-on screen & HTML files are installed here. Also it contains files like JSP, XML etc

Java

- JAVA_TOP points to java directory. This contains oracle application JAR files & ZIP Files.

Portal

- It has portal files. Portal is a webpage that provide access to post install task that is necessary for application

Temp

- Used for caching oracle products like reports.

Util

- This contains JDE, JRE & ZIP utility.

3. Explicate about oracle workflow analysis.

Oracle Workflow provides an infrastructure for the enterprise-wide communication of data related to defined business events, providing the capabilities needed to:

- Manage enterprise business processes that may span trading partners
- Support standard and personalized business rules
- Streamline and automate transaction flows • Manage exceptions without manual intervention
- Acting as a system integration hub, Oracle Workflow can apply business rules to control objects and route them between applications and systems.

Five Major Components of Workflow

- Oracle Workflow Builder
- Workflow Engine
- Workflow Definitions Loader
- Notification System
- Workflow Monitor

Oracle Workflow Builder

- The Workflow Builder is where the user: Creation views and modifies business process definitions graphically.
- Defines the activities and underlying components of business processes
- Graphically creates a process diagram by assembling the activities in the process window
- Workflow Builder allows the user to save a process diagram to a database or a flat file
- The Workflow Builder consists of:
    - Object Navigator (Tree structure)
    - Process Diagram.

Oracle Workflow Engine

- Embedded in the Oracle8/8i server
- Monitors Workflow statuses
- Coordinates the routing of activities for a process
- Contains all packages and procedures

- Used in Workflow
- Workflow process definition MUST be saved to the same database as the Workflow Engine.

Oracle Workflow Definitions Loader

- A utility program that allows the user to transfer process definition between a database and a flat file
- Runs on the server
- Integrated with Oracle Workflow Builder
- Users do not have to be on-line to work.

Workflow Notification System

- Receive and respond to notifications using e-mail or a Web browser
- With available Web functionality, all users with access to the internet can be included in the workflow process
- Access to Notification Workflow from Oracle Applications
- A list of allowed responses if a response is required

  Notification Process Flow
    - Workflow Engine Stops
    - Notification system responds to trigger by sending note
    - Notification system notifies Workflow Engine to continue process.

Workflow Monitor

- Notifications List
    - Displays all current notifications
    - Shows the decision makers in a process as well as the current owner of a process.
- Features
    - Graphically depicts the status of a workflow process
    - Displays detailed information about individual activities

4. **Detail about fundamentals of SQL/PLSQL.**

   SQL:

- SQL (pronounced sequel) is the set-based, high-level declarative computer language with which all programs and users access data in an Oracle database.
- SQL provides an interface to a relational database such as Oracle Database. SQL unifies tasks such as the following in one consistent language:
  - Creating, replacing, altering, and dropping objects.
  - Inserting, updating, and deleting table rows.
  - Querying data.
  - Controlling access to the database and its objects.
  - Guaranteeing database consistency and integrity.

SQL Statements

- All operations performed on the information in an Oracle database are run using SQL statements.
- A SQL statement is a computer program or instruction that consists of identifiers, parameters, variables, names, data types. The types of statements are
  - Data Definition Language
  - Data Manipulation Language

Data Definition Language (DDL) Statements:

- Data definition language (DDL) statements used to Create, alter, and drop schema objects and other database structures, including the database itself and database users. Most DDL statements start with the keywords CREATE, ALTER, or DROP.
- Delete all the data in schema objects without removing the structure of these objects (TRUNCATE).
- Grant and revoke privileges and roles (GRANT, REVOKE).
- Turn auditing options on and off (AUDIT, NOAUDIT).
- Add a comment to the data dictionary (COMMENT).

Data Manipulation Language (DML) Statements:

- Data manipulation language (DML) statements query or manipulate data in existing schema objects.
- DML statements are the most frequently used SQL statements and enable to:
- Retrieve or fetch data from one or more tables or views (SELECT).

- Add new rows of data into a table or view (INSERT) by specifying a list of column values or using a subquery to select and manipulate existing data.
- Change column values in existing rows of a table or view (UPDATE).
- Update or insert rows conditionally into a table or view (MERGE).
- Remove rows from tables or views (DELETE).
- View the execution plan for a SQL statement (EXPLAIN PLAN). See "How Oracle Database Processes DML".
- Lock a table or view, temporarily limiting access by other users (LOCK TABLE).

PL/SQL

- PL/SQL stands for Procedural Language extension of SQL. PL/SQL is a combination of SQL along with the procedural features of programming languages.
- It was developed by Oracle Corporation in the early 90's to enhance the capabilities of SQL.

PL/SQL Engine:

- A PL/SQL language code can be stored in the client system (client-side) or in the database (server-side).
- The concepts like cursors, functions and stored procedures can be used in other database systems like Sybase, Microsoft SQL server etc, with some change in SQL syntax.

PL/SQL Block consists of three sections:

- The Declaration section (optional).
- The Execution section (mandatory).
- The Exception Handling (or Error) section (optional).

```
Declare
        Variable declaration
Begin
        Program execution
Exception
        Exception handling
End
```

PL\SQL block

Declaration Section:

- The Declaration section of a PL/SQL Block starts with the reserved keyword DECLARE.

- It is optional and is used to declare any placeholders like variables, constants, records and cursors used to manipulate data in execution section.

Execution Section:

- The Execution section of a PL/SQL Block starts with the reserved keyword BEGIN and ends with END.
- This is a mandatory section and is the section where the program logic is written to perform any task

Exception Section:

- The Exception section of a PL/SQL Block starts with the reserved keyword EXCEPTION.
- It is optional, any errors in the program can be handled in this section, so that the PL/SQL Blocks terminates gracefully.
- If the PL/SQL Block contains exceptions that cannot be handled, the Block terminates with errors.

PL/SQL Subprograms

- A PL/SQL subprogram is a named PL/SQL block that permits the caller to supply parameters that can be input only, output only, or input and output values.
- A subprogram is either a procedure or a function.
- Procedures and functions are identical except that functions always return a single value to the caller, whereas procedures do not. The term procedure in this chapter means procedure or function.

Advantages of PL/SQL Subprograms

- Improved performance
- Memory allocation
- Improved productivity
- Integrity
- Security with definer's rights procedures.

5. Describe in detail about creating forms and oracle reports.

FORMS:

- Oracle Forms provides a mechanism to store procedures (called Program Units) written in the PL/SQL language within a form. By storing a Program Unit within a form, many blocks in the form can take advantage of the same procedure which reduces code duplication and improves the maintainability of the code.

- Procedures written in PL/SQL may also be stored within the Oracle Database itself as an object in the schema. Such Stored Program Units (also called Stored Procedures) are ideal for situations where highly standardized business rules or applications logic must be implemented across many forms or applications.
- In addition, procedures that require several queries be made to the database may be best implemented in the database as this reduces the network traffic and can significantly improve performance.
- Oracle Forms can make calls to both internal Program Units as well as to Stored Program Units. This flexibility allows application designers extensive control over the execution of applications and facilitates performance tuning.

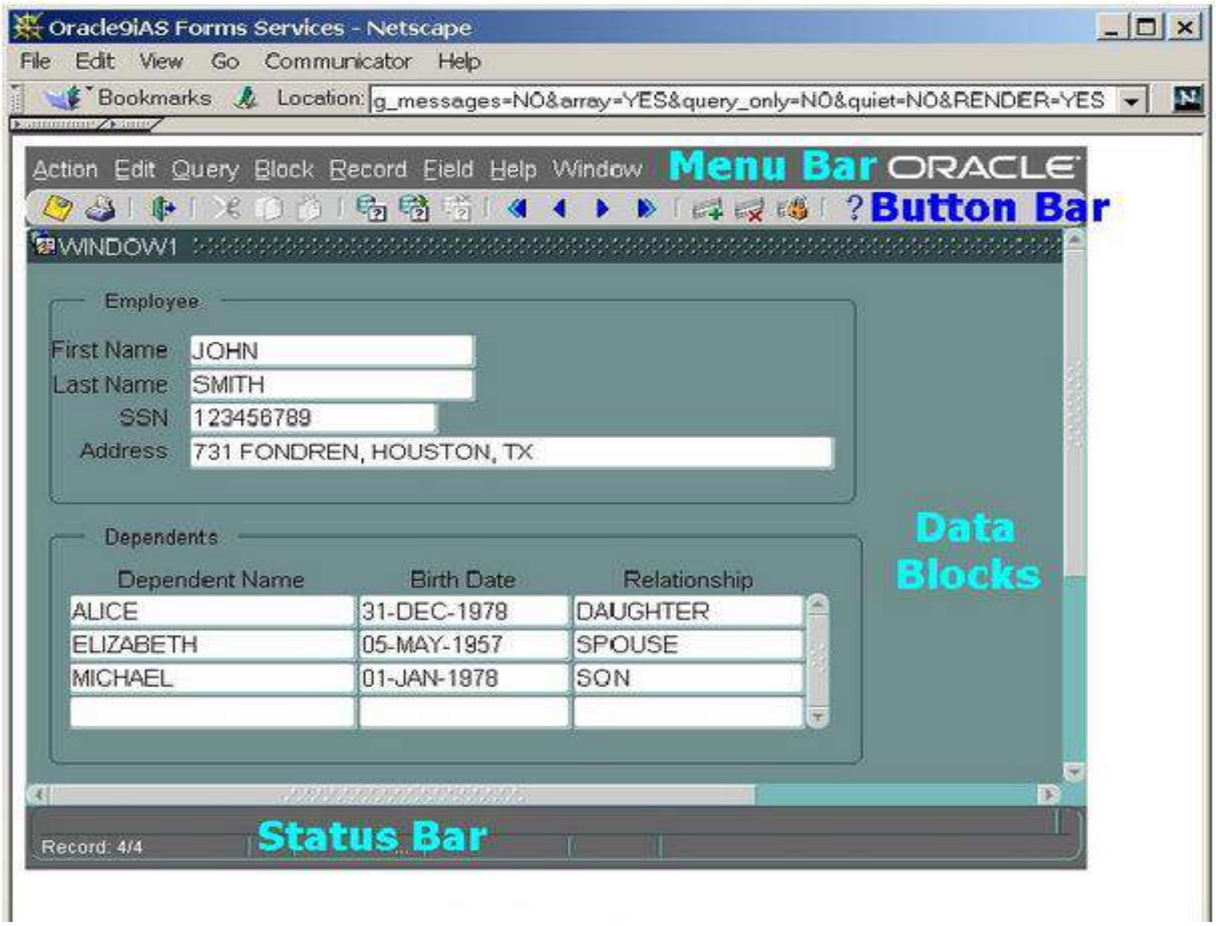
ORACLE FORMS DESIGN

- In Oracle Forms, a form (or data entry form) acts like a window into the database schema. An individual form focuses the attention of the user to one or a few of the tables at a time.
- A form can give prompts so the user knows what kind of input is expected and how data is to be entered and manipulated.
- We can program forms with additional business logic that can help detect bad data or mistakes, make automatic calculations, and suggest appropriate data to be entered and many other helpful techniques that can ensure high quality data can be easily entered into the database.
- By default, every form in Oracle Forms has the capability to query existing data in a table, modify existing data and add new data (records) to the table. A form is built up using one or more data blocks that correspond to tables in the database.
- Fields within the data block correspond to columns in the database table. A data block is similar to a Data Window object.

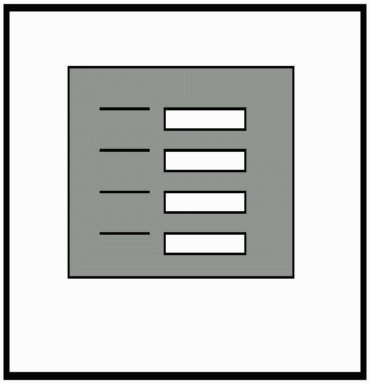
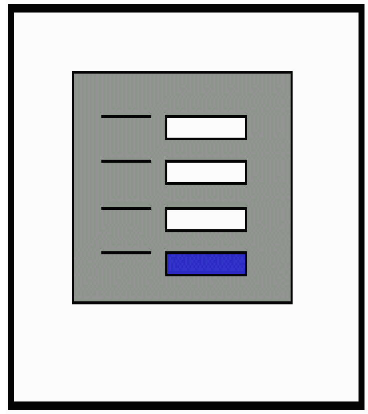
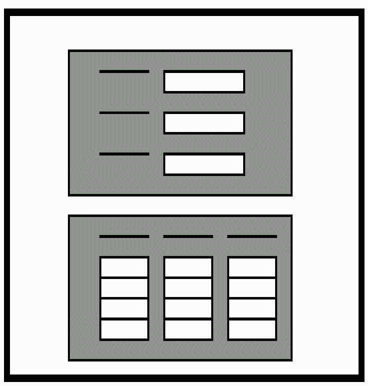
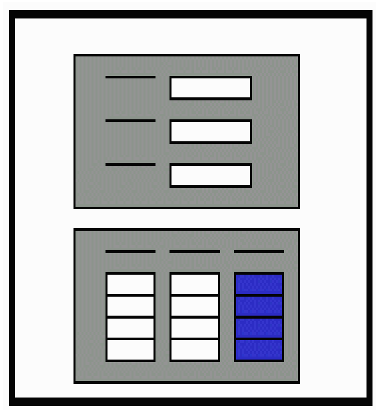
EXAMPLE

- The Example form (in fig) has two data blocks, one for the EMPLOYEE table, and one for the DEPENDENT table. These data blocks are arranged in a Master/Detail setup where a single Employee record (the master) is associated with one or more Dependents records (the details). In this database, Dependents are the spouse, sons and daughters of the employee.

- By default, forms also give a button bar and a menu. These can be used to scroll through the records in a block, navigate between blocks, set up and perform queries against the tables, insert, and update and delete records, clear the form and exit the form. Finally, at the bottom of each form is a status bar that displays any relevant prompts or error messages and an indication of the records in the current data block.



There are four main types of forms that can be designed.

 Single Block Form. This form contains a single data block corresponding to a single database table.	 Single block Form with lookup field. This form contains a single data block corresponding to a single database table with the addition of one or more fields that display data from some other tables. Such data is “looked up” when the form runs.
 Master/Detail Form. This form contains two data blocks that are arranged in a master/detail (one to many) relationship.	 Master/Detail Form with lookup fields. This form is similar to the Master/Detail but has the additional lookup fields in the master and/or detail blocks.

Connect Oracle Forms 11g to database 11g

1. First you need to copy the file tnsnames.ora from database folder to config folder from Oracle Middleware.
2. Open the Oracle Form Builder, go to File Menu and press Connect or just execute the command Ctrl+J.
3. On the last step, you need to enter the user name, password and the database name.
4. Change Module Name in Oracle Forms 11g

Right-click on Form Module name, then choose Property Palette. On Property Palette just click on the name and write your module name.

5. Open the Oracle Form Builder, go to File Menu and press Connect or just execute the command Ctrl+J.
6. Create New Data Block Wizard in Oracle Forms 11g

   First you must click on Data Blocks and the press the button Create. An alert with two option will appear, and here you choose: Use the Data Block Wizard.
7. Welcome to the Data Block Wizard! - Click Next to begin creating your data block.
8. Select the type of Data Block: Table/View or Stored Procedure.
9. Create New Data Block Wizard in Oracle Forms 11g

   Press Browse button and will appear a small window with more choices: Tables, Views, Current user, Synonyms, Other users.
10. After you choose the table name, oracle forms will show all Available Columns from the table. Here you can bring all columns from the table in your form or you can select just a few columns.
11. After you moved the items press Next.
12. Create New Data Block Wizard in Oracle Forms 11g

    Enter a name for your Data Block.
13. You have finished describing you data block. Now you will have two options:
    - Continue with: Call the Layout Wizard
    - Stop: Just create the data block(is recommended when you use multiple data block on the same canvas)
14. Create Wizard Layout Oracle Forms 11g: Display in a frame on a canvas
15. Here you have to choose the canvas name (can be a new one or an existing canvas) and the type of canvas.

    Types of canvas:
    - Content
    - Stacked
    - Vertical Toolbar
    - Horizontal Toolbar
    - Tab
16. Create Wizard Layout Oracle Forms 11g

    Here you select and move item that you want to be displayed in the frame.

For each each item displayed you can choose the item type from the list.

17. Enter a prompt, width and the height for each item.
18. Create Wizard Layout Oracle Forms 11g

    Select the Layout Style for your frame
19. On this step you can choose the Frame Title, you can insert how many records to be displayed in the field Records Displayed, also Distance Between Records and Display Scrollbar.
20. Finish the Wizard Layout and display the new frame in the Layout Editor

    You Finish the Wizard Layout.
21. Display the new frame in the Layout Editor.

REPORTS:

Oracle Reports is the enterprise reporting component of Oracle's BI offering. It enables an organization to convert data into information, and publish the information secure and reliable. The Oracle Reports Builder allows the developer to create sophisticated reports in a variety of layouts and contains many customization features.

- Oracle Reports provides an access to multiple data sources. E.g: manufacturing organization
- It supports SQL, PL/SQL, XML, JDBC, Oracle OLAP, and text files. JDBC data source can be used to access data from different JDBC-compliant data sources, and JDBC-ODBC Bridge can be used to connect to any data that can be exposed as an ODBC data source.
- With Oracle Reports, the source of data is transparent, because data coming from any data source is available to you in exactly the same representation.

Features of Oracle Reports

- The most flexible enterprise-reporting tool in the market today
- Reliable
- Integrated
- Extensible

The Main section of Object navigator is Reports

Reports which include:

1. **Data Model** – Contains information about queries used for a report.
2. **Web Source** – A web page or web service that returns a data set.

3. **Paper Layout** – Contains information about how a paper or screen a report is formatted including headers, footers, margins, fonts, etc.
4. **Paper Parameter Form** – Contains information about the initial screen that is displayed when a form first runs.
5. **Report Triggers** – PL/SQL code that can be executed before, during or after a report has been executed.
6. **Program Units**
7. **Attached Libraries**

Creating reports follows a 4 step process:

1. **Define the Data Model –** This step specifies which queries should be run on the database including how multiple queries are related and how they are grouped. This step must be done by hand. Queries that have been created elsewhere can be imported into Oracle Reports.
2. **Define the Layout** – This step specifies the layout of the report including the overall orientation of query results and the suppression of repeating groups. There are a number of default report layouts that can automatically be applied to a data model.
3. **Create and/or Customize the Parameter Form** – If some user input is required in order to run the report, then a parameter form must be customized. All reports have a default parameter form.
4. (optional) **Create any triggers or program units that will be executed with the report**.

How to generate Report?

1. Connect Oracle Reports 11g to database 11g

   Open the Oracle Reports Builder, go to File Menu and press Connect or execute the command Ctrl+J.
2. On the second step, you need to enter the user name, password and the database name.

   Create Report Wizard in Oracle Reports 11g

   First you must click on Reports and the press the button Create. An alert "Create a new report" with two option will appear, and here you choose: Use the Report Wizard.
3. Welcome to the Report Wizard! - Click Next to begin creating your report.
4. Choose The Type of Layout and Style Reports in Oracle 11g

   Here you must choose the type of layout you would like to generate: Create both Web and Payper Layout, Create Web Layout only, Create paper Layout only.

5. On the next step you have to choose the report style and give a title name for your report. The report style option are: Tabular, Group Left, Group Above, Matrix, Matrix with Group.
6. Choose a Data Source and Select Sql Query Builder for Reports in Oracle 11g
   On this step you have to choose a Data Source for your report. The Data Source can be from JDBC Query, SQL Query, Text Query, XML Query.
7. You can do write the query manually, you can use Query Builder or Import Builder.
8. Move Available Fields in Displayed Fields in R reports Oracle 11g
   Select the fields that you would like to display in your report. You can move one item or all items to target from Available Fields in Displayed Fields.
9. After you moved the items press Next.
10. Select Available Fields in Reports Wizard to calculate totals
    Select the fields that you would like to calculate totals in your report. Also you can do choose to calculate: sum, count, minimum, maximum, average.
11. Here you can modify the labels of the fields and also the width for each field.
12. Choose a Template for your Report Wizard in Oracle Reports 11g
    Select the template for your report. You have multiple options: Predefined Template, Template File or No Template.
13. You have finished describing your report!
14. Run Module Report Editor Paper Design Oracle Reports 11g
    Select the template for your report. You have multiple options: Predefined Template, Template File or No Template.

6. Explain about oracle Electronic Data Interchange (EDI) and its functions.

- EDI (Electronic Data Interchange) is the <u>electronic exchange of business documents between the computer systems of business partners, with a standard format over a communication network</u>.
- It can also be called as paperless exchange.
- EDI, electronic funds transfer (EFT), electronic mail and fax are increasingly being used to schedule operations, streamline order fulfillment, and optimize cash flow. The National Institute of Standards and Technology in a 1996 publication defines Electronic Data Interchange as "the computer-to-computer interchange of strictly formatted messages that represent documents other than monetary instruments.

- EDI can be transmitted using any methodology agreed to by the sender and recipient. This includes a variety of technologies, including modem (asynchronous, and bisynchronous), FTP, Email, HTTP etc.

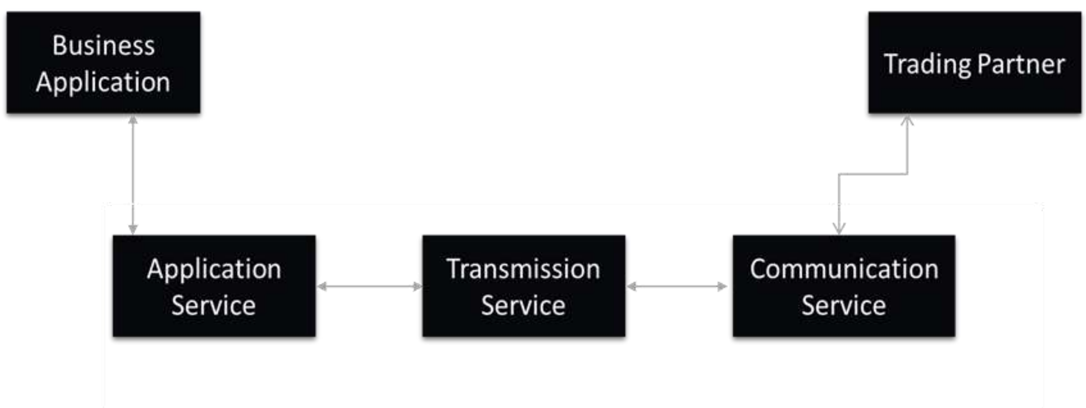
EDI Standard Formats

- The business documents that are been exchanged between business partners need to be in a standard format. ANSI X12 (American National Standards Institute) or EDIFACT (Electronic Data Interchange For administration, Commerce, and Transport) are two standards, which supply a common language for formatting the information content that is been exchanged.

Components

Three major Components in the EDI Process:

1. Application service
2. Translation service
3. Communication service



Application Services:-

- It provides the link between application and EDI. It allows you to send documents from an EDI system.
- The set of callable routine is used to transfer document from the business application into EDI document destination can be either intra-company or to the external companies.

Translation service:-

- Converts the outgoing documents from an internal format file to an agreed external format. Translates internal document from external format to EDI internal format file.

Communication service:-

- The communication service sends and receives transmission files to and from the trading partners either directly or by using party service called a valued added network (VAN).

File Types

- **Internal format file (IFF):-**It contains single document for single trading partner.
- **External format file (EFF):-**It contains same data as the internal format file translated into the appropriate standard document format.
- **Transmission file:-** it contains one or more document for the same trading partner. Documents of same format are packed into functional groups. The functional groups going to one trading partner are packaged into an interchanged set. An interchanged set contains one or more functional groups of documents with the same sender and receiver.

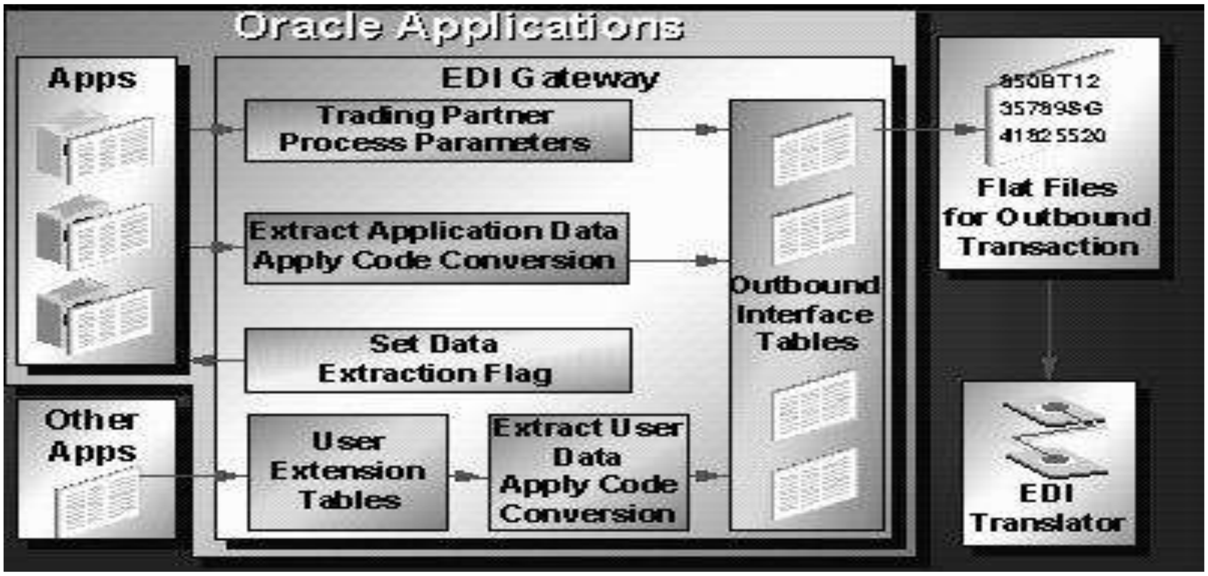
Advantages

- One Time Data Entry
- Reduced Errors
- Online Data Storage
- Faster management Reporting
- Automatic Data Reconciliation

FUNCTIONS OF EDI

- Define Trading Partner Relationships
- Define Code Conversions
- Customize Flat File Formats
- Use Standard Report Submission
- Initiate Extract Process
- Initiate Import Process
- View Extract/Import Process Status

A typical flow can be understood as:



Step 1. Define Data File Directories

- To use EDI Gateway, you must first create directories where data files, for both inbound and outbound transactions, are stored.

Step 2. Define Trading Partner Data

- First, define a trading partner group and associate all trading partner location (address) entities to this group. Define basic information about the trading partner and associate that to a specific trading partner's location.

Step 3. Define Code Conversions

- The Oracle EDI Gateway code conversion function provides a method by which trading partner and standard codes can be converted to Oracle Application codes.
- Define code conversion categories to specify a subset of codes within the code conversion table.
- For each data element in a transaction that requires code conversion by EDI Gateway, assign one code conversion category.
- Enter the actual code conversion values from the internal codes to external codes into the code conversion table.

Step 4. Customize Data File Formats

- Oracle delivers a standard predefined data file format containing all the data that may be required to support a particular EDI transaction.

Step 5. Initiate EDI Transactions

- Once data file directories, trading partner information, code conversions, and optional customizations of data files have been performed, use Oracle Applications Standard Request Submission to run extract programs for outbound transactions and import programs for inbound transactions.

7. Elucidate in detail about oracle data and oracle database.

Oracle database

Database Management System (DBMS)

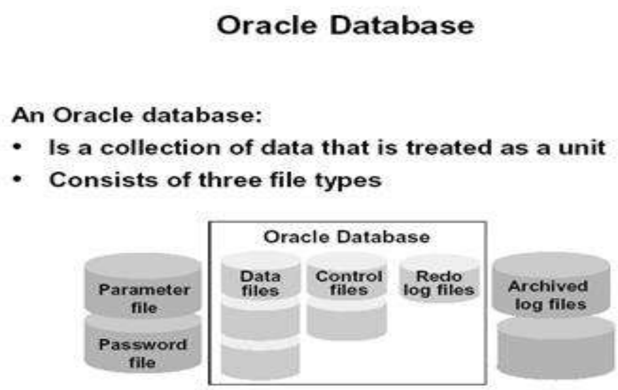
- A database management system (DBMS) is software that controls the storage, organization, and retrieval of data.
- A database application is a software program that interacts with a database to access and manipulate data.
- A relational database stores data in a set of simple relations. A relation is a set of tuples. A tuple is an unordered set of attribute values.
- A table is a two-dimensional representation of a relation in the form of rows (tuples) and columns (attributes).

History of Oracle Database

- Founding of Oracle: In 1977, Larry Ellison, Bob Miner, and Ed Oates started the consultancy Software Development Laboratories, which became Relational Software, Inc. (RSI). In 1983, RSI became Oracle Systems Corporation and then later Oracle Corporation.
- First commercially available RDBMS: In 1979, RSI introduced Oracle V2 (Version 2) as the first commercially available SQL-based RDBMS.
- Portable version of Oracle Database: Oracle Version 3, released in 1983, was the first relational database enabling the database to be ported to multiple platforms.
- Enhancements to concurrency control, data distribution, and scalability
- Version 4 introduced multiversion read consistency. Version 5, released in 1985, supported client/server computing and distributed database systems. Version 6 brought enhancements to disk I/O, row locking, scalability, and backup and recovery.

ORACLE DATABASES

- Oracle Database (commonly referred to as Oracle RDBMS or simply as Oracle) is an object-relational database management system produced and marketed by Oracle Corporation.
- Larry Ellison and two friends and former co-workers, Bob Miner and Ed Oates, starteda consultancy called Software Development Laboratories (SDL) in 1977.SDL developed the original version of the Oracle software



Physical Structures

- **Datafiles (*.dbf)**

  The datafiles contain all the database data. The data of logical database structures, such as tables and indexes, is physically stored in the datafiles allocated for a database
- **Control Files (*.ctl)**

  A control file contains entries that specify the physical structure of the database such as Database name and the Names and locations of datafiles and redo log files
- **Redo Log Files (*.log)**

  The primary function of the redo log is to record all changes made to data. If a failure prevents modified data from being permanently written to the datafiles, then the changes can be obtained from the redo log, so work is never lost
- **Archive Log Files (*.log)**

  Oracle automatically archives log files when the database is in ARCHIVELOG mode. This prevents oracle from overwriting the redo log files before they have been safely archived to another location.
- **Parameter Files (initSID.ora)**

  Parameter files contain a list of configuration parameters for that instance and database.
- **Alert and Trace Log Files (*.trc)**

Each server and background process can write to an associated trace file. When an internal error is detected by a process, it dumps information about the error to its trace file.

Logical Structures

- **Tablespace**

  A database is divided into logical storage units called tablespaces, which group related logical structures together. One or more datafiles are explicitly created for each tablespace to physically store the data of all logical structures in a tablespace.

- **Data Blocks**

  At the finest level of granularity, Oracle database data is stored in data blocks. One data block corresponds to a specific number of bytes of physical database space on disk.

- **Extents**

  An extent is a specific number of contiguous data blocks, obtained in a single allocation, used to store a specific type of information.

- **Segments**

  A segment is a set of extents allocated for a certain logical structure. The different types of segments are :

  1. Data segment – stores table data
  2. Index segment – stores index data
  3. Temporary segment – temporary space used during SQL execution
  4. Rollback Segment – stores undo information

- Schema

  A schema is a collection of database objects. A schema is owned by a database user and has the same name as that user. Schema objects are the logical structures that directly refer to the database's data. Schema objects include structures like tables, views, and indexes

Oracle Instance

- Logical memory structures that are collectively called an Oracle instance
- Physical file structures that are collectively called a database .It is the actual execution of DBMS software that manages data in the databases tablespace. Two Main Components:

  SGA (System Global Area)

  - A group of shared memory structures that contain data and control information for one Oracle database instance.

- The data in the instance's SGA is shared among the multiple concurrent users.
- Allocated when you start the database instance.
- De-allocated when the instance is shutdown.

PGA (Program Global Area)

- Each server process has a PGA allocated that is a private area for each server
- Work area for each application.

8. Differentiate DW vs OLTP.

Characteristics	Data Warehouse	OLTP
Source of data	Consolidation data; OLAP data comes from the various OLTP Databases	Operational data; OLTPs are the original source of the data.
Purpose of data	To help with planning, problem solving, and decision support	To control and run fundamental business tasks
What the data	Multi-dimensional views of various kinds of business activities	Reveals a snapshot of ongoing business processes
Inserts and Updates	Periodic long-running batch jobs refresh the data	Short and fast inserts and updates initiated by end users
Queries	Often complex queries involving aggregations	Relatively standardized and simple queries Returning relatively few records
Processing Speed	Depends on the amount of data involved; batch data refreshes and complex queries may take many hours; query speed can be improved by creating indexes	Typically very fast
Space Requirements	Larger due to the existence of aggregation structures and history data; requires more indexes than OLTP	Can be relatively small if historical data is archived
Database Design	Typically de-normalized with fewer tables; use of star and/or snowflake schemas	Highly normalized with many tables
Backup and Recovery	Instead of regular backups, some environments may consider simply reloading the OLTP data as a recovery method	Backup religiously; operational data is critical to run the business, data loss is likely to entail significant monetary loss and legal liability
Workload	Designed to accommodate adhoc queries. We may not know the workload of our data warehouse in	Supports only predefined operations. Our applications might be specifically tuned or designed to support only these

	advance, so it should be optimized to perform well for a wide variety of possible queries	operations.
Data Modifications	Updated on a regular basis by the ETL process using bulk data modification techniques. End user of a data warehouse do not directly update the database	Subjected to individual DML statements routinely issued by end users. The OLTP database is always up to date and reflects the current state of each business transcation.
Scheme design	Uses denormalized or partially denormalized schemes (such as star scheme) to optimize query performance.	Uses fully normalized schemes to optimize DML performance and to guarantee data consistency

9. Discuss in detail about DW Connectors.

Teradata OLAP Connector

- Teradata OLAP Connector is the one tool available that makes the direct connection between the Teradata system and Excel applications.

Overview

- Teradata OLAP Connector provides self-service analytics for Excel, Tableau, and arcplan users by enabling them to access the corporate data stored in the Teradata Data Warehouse.
- This eliminates the inconsistencies and security concerns associated with the traditional approach of extracting and storing corporate data locally.

Challenges

Desktop-based BI solutions allow business users to extract data from the data warehouse for desktop analysis but this approach is challenging:

- Creates spread marts all over the organization
- Static view of the data of when it was extracted which leads to inconsistencies with the reporting
- Limited data leads to limited analysis
- Causes security challenges for leaving sensitive data on the users's desktops
- Data extractions scripts are difficult to maintain

ORACLE BIG DATA CONNECTORS

- Oracle Big Data Connectors is a software suite that integrates processing in Hadoop with operations in a data warehouse.

- Designed to leverage the latest features of Apache Hadoop, Big Data Connectors connect Hadoop clusters with database infrastructure to harness massive volumes of structured and unstructured data for critical business insights.
- Big Data Connectors greatly simplify development and are optimized for efficient connectivity and high-performance between Oracle Big Data Appliance and Oracle Exadata. Oracle Big Data Connectors 3.0 delivers a rich set of new features, increased connectivity, enhanced performance, and security for Big Data applications.

Oracle Big Data Connectors

- Large volumes of data are increasingly collected and processed in Hadoop, while enterprise IT systems are centered on relational data warehouses.
- Oracle Big Data Connectors bridges data processing in Hadoop with Oracle Database, providing the crucial ability to unify data across these systems.
- Combining pre-processing of large data volumes of raw and unstructured data in Hadoop with the advanced analytics, complex data management, and real-time query capabilities of Oracle Database, Oracle Big Data Connectors deliver features that support information discovery, deep analytics and fast integration of all data in the enterprise.
- The components of this software suite are:
    - Oracle SQL Connector for Hadoop Distributed File System
    - Oracle Loader for Hadoop
    - Oracle Data Integrator Application Adapter for Hadoop
    - Oracle R Advanced Analytics for Hadoop
    - Oracle XQuery for Hadoop
    - Oracle Big Data Connectors work with Oracle's engineered systems - Oracle Big Data Appliance and Oracle Exadata - as well as with supported Hadoop distributions and database versions on non-engineered systems.

KEY FEATURES

- Tight integration with Oracle Database
- Leverage Hadoop compute resources for data in HDFS
- Enable Oracle SQL to access and load Hadoop data
- Fast and very efficient load from Hadoop into Oracle Database

- Partition pruning of Hive tables during load and query
- Graphical user interfaces of Oracle Data Integrator drive data transformation workflows on.

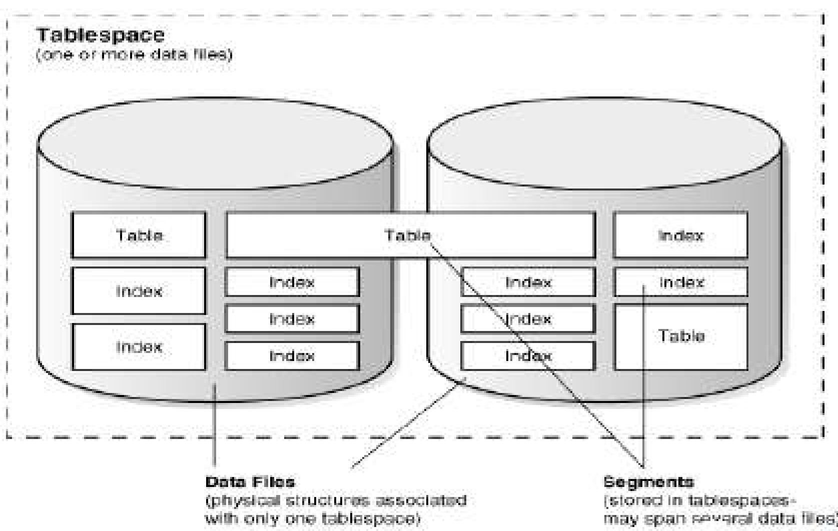
10. Explain in detail about Data file structure.

<u>Overview of Data Files</u>

- At the operating system level, Oracle Database stores database data in data files.
- Every database must have at least one data file.

Use of Data Files

- Oracle Database allocates space for user data in tablespaces, the data for a nonpartitioned table is stored in a single segment, which is turn is stored in one tablespace..
- Oracle Database physically stores tablespace data in data files
- Each tablespace consists of one or more data files, which conform to the operating system in which Oracle Database is running.
- The data for a database is collectively stored in the data files located in each tablespace of the database.
- A segment can span one or more data files, but it cannot span multiple tablespaces.
- A database must have the SYSTEM and SYSAUX tablespaces. Oracle Database automatically allocates the first data files of any database for the SYSTEM tablespace during database creation.



Relationship between table spaces, data files, and segments.

Permanent and Temporary Data Files

- A permanent tablespace contains persistent schema objects. Objects in permanent tablespaces are stored in data files.
- A temporary tablespace contains schema objects only for the duration of a session.

Online and Offline Data Files

- Every data file is either online (available) or offline (unavailable). You can alter the availability of individual data files or temp files by taking them offline or bringing them online.
- Offline data files cannot be accessed until they are brought back online.
- Administrators may take data files offline for many reasons, including performing offline backups, renaming a data file, or block corruption.
- The database takes a data file offline automatically if the database cannot write to it.

Data File Structure

- Oracle Database creates a data file for a tablespace by allocating the specified amount of disk space plus the overhead for the data file header.
- The operating system under which Oracle Database runs is responsible for clearing old information and authorizations from a file before allocating it to the database.
- The data file header contains metadata about the data file such as its size and checkpoint SCN. Each header contains an absolute file number and a relative file number. The absolute file number uniquely identifies the data file within the database.
- The relative file number uniquely identifies a data file within a tablespace. When Oracle Database first creates a data file, the allocated disk space is formatted but contains no user data.
- However, the database reserves the space to hold the data for future segments of the associated tablespace. As the data grows in a tablespace, Oracle Database uses the free space in the data files to allocate extents for the segment. Over time, updates and deletions of objects within a tablespace can create pockets of empty space that individually are not large enough to be reused for new data.
- This type of empty space is referred to as fragmented free space.

Space in a Data File

DEPARTMENT OF CSE

CS T61 ENTERPRISE SOLUTIONS

Unit-IV – PEOPLESOFT

QUESTION BANK WITH ANSWER

PeopleSoft: Basic PeopleSoft Functionality – Opening Multiple Windows - Database structure – Understanding People Soft Data Mover – Records - Pages vs. Forms. **PeopleSoft HRMS:** Introduction to PeopleSoft HRMS database - PeopleSoft products - Functional PeopleSoft - financial management system - PeopleSoft Enterprise HRMS.

PART A (2 MARKS)

1. **Define people soft.**

   PeopleSoft, Inc. was a company that provided Human Resource Management Systems (HRMS), Financial Management Solutions (FMS), Supply Chain Management (SCM), Customer Relationship Management (CRM), and Enterprise Performance Management (EPM) software, as well as software for manufacturing, and student administration to large corporations, governments, and organizations.

2. **When PeopleSoft is founded?**

   - Founded in 1987 by Ken Morris and David Duffield, PeopleSoft was originally headquartered in Walnut Creek, California before moving to Pleasanton, California.
   - Duffield envisioned a client–server version of Integral Systems' popular mainframe HRMS package.

3. **Draft a note on evolution of PeopleSoft.**

   - 1987 – Addressing organizational Human Relations departments, from recruitment to retirement. Managing the payroll, benefits, travel, expenses, and vacation of all employees was the key focus of the first PeopleSoft product.
   - 1997 – Release of PeopleSoft 7, they upgraded the supply chain and finance modules, and added the functionality for analyzing business data.

- January 2000 – PeopleSoft 8 saw the addition of a module for customer relationship management through the acquisition of Vantive Corporation.

4. **Which corporation control PeopleSoft?**
    - It existed as an independent corporation until its acquisition by Oracle Corporation in 2005.
    - The PeopleSoft name and product line are now marketed by Oracle.

5. **List out the applications of PeopleSoft.**

   The applications are
    - Comprehensive business and industry solutions
    - Enabling organizations to increase productivity
    - Accelerate business performance and
    - Provide a lower cost of ownership

6. **Mention the benefits of PeopleSoft.**
    - End-user initiate transactions and date in to the system once and make it accessible for multiple purposes (Finance, Human Resources, and Reports)
    - Tracking of transaction status is available online, almost eliminating the need to make multiple phone calls regarding the transaction's status.
    - Various transactions processing are online
    - Transaction approvals have been simplified
    - Some transactions can be approved and submitted online, while meeting internal control, and presenting less need to hand carry papers from office to office.

7. **Define Multiple Windows.**
    - IE 8 can open multiple unique sessions (but only when using the new IE 8 menu option called "New Session" or when creating a new shortcut with a special -no merge instruction sent to IE executable)
    - Do not use Windows Start > Internet Explorer if you want a new session - IE 7 can open as well new unique sessions but each time one should use the Windows Start > Internet Explorer to start a new browser instance (this is opposite to IE 8 new behavior - see above)

- Using New Window (from any IE version or Firefox ...from menu or using CTRL+N ...or from PIA "New Window" links) will always open a new window tied to the same browser session
- This will share the same cookies with previous windows.

8. What is People soft Database structure?

- PeopleSoft has a long history of offering customers a choice of market-leading databases.
- People Tools provides an abstraction layer, which insulates application developers from the intricacies of each of the specific database platforms. Customer databases choices include:
    - Oracle
    - IBM DB2
    - Microsoft SQL Server
    - Informix
    - Sybase

The following image describes three distinct yet integrated database layers for PeopleSoft system.

9. Write a note on PeopleSoft Database Layer.

There are three types of layers in PeopleSoft

- System Tables
- People Tools metadata
- PeopleSoft application data tables

10. Define PeopleSoft Forms.

- The Forms and Approval Builder enables you to design online forms, specify the approval process they require, and deploy them to users within your organization. Use this feature to convert manual procedures within your organization to paperless processes that include workflow-based approvals and an audit trail for tracking progress.

- No coding is required on your part, and future upgrades to your PeopleSoft system will not require you to update these forms, since the forms you create are not customizations

11. Write a short notes on Data Mover.

- PeopleSoft Data Mover, is used to import / export data between databases. Apart from this, Data Mover can be used for a variety of functions like database security management, running SQL scripts etc.
- On Windows machines, you can use the Data Mover using the GUI as well as from the command prompt. While UNIX variants support only command line operation.

12. Delineate Records.

- Fields that are grouped together as a unit are records.
- PeopleSoft records can be defined as the collection of fields on a page. It's similar what we use tables in RDBMS.

13. Define PAGES.

- Pages are the graphical interface between your users and your application database.
- As a system designer, you configure or build pages that meet the data requirements of the application and that are easy to use and understand.
- Using PeopleSoft Application Designer, you can create, modify, and delete page definitions in your PeopleSoft system.

14. What are the types of records?

- Table
- View
- Derived/Work
- Sub record
- Dynamic View
- Query View
- Temp table

15. Write a note on Multiple Occurrence of Data.

- On some pages, you may want a few of the field controls to display multiple rows or occurrences of data. To do this, you can add a level-based control: a scroll area, a grid, or a scroll bar.
- Users can then add, edit, delete, find, and scroll through multiple occurrences of data in a page control or group of controls using action buttons, links, or the browser scroll bar, depending on how you set the occurs count.
- Using a scroll area or a grid, rather than a scroll bar, is the preferred page design to show multiple occurrences of data.

16. Define Sub records.

- PeopleSoft sub records are the collection of fields that are grouped together to form an entity so that they can be used for building blocks for multiple records.
- Since sub records are not build through application designer they don't have their existence in the database.

17. How a Form is created?

The steps used to create a form:

- Defining the basic information for a form, including the form ID and form owner.
- Providing instructions that describe how to use the form.
- Specifying the fields that appear on the form.
- Attaching files to the form (optional).
- Defining the menu item that is used to access the form.
- Specifying the approval workflow that is required for the form.
- Previewing, testing, and activating the form.

18. List the types of fields in forms.

- Numeric
- Text
- Date
- Time
- Yes/No
- Prompt

19. Define PeopleSoft HRMS.

- Human Resources Management System (HRMS) or Human Resources Information System (HRIS), refers to the systems and processes at the intersection between human resource management (HRM) and information technology.
- It merges HRM as a discipline and, in particular, it's basic HR activities and processes with the information technology field.

20. Mention the types of PeopleSoft Enterprise HRMS.

- PeopleSoft Enterprise HRMS integrations
- PeopleSoft Enterprise HRMS implementation

21. What is PeopleSoft Enterprise HRMS Integration?

- PeopleSoft HRMS integrates with other PeopleSoft applications, such as PeopleSoft Enterprise Financials, PeopleSoft Enterprise Workforce Analytics, and PeopleSoft Enterprise Learning Management.

22. Write a note on PeopleSoft Enterprise HRMS Implementation.

- PeopleSoft Setup Manager enables you to generate a list of setup tasks for your organization based on the features that you are implementing.
- The setup tasks include the components that you must set up, listed in the order in which you must enter data into the component tables, and links to the corresponding People Book documentation.

23. Draft a note on Service Automation.

- Oracle's PeopleSoft Enterprise Services Automation applications enables project-centric organizations and departments to establish core operational processes that support full project lifecycle management—across operations and finance—from project selection, planning and staffing, through execution, cost control, and analysis

24. Define Enterprise Performance Management.

- Oracle's PeopleSoft Enterprise Performance Management enables organizations to achieve world-class performance by aligning the right information and resources to strategic objectives.

- PeopleSoft Enterprise Performance Management offers performance management solutions for every budget and every phase of the management cycle, helping managers formulate strategies for profitable growth, align strategies with operational plans, and actively monitor day-to-day operations.

25. What is meant by Supplier Relationship Management?

- PeopleSoft Supplier Relationship Management provides packaged integration of procurement functions with financial management, human capital management, and ERP suites
- PeopleSoft Supplier Relationship Management gives customers the flexibility to leverage applications on-demand, on-premises, or in any combination needed to achieve procurement objectives.
- PeopleSoft Supplier Relationship Management fully integrates procurement with HR in order to provide total workforce management of both contingent and full-time workers.

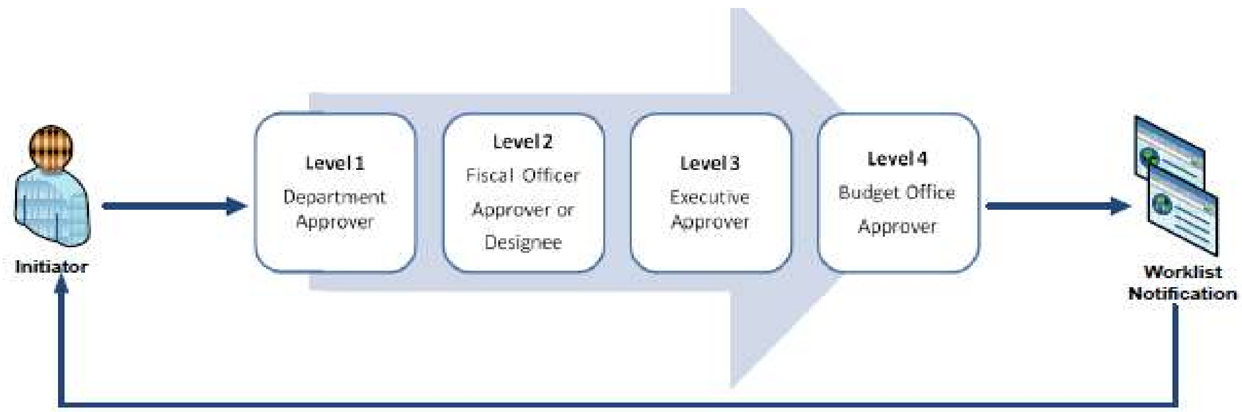
26. Delineate Supply Chain Management.

- Oracle's PeopleSoft Supply Chain Management (SCM) provides a cohesive yet flexible solution for the synchronized supply chain, driving efficiencies in cost savings over your entire supply chain—including your plan-to-produce and order-to-cash business processes.

27. List out the FMS Modules.

- Financial Management System (FMS) encompasses modules including: accounts payable, receivable, billing, general ledger.
- Each module is used in a Cost Center to manage a budget and to organize transactions.

28. Draw the diagram of PeopleSoft workflow process.



29. What is Business Reports?

- Using the Business Intelligence Reporting application, you can view detailed or summary reports of financial activity
- Review monthly or annual expenses and revenue by Cost Center
- Review project activity
- Review salary and wages for a Cost Center.

30. List out the products of PeopleSoft.

PeopleSoft Products

- Campus Solutions
- Customer Relationship Management
- Enterprise Performance Management
- Financial Management
- Human Capital Management
- Service Automation
- Supplier Relationship Management
- Supply Chain Management

31. What is meant by PeopleSoft accelerator?

- A small change can have a disastrous effect on your PeopleSoft system.

- Every update or change requires testing to make sure it doesn't bring your business down. All that testing takes time and money away from your business.
- For businesses that rely on HP Software Quality Management and PeopleSoft, we've helped you get started with a platform and have prebuilt the business-process tests for you.
- Validate your critical cross-platform business processes seamlessly, reducing risk while being more responsive.

32. Define Financial Management Systems.

- New software used to manage the financials of UT Dallas Product name is PeopleSoft Financials Management System. Covers the following
- Financials (account receivable, account payable, asset management, budget, account reconciliation, IDTs, etc.)
- Human resources (payroll)
- eProcurement (all supply ordering, check requests)
- Business Intelligence reports

PART – B (11 MARKS)

1. Explain in detail about basic functionality of PeopleSoft.

PEOPLESOFT

- PeopleSoft, Inc. was a company that provided Human Resource Management Systems (HRMS), Financial Management Solutions (FMS),Supply Chain Management (SCM), Customer Relationship Management (CRM), and Enterprise Performance Management (EPM) software, as well as software for manufacturing, and student administration to large corporations, governments, and organizations.
- Founded in 1987 by Ken Morris and David Duffield, PeopleSoft was originally headquartered in Walnut Creek, California before moving to Pleasanton, California. Duffield envisioned a client–server version of Integral Systems' popular mainframe HRMS package.

BASIC PEOPLESOFT FUNCTIONALITY

Evolution of PeopleSoft

- 1987 – Addressing organizational Human Relations departments, from recruitment to retirement. Managing the payroll, benefits, travel, expenses, and vacation of all employees was the key focus of the first PeopleSoft product.
- 1997 – Release of PeopleSoft 7, they upgraded the supply chain and finance modules, and added the functionality for analyzing business data.
- January 2000 – PeopleSoft 8 saw the addition of a module for customer relationship management through the acquisition of Vantive Corporation.

Applications of PeopleSoft

- The applications are comprehensive business and industry solutions, enabling organizations to increase productivity, accelerate business performance, and provide a lower cost of ownership.

Benefits of PeopleSoft

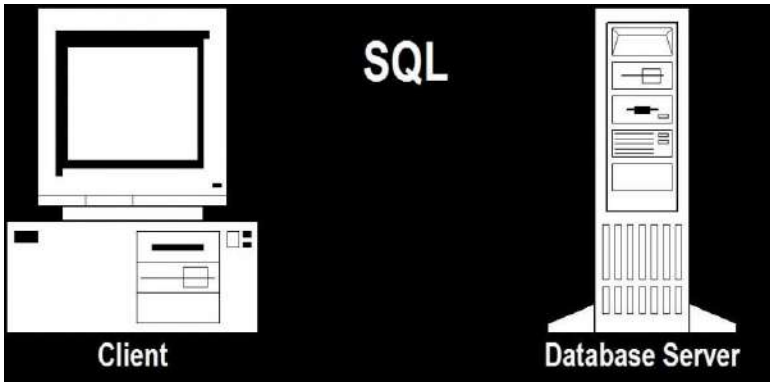
- End-user initiate transactions and date in to the system once and make it accessible for multiple purposes (Finance, Human Resources, Reports)
- Tracking of transaction status is available online, almost eliminating the need to make multiple phone calls regarding the transaction's status.
- Various transactions processing are online
- Transaction approvals have been simplified
- Some transactions can be approved and submitted online, while meeting internal control, and presenting less need to hand carry papers from office to office.

PeopleSoft Architecture Core Components

- PeopleSoft Enterprise Solution consists of several core components which are required for functioning of PeopleSoft Software.
- All these core components are interconnected and send/receive data and information required for processing of business needs.
- **Database Server**
  - Oracle (8i, 9i ,10g, 11i)
  - MS Sql Server
  - DB2 , Informix , Sybase
- **Web server**
  - BEA Web Logic
  - IBM WebSphere
- **Application server**
  - It is known as the heart of PeopleSoft and processes all the business logic.
- **NT File server**
  - Batch server – used for running PeopleSoft processes and reports e.g. application engine ,sqr reports etc.
- **Client (Any Internet Browser)**
  - PeopleSoft Internet Architecture is accessed via a URL and can be easy browsed from any operating system and device which has a browser. You can access Peoplesoft from your iPhone, iPad or any other device that has Peoplesoft supported browsers.
  - Internet explorer
  - Firefox
  - Chrome
  - Netscape navigator

<u>PeopleSoft 2 Tier Architecture</u>

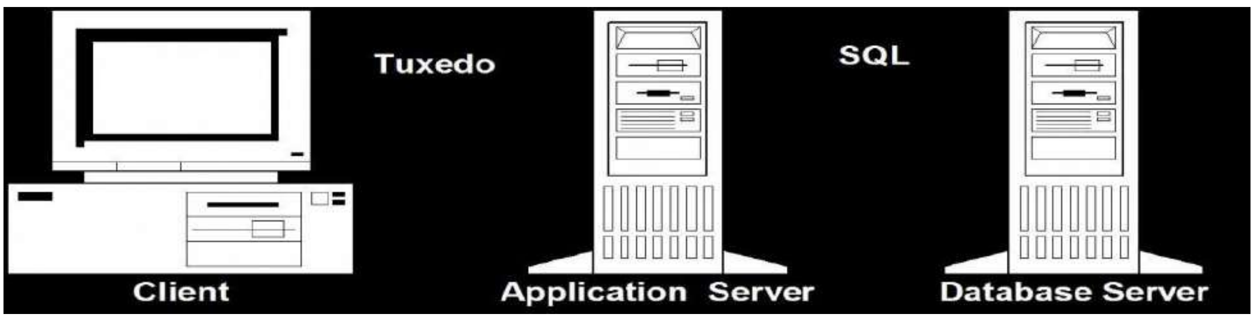
- 2 tier architecture of PeopleSoft Application depicts direct connection between People Tools and Database. Best example of this 2 tier architecture is PeopleSoft Tools like PeopleSoft Application Designer and PeopleSoft Data mover Tool.
- These tools make direct connection with PeopleSoft Database and are used by PeopleSoft developers and administrators.



PeopleSoft 2 Tier Architecture

<u>PeopleSoft 3 Tier Architecture (Logical)</u>

- In 3 tier setup of PeopleSoft architecture, PeopleSoft Application Designer doesn't connect to PeopleSoft Database directly.
- Instead it connects to Application Server using Tuxedo and then Application Server in turn connects with Database.
- Application Server communicates with Database using SQL queries



PeopleSoft 3 Tier Architecture

2. **Sketch and explain in detail about PeopleSoft database structure.**

- PeopleSoft has a long history of offering customers a choice of market-leading databases. People Tools provides an abstraction layer, which insulates application developers from the intricacies of each of the specific database platforms. Customer databases choices include:
  - Oracle
  - IBM DB2
  - Microsoft SQL Server
  - Informix
  - Sybase

- After you install your database engine there are three distinct layers within the database that work in concert to store and manage data for your PeopleSoft system.
- The database system tables manage both the PeopleTools and PeopleSoft application database objects, while the PeopleSoft application tables reside within the infrastructure defined by the PeopleTools metadata.

The following image describes three distinct yet integrated database layers for PeopleSoft system.

- **System tables**, also called system catalog tables, are analogous to a table of contents for a book or to file allocation tables on a hard drive. The structure and table names vary depending on which RDBMS you use. System catalog tables:
  - Keep track of all of the objects that reside in the database instance.
  - Are created by and owned by the RDBMS.
  - Are often described as system metadata.
- **People Tools tables** provide the infrastructure for PeopleSoft applications by storing and managing PeopleSoft application metadata. This metadata consists of information that defines the application, such as records, fields, pages, PeopleCode, and security. People Tools tables:
  - Define the structure of all object definitions that make up an application.
  - Use the same table structure for all applications.
  - Contain data that is added and updated only when the application is installed, or when using development tools such as PeopleSoft Application Designer or Data Mover.
- **Application data tables** store data entered through a PeopleSoft application. The specific tables and their structures vary by application. Application data tables:
  - Contain transactional data entered by users.

PeopleSoft database architecture consists of three categories as follows:

- SYSTEM CATALOG TABLE

- PEOPLETOOLS TABLE
- APPLICATION TABLE

SYSTEM CATALOG TABLE

- This tables are handled by DBA's since it has the DB related information along with it is used to check the performance of the PeopleSoft DB.
- Any change in the application tables updates the system catalog tables.
- Example of system catalog tables are: SYSCOLUMS, SYSTABLES

PEOPLETOOLS TABLES

- All delivered tables can be found in Peopletools Tables
- The metadata is stored in Peopletools tables like fields, records, page, component, menu, AE.
- Example of peopletools table are: PSRECDEFN, PSDBFIELD

APPLICATION TABLES

- The tables we develop are stored in application tables.PeopleSoft has preferred the naming convention of application tables to be starting with PS_
- Example of application tables are: PS_JOB
    - PS_EMPL_TBL
    - PSRECDEFN
    - PSDBFIELD
    - PS_JOB
    - PS_EMP_TBL
- Application tables are further categorized into three parts as follows:
    - Control Table
    - Transaction Table
    - Run control Table

3. Describe in detail about Data Mover.

- PeopleSoft Data Mover, is <u>used to import / export data between databases</u>.
- Apart from this, <u>Data Mover can be used for a variety of functions like database security management, running SQL scripts etc</u>. On Windows machines, you can use the Data Mover using

the GUI as well as from the command prompt. While UNIX variants support only command line operation.

- PeopleSoft Data Mover enables the database administrator or system programmer to perform a number of functions, including:
    - Transfer application data between PeopleSoft databases.
    - Move PeopleSoft databases across operating systems and database platforms.
    - Execute Structured Query Language (SQL) statements against any PeopleSoft database, regardless of the underlying operating system or database platform.
    - Create, edit, and run scripts.

PeopleSoft Data Mover Environment

- **Using the Data Mover development environment**.
  This a graphical user interface (GUI), which runs only in Microsoft Windows. Use the Data Mover shortcut in the PeopleSoft program group. Select Start, Programs, your_PSFT_program_group, Data Mover.
- **Using the Data Mover command-line interface.**
  The command-line interface is intended mainly for UNIX servers. You run PeopleSoft Data Mover from a console in Microsoft Windows and from a telnet session in UNIX.

PeopleSoft Data Mover Operating Modes

Operating modes determine how you are connected to the database. PeopleSoft Data Mover operating modes are:

- **Regular mode.**
    - Most of the time, you use regular mode. To sign in to regular mode, enter your PeopleSoft user ID and password during sign-in. In regular mode, all commands are valid.
- **Bootstrap mode.**
    - In bootstrap mode, you use a database access ID and password when signing in. Typically, you use bootstrap mode for database loading, because no PeopleSoft security tables are established yet. You also use bootstrap mode for running some security commands, such as ENCRYPT_PASSWORD.

Data Mover script commands:

- PeopleSoft Data Mover commands are platform-independent and are unique to PeopleSoft Data Mover.
- You can use PeopleSoft Data Mover commands for importing, exporting, and other tasks, such as controlling the run environment, renaming fields and records, administering database security, and denoting comments.

Supported SQL commands:

- With PeopleSoft Data Mover, you can use supported SQL commands in scripts on any supported database platform.
- Except as noted in the following discussion regarding standard SQL commands, you can use all of the supported SQL commands with the following Data Mover SET statements:
  - SET LOG
  - SET NO COMMIT
  - SET NO TRACE

Standard SQL Commands with DMS Scripts

- PeopleSoft Data Mover supports the following standard SQL commands:
  - ALTER
  - COMMIT
  - CREATE
  - DELETE

4. Explicate in detail about Records and its types.

- Fields that are grouped together as a unit are records.
- PeopleSoft records can be defined as the collection of fields on a page. It's similar what we use tables in RDBMS.
- The difference in PeopleSoft records is the structure of PeopleSoft records is stored in metadata tables that can be defined or edited through PeopleSoft application designer.
- Seven types of PeopleSoft Records:
  - Table
  - View
  - Derived/Work
  - Sub record

- Dynamic View
- Query View
- Temp table

Table:

- A table in PeopleSoft is similar like other database tables which has fields and that can be built and alter using application designer in PeopleSoft.

View:

- PeopleSoft View is a query written specially to retrieve/modify data on the records and also view's are used to show transaction data on reports and much more.

Derived/Work:

- Derived record are not built by application designer since there is no option to do that.

- It is just used for temporary storage and display purpose and the derived/work record data is saved never anywhere.

Subrecord:

- PeopleSoft subrecords are the collection of fields that are grouped together to form an entity so that they can be used for building blocks for multiple records.
- Since subrecords are not build through application designer they don't have their existence in the database.

Dynamic View:

- With dynamic view we can define a record definition that can be used like a view in pages and PeopleCode and is not actually stored as a SQL view in the database.

Query View:

- With Query view we can define the record definition as a view that is created using the PS Query. This kind of view when created by PeopleSoft Application Designer it prompts you to save the definition.

Temp table:

- Temporary tables in PeopleSoft are there to avoid the signification risk of data contention and helps to improve performance since the application data is getting processed in temporary table not on the large application data tables.

5. **Discuss in detail about PeopleSoft pages and forms.**

FORMS:

- The Forms and Approval Builder enables to design online forms, specify the approval process they require, and deploy them to users within your organization.
- Use this feature to convert manual procedures within your organization to paperless processes that include workflow-based approvals and an audit trail for tracking progress.
- No coding is required on your part, and future upgrades to your PeopleSoft system will not require you to update these forms, since the forms you create are not customizations.
- Once a form is published, users navigate to the form using the main menu, complete the fields and submit it for approval.
- Each published form includes three tabs: Form, Instructions, and Attachments.
- An audit trail of the approval history and comments is automatically generated as the form goes through the approval process. You can review the audit trail to see the history for each step of the approval chain.

Form Design

- To create forms, use the Design Form Wizard component (FORM_DESIGN_WIZARD), which guides you through the following steps of the form creation process:
    - Defining the basic information for a form, including the form ID and form owner.
    - Providing instructions that describe how to use the form.
    - Specifying the fields that appear on the form.
    - Attaching files to the form (optional).
    - Defining the menu item that is used to access the form.
    - Specifying the approval workflow that is required for the form.
    - Previewing, testing, and activating the form.
- Forms can contain fields arranged into one or two columns. You can specify which fields are required, and define the edits that a field must pass in order for the completed form to be saved. As you design the form, you can use the following field types:
    - Numeric
    - Text
    - Date
    - Time
    - Yes/No
    - Prompt:

Enables form users to select values from existing PeopleSoft records to complete the field.

- Code:

  Enables you to build a list of values that appear in a drop-down list. Form users can select a value from the list to complete the field.

- Section:

  Enables you to organize the form into multiple sections.

Form Status

- It is important to understand the distinction between a form and a form instance. For the purposes of this documentation, when we use the term form, we are referring to the “master” form, or template, that form designers create using the Form Design Wizard component.
- A form instance is a deployed form that has been completed by a form user. Each form instance is automatically assigned a unique sequence number, so all completed forms can be tracked and managed independently.

The possible status values for a *form* are:

In Design	Indicates a form that is being designed, and is not active.
Activated	Indicates a form that is active and available for form users to complete. Depending on your user role, you can activate a form using the following pages: • Design Form: Complete page of the Form Design Wizard component. • Manage Forms page.
Inactive	Indicates a form that has been inactivated by either the form's owner or a form administrator. Form administrators or form owners can inactivate forms by using the Manage Forms page. When a form is inactive, form users are not permitted to complete the form.

The possible status values for a *form instance* are:

Initial	When a form user accesses an activated form and begins to complete it, the status of the form instance is set to *Initial.*
Pending	When the form user finalizes the form instance and submits it for approval, the status changes to *Pending.*
On Hold	When an approver or reviewer requests more information about a submitted form, the status changes to *On Hold.*
Approved, Denied, or ***Cancelled***	As the form instance flows through the required approval chain, the status subsequently updates to approved, denied, or cancelled.

Security and Delivered Roles

- User roles determine who has permission to access, design, and administer forms.
- The following table lists the delivered roles and associated permission lists for Form and Approval Builder.

FORM_USER	Form user: Can complete and submit forms.
FORM_DESIGNER	Form designer: Can complete and submit forms, design forms
FORM_ADMIN	
FORM_CI_DEVELOPERS	Form to component interface (CI) developer

PAGES

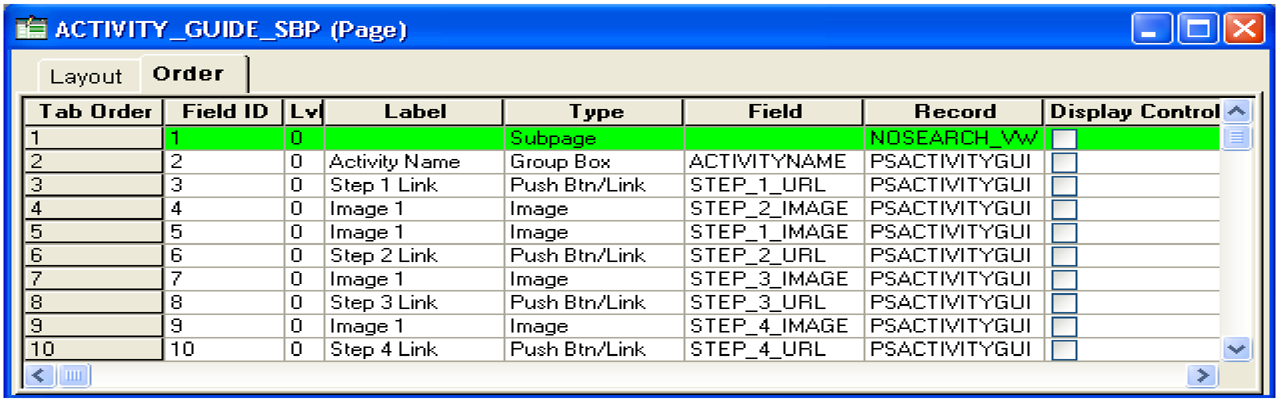
- Pages are the graphical interface between the users and the application database.
- As a system designer, we configure or build pages that meet the data requirements of the application and that are easy to use and understand.
- Using PeopleSoft Application Designer, we can create, modify, and delete page definitions in your PeopleSoft system.
- The page design based on the type of data that we plan to access and maintain. In some cases, a page references a single record definition; in others, you may want to reference multiple records.

- To accommodate a variety of page designs, PeopleSoft Application Designer uses level-based controls.
- The three level-based controls are:
  - Grids
  - Scroll areas
  - Scroll bars
- You specify the level of these controls by setting occurs levels on the Record tab of the page field properties dialog box for the level-based control that you are configuring.
- Level-based controls have four levels:
- **Level 0**
  - This level directly corresponds to the high-level key information of the underlying record.
  - Level 0 fields are usually physically located at the top of the page, set to be display-only, and display the data that the user entered on the initial search page.
- **Levels 1−3**
  - These levels correspond to lower level key information that is subordinate to the Level 0 key values.
  - Level 1 is subordinate to Level 0; Level 2 is subordinate to and nested within Level 1; and Level 3 is subordinate to and nested within Level 2.

- The first occurs level on a page is Level 0.
- In general, this level is reserved for the primary key fields that are used to search for data.
- A page might not contain any level-based controls; in such a case, all fields are set to Level 0.
- This case is particularly true for secondary pages or subpages that contain few data entry fields, as shown in the following example.
- The third column, Lvl (Level), that appears in the grid on the Order tab indicates that all fields on the page are at Level 0:

Example of page definition Order tab

This denotes the fields and controls on the Example of page definition Order tab.



ACTIVITY_GUIDE_SBP (Page)

Layout | Order

Tab Order	Field ID	Lvl	Label	Type	Field	Record	Display Control
1	1	0		Subpage		NOSEARCH_VW	
2	2	0	Activity Name	Group Box	ACTIVITYNAME	PSACTIVITYGUI	
3	3	0	Step 1 Link	Push Btn/Link	STEP_1_URL	PSACTIVITYGUI	
4	4	0	Image 1	Image	STEP_2_IMAGE	PSACTIVITYGUI	
5	5	0	Image 1	Image	STEP_1_IMAGE	PSACTIVITYGUI	
6	6	0	Step 2 Link	Push Btn/Link	STEP_2_URL	PSACTIVITYGUI	
7	7	0	Image 1	Image	STEP_3_IMAGE	PSACTIVITYGUI	
8	8	0	Step 3 Link	Push Btn/Link	STEP_3_URL	PSACTIVITYGUI	
9	9	0	Image 1	Image	STEP_4_IMAGE	PSACTIVITYGUI	
10	10	0	Step 4 Link	Push Btn/Link	STEP_4_URL	PSACTIVITYGUI	

- Nesting controls involve two or more level-based controls on a page, such as two scroll areas, when the second scroll area has occurs level of 2.
- You nest controls when the new data that you want to add is a repeating set of data for each entry in your first level-based control. In doing so, you create a hierarchical, or parent and child, relationship between the controls and the processing of the record definitions.
- The Level 2 control is the child of and is subordinate to the Level 1 control.

Example of Compensation page with nested grid in scroll area

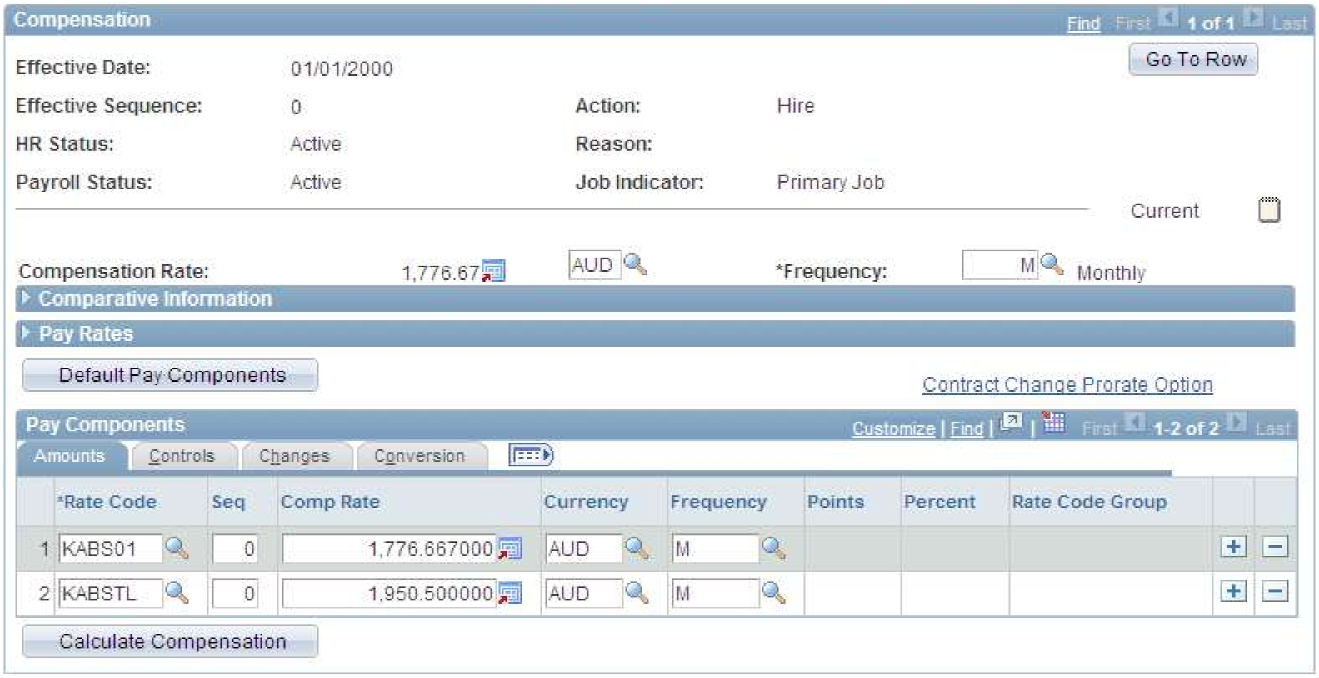
- This example illustrates the fields and controls on the Example of Compensation page with nested grid in scroll area.

Levels and Runtime Processing

- Levels play an important role in runtime processing.
- The component processor relies on the level at which you place a field on a page to determine how to process any PeopleCode attached to the field in the record definition.

Effective Dates and Level-Based Controls

- The EFFDT (effective date) field must be the only key field that is controlled by level-based controls that you create to help users maintain multiple occurrences of data that is keyed by effective date. Otherwise, the effective date processing for update actions does not function correctly.



Prompt fields:

- You can enable your users to look up the valid values that they can enter in a field.
- For this, PeopleSoft Application Designer provides prompts, or lookup buttons. Your PeopleSoft applications use the following three types of prompts:

  Drop-down list box

  - This type of prompt generates a list of values, which appears directly below the prompt field when the user clicks the down arrow that appears beside the field.

  Calendar prompt

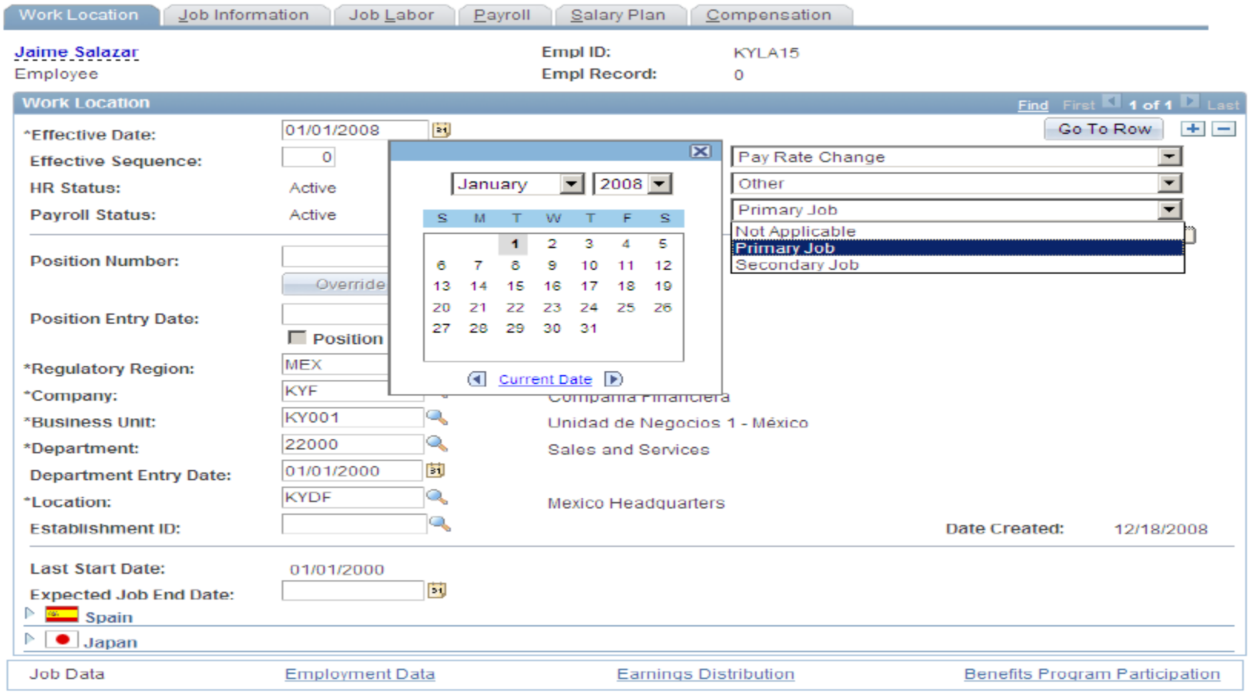
  - This type of prompt generates a small calendar, which appears next to the date field, when the user clicks the calendar button. Users can navigate the calendar to find and select a date.

  Lookup button

  - A prompt or lookup button opens a lookup page in the user's browser populated with up to 300 available values for that field. The user can then either select the desired value or refine their search further.

Work Location page showing sample prompt fields:

This example illustrates the fields and controls on the Work Location page showing sample prompt fields.



Hidden pages

- Hidden pages are <u>work pages that are associated with derived or work records; they are often used in work groups. You can store all of your work field controls on hidden pages</u>.
- Create these pages when you want PeopleCode calculations to be performed in the background so that the user does not see them. As a convention, the names of work pages that are delivered with your application names end with the suffix _WRK; you should follow the same naming convention.
- An example name is: MC_TYPE_WRK.

6. **Elucidate about PeopleSoft HRMS database.**

<u>INTRODUCTION TO PEOPLESOFT HRMS DATABASES</u>

- Functional flexibility, <u>the ability to perform human resources feats in a single click, is a cornerstone of the PeopleSoft product</u>.
- On the other hand, PeopleSoft is a software product with many programmers behind it. These programmers work together, writing separate components and improving technical benchmarks, like access time and storage capabilities.
- To prevent a tangled web of data, PeopleSoft stores data using a database organizational concept called “data integrity.” The philosophy of data integrity is simple: a database stores data efficiently by storing it once, and only once.

- Duplication equals inefficiency and creates a chance that data will be changed in one place and not another. With each new PeopleSoft release, data is spread thinner and thinner, to more and more tables.
- Tables contain the most commonly used data in PeopleSoft HRMS. These tables are not all essential, but they are accessed often and provide the foundation for the database.

Types of Tables:

There are six basic types of tables

- **Base Table:**
  - A base table is the place where nearly every query starts. These tables store information about an employee and contain data about the employee.
  - A base table stores live data that is continually changing.
  - The table could store information about employees, their dependents, their earnings, taxes, deductions, or benefits. In short, these tables hold the real data, the non-static data.
- **Control Table**
  - A control table contains a short list of values that classify and categorize.
  - For example, a table that contains all of the possible earnings codes (regular, bonus, overtime, etc.) is a control table, whereas the table that contains the actual earning amounts is a base table.
  - Each control table has a key (the code) which ties to a field on a base table.
- **Views**
  - Views are timesavers; they are the result set of an SQL statement.
  - For example, the benefits view table takes fields from several tables, links them together correctly, and presents the result as a new table.
- **Reporting Tables**
  - These tables are similar to views, but are not dynamic.
  - Their data is only current after a program is executed every night.
  - Their chief benefit is performance. Instead of joining 10 tables every time you look something up, the tables are joined once at night and then used throughout the next day as a single table.
- **Application Tables**
  - The PeopleSoft application stores application rules and definitions in application tables. Occasionally these tables temporarily store data in the middle of a process.

- With few exceptions, these tables store data that is not relevant to the organization.

- **The Non-Table: Sub-Records**
    - PeopleSoft is intent on making sure no data or effort is ever duplicated. For this reason, they created sub records.
    - Sub-records are like a mini-table, since they are a definition of a group of fields.
    - The sub-record is a single definition that is used in many different records.
    - Addresses are consistently stored, and if a change is required, such as an extra digit added to the postal code definition, that change needs to be made in only one place—that is, from an application standpoint.
- **Identifying the Elements in table:**
    - Once we find the table we're looking for, there are several ways to look at it.
    - Most people find it quicker and more convenient to print out the table.
    - PeopleSoft prints the table with the record name, field names, definitions, formats, descriptions, and other technical information.

Keys

- Identifying the keys in a table is a critical step to retrieving accurate data.
- Keys are the unique identifiers of a row in the table. Sometimes there is one key, such as employee ID, but more often there are several.
- When there is more than one key, the values in a certain key could be duplicated, but there are never two rows with the same values in every key.
- In other words, the combination of the key fields must be unique.

EMPLID	COMPANY	EFFDT	STATE
-------	--------	------------	------
8121	CCB	01-JAN-1991	CA
8121	CCB	01-JAN-1991	PA
8200	PST	01-OCT-1993	CT
8200	PST	01-JAN-1991	CA.

- When selecting data, always consider every key. To retrieve the appropriate row, each key must be limited to exactly the values you are looking for.

Required Fields

- The column labeled “Req” tells you whether PeopleSoft requires a field.
- A required field, from an application perspective, must be filled in. When inserting data using a SQL statement the required flag tells you whether to expect blanks in the data returned.
- If it isn’t required, there is a good chance some fields will be blank.
- If it is required, there is a slim chance that the field will be blank.

Translation Values

- Many fields have only a few possible values. These values are stored in a common table.
- The table printout provides a nice way to look at the possible values for certain fields, for example, status field can only have three possibilities: E, G, or N. Hard-coding the criteria of ‘E’ for exempt into your query would return employees who are exempt from state taxes.

Prompt Tables

- The prompt table column lists the tables that provide values for a certain field.
- Often this is a control table. In PeopleSoft, when typing state tax data into a panel, you can prompt for a list of possible companies.

7. Explain in brief about PeopleSoft Products.

<u>PeopleSoft Products</u>

PeopleSoft Accelerator

- A small change can have a disastrous effect on your PeopleSoft system.
- Every <u>update or change requires testing to make sure it doesn't bring your business down</u>. All that testing takes time and money away from your business. A Full Automation Framework
- Our Accelerator Platform provides a complete environment with a structured approach ready for fast customization. By the time implementation is over, you'll have a library of test cases that cover a vast array of business processes customized for your unique environment.

Script less Testing For:

- Patches
- Hot Packs
- Upgrades
- Technology Stack Upgrades or Modifications

Generate Thorough Documentation

- No guesswork here. Accelerator documents all tests performed and the results achieved with the meticulous accuracy that manual testers only experience in their nightmares.

- Audit and compliance made easy - reporting can now be done with no additional costs or complexity.

PeopleSoft Products

- Campus Solutions
- Customer Relationship Management
- Enterprise Performance Management
- Financial Management
- Human Capital Management
- Service Automation
- Supplier Relationship Management
- Supply Chain Management

Campus Solutions

- PeopleSoft Campus Solutions partners with higher education industry leaders around the globe to deliver the most modern technology solution designed specifically for higher education. Our investment in PeopleSoft Campus Solutions delivers:
- A global perspective supporting education and research without boundaries
- Support for varied models of instruction and business requirements
- The deepest and broadest set of capabilities to support the student life-cycle
- Innovative technology to enhance, strengthen and adapt capabilities
- The ability to recruit, enroll and manage a flow of students seeking higher education across borders
- Support for multi-institution, multi-currency, multi-campus, and multi-language operations

Customer Relationship Management:

- Setting up installation, basic system tables, and security options.
- Setting up and using features that are common to multiple PeopleSoft CRM applications, such as notes, search collections, alternate character, interactive reports, and diagnostic reports.
- Administering worker information.
- Setting up and using interactions and 360-degree views.
- Setting up and using self-service applications.
- Managing relationships with customers.

Enterprise Performance Management:

- PeopleSoft Enterprise Performance Management offers performance management solutions for every budget and every phase of the management cycle, helping managers formulate strategies for profitable growth, align strategies with operational plans, and actively monitor day-to-day operations.
- Reduce month-end close from weeks to days to achieve best-in-class status against industry standards
- Consolidate the financials from thousands of groups worldwide and report the numbers to Wall Street
- "The tight integration between [performance management] and PeopleSoft applications is very attractive to customers that want a single, standard platform for information."

Financial Management

- Oracle's PeopleSoft Financial Management reduces costs by automating, centralizing, and standardizing global transactional processes.
- Manage risk and reduce compliance costs with end-to-end processes for governance, risk, and compliance.
- Make the right investments in the business with integrated performance management and business intelligence.

Human Capital Management

- Oracle's PeopleSoft Human Capital Management enables you to architect a global foundation for HR data and improved business processes.
- PeopleSoft Human Capital Management delivers a robust set of best-in-class human resources functionality that enables you to increase productivity, accelerate business performance, and lower your cost of ownership.

Service Automation

- Oracle's PeopleSoft Enterprise Services Automation applications enables project-centric organizations and departments to establish core operational processes that support full project lifecycle management—across operations and finance—from project selection, planning and staffing, through execution, cost control, and analysis.

Supplier Relationship Management

- PeopleSoft Supplier Relationship Management provides packaged integration of procurement functions with financial management, human capital management, and ERP suites
- PeopleSoft Supplier Relationship Management gives customers the flexibility to leverage applications on-demand, on-premises, or in any combination needed to achieve procurement objectives.
- PeopleSoft Supplier Relationship Management fully integrates procurement with HR in order to provide total workforce management of both contingent and full-time workers.

Supply Chain Management

- Oracle's PeopleSoft Supply Chain Management (SCM) provides a cohesive yet flexible solution for the synchronized supply chain, driving efficiencies in cost savings over your entire supply chain—including your plan-to-produce and order-to-cash business processes.
- Extends your supply chain in real time by connecting suppliers and customers with company business processes.
- Provides integrated spend management for all categories of goods and services.
- Delivers embedded analytics to monitor supply chain performance and adjust as conditions and business goals shift.

8. Detail about Functional management system.

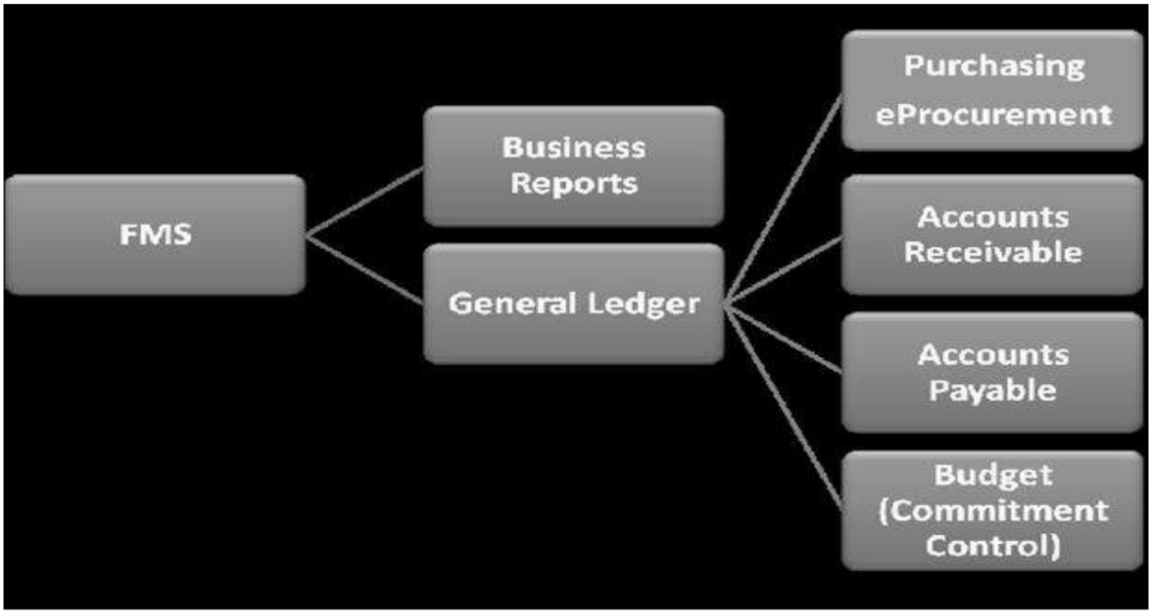
- New software used to manage the financials of UT Dallas Product name is PeopleSoft Financials Management System. Covers the following
    - Financials (account receivable, account payable, asset management, budget, account reconciliation, IDTs, etc.)
    - Human resources (payroll)
    - e Procurement (all supply ordering, check requests)
    - Business Intelligence reports

The FMS Modules

- Financial Management System (FMS) encompasses modules including:
    - Accounts payable,

- Receivable,
- Billing,
- General ledger.

- Each module is used in a Cost Center to manage a budget and to organize transactions.



- The Financial Management System (FMS) is composed of many modules. The modules are designed to perform specific activity.
- Those activities or tasks include accounts payable, accounts receivable, billing, payroll processing, etc.

General Ledger

- Financial transactions are generated in separate systems and loaded into the General Ledger.
- The remaining financial transactions are entered into the General Ledger by you using PeopleSoft Financials applications.
- Examples include daily cash receipts, service center charges and departmental allocations and corrections.

Accounts Payable Integration

- Each application uses details from PeopleSoft transactions.
- This information is used to build reports, which in turn is used to ensure accurate accounting of funds distribution (your yearly budget).

Commitment Control

- Commitment Control is used in PeopleSoft to manage all budget activity. Using Commitment Control, you can
- View all available balances and transactions (revenue or expense) for a given Cost Center
- Move funds between Cost Centers
- Add or reduce funds for a Cost Center

Purchasing Applications

- **eProcurement-**
    - Used to purchase goods and services using a web-based application
    - Available from Galaxy portal
- **eProcurement is designed to:**
    - Reduce Purchase Order cycle time
    - Make purchasing and check requests as easy as possible
    - Provide easy on-line access to most commonly ordered products and services

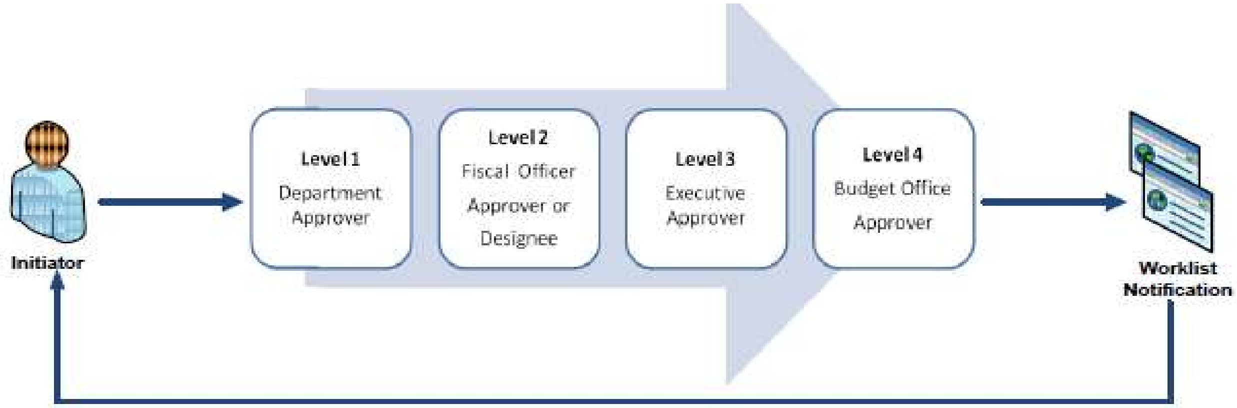
Travel and Expenses

- Used to authorize and record any travel expenses incurred by UT Dallas faculty, administrator, staff or student for university business
- Use paper forms to set up travel and record expenses
- Transactions are captured in General Ledger

Purchasing Card

- Used to make authorized purchases
- Transactions are captured in General Ledger

Workflow Process



- The Creator submits a transaction such as an IDT Journal to Workflow Approval. Each approver (Approver 1 and Approver 2), reviews and approves or denies the journal.
- The Finance or Budget approver finalizes the transaction.

Business Reports:

- Using the Business Intelligence Reporting application, you can view detailed or summary reports of financial activity
- Review monthly or annual expenses and revenue by Cost Center
- Review project activity
- Review salary and wages for a Cost Center.

9. Draft a note on PeopleSoft enterprise HRMS.

PEOPLESOFT ENTERPRISE HRMS

- PeopleSoft HRMS is an integrated suite of applications and business processes that are based on PeopleSoft's Pure Internet Architecture (PIA) and enterprise portal technologies.
- The sophisticated features and collaborative, self-service functionality available in PeopleSoft HRMS enable you to manage your human resources from recruitment to retirement while aligning your workforce initiatives with strategic business goals and objectives.

Types of PeopleSoft Enterprise HRMS is

- PeopleSoft Enterprise HRMS integrations.
- PeopleSoft Enterprise HRMS implementation.

PeopleSoft Enterprise HRMS Integrations

- PeopleSoft HRMS integrates with other PeopleSoft applications, such as PeopleSoft Enterprise Financials, PeopleSoft Enterprise Workforce Analytics, and PeopleSoft Enterprise Learning Management.
- PeopleSoft HRMS also integrates with other third-party applications. PeopleSoft HRMS uses various integration technologies to send and receive data.

PeopleSoft Enterprise HRMS Implementation

- PeopleSoft Setup Manager enables you to generate a list of setup tasks for your organization based on the features that you are implementing.
- The setup tasks include the components that you must set up, listed in the order in which you must enter data into the component tables, and links to the corresponding People Book documentation.
- PeopleSoft HRMS also provides component interfaces to help you load data from your existing system into PeopleSoft HRMS tables.
- Use the Excel to Component Interface utility with the component interfaces to populate tables.

This table lists all of the components documented in the PeopleSoft Enterprise HRMS Application Fundamentals People Soft that have component interfaces:

Component	***Component Interface***
Departments component (DEPARTMENT_TBL)	DEPARTMENT_TBL
Establishment component (ESTABLISHMENT_DATA)	ESTABLISHMENT_DATA
Job Code Task Table component (JOBCODE_TASK_TABLE)	JOBCODE_TASK_TABLE
Job Code Table component (JOB_CODE_TBL)	CI_JOB_CODE_TBL
Job Tasks component (JOB_TASK_TABLE)	JOB_TASK_TABLE
Location component (LOCATION_TABLE)	LOCATION_TABLE
FLSA Calendar Table (FLSA_CALENDAR)	FLSA_CALENDAR
Pay Run Table (PAY_RUN_TABLE)	PAY_RUN_TABLE
Tax Location Table component (TAX_LOCATION_TBL)	TAX_LOCATION_TBL

Department of Computer Science and Engineering

Subject Name: **ENTERPRISE SOLUTIONS** Subject Code: **CS T61**

UNIT – V

Siebel Enterprise Applications - Siebel eBusiness Applications – Siebel Tools – Tables and Columns – Business Component – Business Objects – Applets – Joins – Links – Views – Screens – Configuring applications.

2 MARKS

1. What is Siebel?

Siebel is a prominent vendor of interoperable e-business software. They also call themselves Siebel Systems. The company's customer relationship management (CRM), enterprise resource management (ERM), and partner relationship management (PRM) applications are designed to automate those aspects of business and allow an enterprise to perform and coordinate associated tasks over the Internet and through other channels, such as retail or call center networks.

2. Name some customers of Siebel?

Siebel's customers include Chase Manhattan Bank, Deutsche Telecom, IBM, Lucent, Yahoo, and Microsoft.

3. What is Siebel Enterprise Application Integration?

*Siebel Enterprise Application Integration (*Siebel EAI*)* is the set of products on the Siebel Business Platform that includes tools, technologies, and prebuilt functional integrations that facilitate application integration.

4. What are the Features of Siebel EAI?

Some of the features provided by Siebel EAI include:

- Provides components to integrate Siebel Business Applications with external applications and technologies within your company
- Works with third-party solutions, such as solutions from IBM, TIBCO, WebMethods, and so forth
- Provides bidirectional, real time and batch solutions to integrate Siebel applications with other applications

- Provides a set of interfaces that interact with each other and with other components within a Siebel application.

5. What is Siebel e-Business applications?

Siebel e-Business applications synchronize the data through one central repository regardless of the contact points (e.g., web, e-mail, call center, field visits and resellers). By having everyone in the organization working with the same repository (data are stored only once), a complete 360-degree view of the customer can be provided to every department or function on an as-needed basis with consistent and up-to-date information available throughout the organization.

6. What is CRM?

Customer Relationship Management (CRM) processes is an approach to managing a company's interaction with current and future customers. It often involves using technology to organize, automate, and synchronize sales, marketing, customer service, and technical support.

7. Write the three levels of CRM?

- Operational CRM (communication strategy level)
- Analytical CRM (marketing strategy level)
- Collaborative CRM (business strategy level)

8. What is Siebel Tool?

Siebel Tools is an integrated environment for configuring all aspects of a Siebel application so a single configuration can be:

- Deployed across HTML and wireless clients
- Used to support multiple Siebel applications and languages
- Easily maintained
- Automatically upgraded to future Siebel product releases

9. Whether Siebel Tools is a programming environment?

Siebel Tools is not a programming environment

10.What are the two windows of Navigation in Siebel Tools ?

- Object Explorer window
- Object List Editor window

11. Define Tables?

A table object definition is the direct representation of a physical database table in a DBMS. Siebel applications provide a set of standard tables that are included in standard Siebel applications. These tables have predefined names and structures.

12. Define Base Table?

The term base table is used in two different contexts:

- The base table for an extension table is the table it extends. This is specified in the Base Table property of the extension table’s object definition.

- The base table for a business component is the table that provides most of its essential fields. This is specified in the Table property of the Business Component object definition.

13. Write the Properties of the Table Object Type?

- Name
- Type
- Base Table
- User Name
- Alias

14. Define Data tables?

Data tables comprise most of the tables in Siebel applications. They serve as base tables for business components, and their columns provide the data for fields. Data tables can be public or private.

15. Define an Extension table?

An extension table provides additional columns to a data table that cannot be directly added to the original table because the underlying DBMS may support only a limited number of columns, or will not allow adding a column to a table once it is populated with data. An extension table allows you to provide additional columns for use as fields in a business component without violating DBMS or Siebel application restrictions.

16. Define an Interaction Table?

An intersection table implements a many-to-many relationship between two business components.

17. What is Column object?

A Column object definition is the direct representation of a database column in a DBMS. The name, data type, length, primary key and foreign key status, alias, and other properties of the database column are recorded as properties in the corresponding column object definition.

18.What are the types of Column Objects ?

- Name
- Default
- Physical Type (Physical Type Name)

19.What are the types of Columns?

- Data Columns
- Extension Columns
- System Columns

20. Define Data Columns?

Data columns comprise most of the columns in Siebel applications. They are sometimes referred to as base columns. Data columns provide the data for fields, or serve as foreign keys that point to rows in other tables. The developer cannot modify the properties of data columns, unlike extension columns. Data columns can be public or private.

21. Define Extension Columns?

An extension column is a column that is not used by standard Siebel applications. There are three kinds of extension columns:

- Standard extension columns.
- Custom extension columns in an extension table
- Custom extension columns in a base table

22. Define System Column?

System columns have a value of System in their Type property. System columns appear in all tables in Siebel applications, although the same set of system columns does not appear in every table. You can use the data in system columns for various purposes; for example, the ROW_ID column in tables is used in the construction of joins.

23. What is meant by Business Component?

A business component is a logical entity that associates columns from one or more tables into a single structure.

24. Define Business Object?

A business object implements a business model (as represented in a logical database diagram), tying together a set of interrelated business components using links. The links provide the one-to-many relationships that govern how the business components interrelate in the context of this business object.

25. Define Applets?

An Applet is a data entry form, composed of controls, that occupies some portion of the Siebel application user interface. An applet can be configured to allow data entry, provide a table of data rows, or display business graphics, or a navigation tree. It provides viewing, entry, modification, and navigation capabilities for data in one business component.

26. Write the types of Applets?

- Form Applet
- List Applet

27. Define Form Applet?

A form applet displays data in a data entry form. Fields in the business component appear on the form applet as text boxes, check boxes, and other standard controls

28. Define List Applet?

A list applet allows the simultaneous display of data from multiple records. A list applet displays data in a list table format, much like a spreadsheet or word processor table.

29. Define Joins?

A Join object definition creates a relationship between a business component and a table other than its base table. The join allows the business component to use columns from that table. The join uses a foreign key in the business component to obtain rows on a one-to-one basis from the joined table, even though the two do not necessarily have a one-to-one relationship.

30. Define Links?

A link implements a one-to-many (or master-detail) relationship between business components based on their base tables. The Link object type makes master-detail views possible, in which one record of the master business component displays with many detail business component records that correspond to the master.

31. What is Multi value Link?

A multi-value link implements a special use of the Link object type, which is the maintenance by the user of a list of records attached to a control or list column in an applet. The group of attached detail records is called a multi-value group.

32. Define Views?

A view is a collection of applets that appear at the same time on the same screen. A view can be thought of as a single window's worth of data forms (applets). Generally, a Siebel application window displays one view at any one time. The currently active view is changed by selecting a different view from the view tabs.

33. Define List Form View?

A list applet and a form applet display data from the same business component. The list applet appears above the form applet. The form applet presents the same information as the currently selected record in the list applet, with a different arrangement that may include more fields.

34. Define Master-detail view?

In a master-detail view, a form applet and a list applet display data from two business components related by a link. The form applet appears above the list applet. The form applet displays one record from the master business component in the master-detail relationship. The list applet displays all of the records from the detail business component that have as their master record the record currently displayed in the form applet.

35. Define Thread Bar?

The thread bar is a navigational tool for the user. It provides the means to navigate from view to view among the views previously visited in the current screen.

36. What is Drilldown Behavior in a View?

It allows the user to drill down from a cell in a list applet (or using a pop-up menu in either a form or list applet) to a particular view. Drilldown controls or list columns in a list applet in

Siebel applications consist of colored, underlined text, much like a hypertext link in a Web browser.

37. Give the types of Drilldown Behavior?

- Static Drilldown Behavior
- Dynamic Drilldown Behavior

38. Define screens?

A screen is a logical collection of views. It is not a visual construct in itself; rather, it associates views so that other visual constructs, such as the Site Map and tab bar, can reflect the list of views contained in the currently active screen.

39. Write any Configuring applications for deployment in High Interactivity mode.

- Browser scripting is fully supported in High Interactivity mode.
- You cannot modify the appearance of the rich text editor.
- You cannot modify the background and text color of list applets.
- You cannot place method-invoking controls.

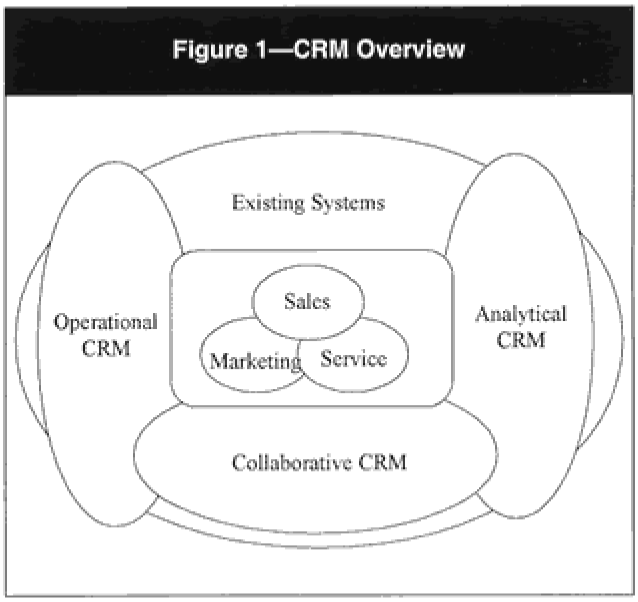
11 MARKS

1. Explain CRM Processes in detail.

Companies, both large and small, are looking to gain resources like automating finance, manufacturing, distribution, human resources management and general office operations, many businesses through CRM strategies that enable them to manage all customer contact points, including person-to-person contact, the telephone, e-mail and the Internet.

Different levels of CRM3 (see **figure 1**) include:

- **Operational CRM** (communication strategy level) is the automation of the customer-oriented front-end marketing, sales and service processes and their integration with back-office systems. An example of operational CRM is a customer database maintained purely for direct mailing purposes.
- **Analytical CRM** (marketing strategy level) is the analyses of the data created with operational CRM to better meet the needs and desires of customers. For example, relations between events and buying behaviors are analyzed, typically in a data warehouse (possibly with neural network capabilities).
- **Collaborative CRM** (business strategy level) is the overall communication and coordination model of a customer life cycle between the channels and contact points. In this collaborative CRM approach, business processes and a central customer database are drivers for success.



Siebel is designed to deal with all the different levels of CRM and supports CRM as a business strategy. It should be taken into account that the way CRM is implemented and effectively utilized will vary between organizations. It is dependent on many variants, including the way of doing business, buying habits of customers (e.g., convenience vs. specialty goods), industry (e.g., telecom with many consumers vs. forestry with a few business customers), company size, existing systems and organization structure.

Organization-specific aspects need to be incorporated in the process designs for marketing (e.g., campaign management, lead generation), sales (e.g., opportunity management, contact and account management and quotation management) and service (e.g., call center processes, service agreement management). These processes have to be related to the relevant business units and Siebel products. As a result of this mapping process, changes or improvements may be identified for the business units as well as gaps between the process design and the Siebel standard solution.

It is important for the IT auditor to be involved in the process design to ensure integrity, security, auditability, controllability, availability, efficiency and effectiveness requirements are defined for the core marketing, sales and service processes as well as for the management control and support activities surrounding these processes (e.g., interfaces with the backbone ERP software).

2. Explain Siebel Tool in detail?

Siebel Tools is an integrated environment for configuring all aspects of a Siebel application so a single configuration can be:

- Deployed across HTML and wireless clients

- Used to support multiple Siebel applications and languages
- Easily maintained
- Automatically upgraded to future Siebel product releases

Siebel Tools is not a programming environment; it is a declarative application configuration tool. Standard Siebel applications provide a core set of object definitions that you can use as a basis for your tailored application. Using Siebel Tools and other configuration tools that are part of a Siebel solution, Siebel application developers, system administrators, and database administrators can customize a standard Siebel application without modifying source code or SQL. Some of the configuration tools are accessed through the Siebel applications. Siebel Tools, however, is a separate product with its own user interface.

Navigation in Siebel Tools is done mainly in two windows:

- Object Explorer window
- Object List Editor window

3. Explain in details about Tables.

Tables:

A table object definition is the direct representation of a physical database table in a DBMS. Siebel applications provide a set of standard tables that are included in standard Siebel applications. These tables have predefined names and structures, and typically begin with one of the following prefixes:

S_. Table names starting in S_ are standard tables for supplied with Siebel Sales and Siebel Service. For example, the S_CONTACT table stores contact information, and S_OPTY stores opportunity information. Nearly all standard tables are of this type.

W_. Table names starting in W_ are Data Warehouse tables used in Siebel Analytics to demoralize data used in the S_ tables.

Base Tables

The term base table is used in two different contexts:

■ The base table for an extension table is the table it extends. This is specified in the Base Table property of the extension table’s object definition.

■ The base table for a business component is the table that provides most of its essential fields. This is specified in the Table property of the Business Component object definition. The set of fields supplied by the base table is supplemented by non updateable fields that are obtained from joins.

Properties of the Table Object Type:

The following are the key properties in a table object definition:

1. Name: Provides the name of the table in the DBMS.

2. Type: Indicates which of the following styles describes the table.

- **Data (Public).** Public data tables are among the original set of tables implemented in Siebel applications. They hold data that is made available through business components to developers and users. Public data tables can be extended using extension tables and extension columns. These extension are subject to database restrictions. Data tables are discussed in “Data Tables” on page 196.
- **Data (Private).** Private data tables are similar to public data tables, but cannot have extension columns.
- **Data (Intersection).** Identifies an intersection table. An intersection table implements a many-to-many relationship between two data tables. Intersection tables are discussed in “Intersection Tables” on page 204.
- **Extension**. An extension table adds additional columns to a data table that the original data table is unable to hold due to DBMS platform or Siebel application design restrictions. Extension table names have an _X suffix, or _XM or one-to-many, or _T for TAS extension tables. NOTE: Extension tables that have _XM suffix have table type of “Data (Public).” Extension tables are discussed in “Extension Tables” on page 196.
- **Interface.** Interface tables are used by Siebel Enterprise Integration Manager (EIM) to import initial data for populating one or more base tables and subsequently to perform periodic batch updates between Siebel applications and other enterprise applications. Interface table names end in _IF or _XMIF. Interface tables are discussed in “Column Objects” on page 215.
- **Database View**, Dictionary, Journal, Log, Repository, Virtual Table, and Warehouse styles. These are all table types that are reserved for Siebel internal use.
- **Extension (Siebel).** These tables are reserved for Siebel use only. They are usually extensions from S_PARTY. If customers want to extend person- and organization-related tables they need to extend from S_PARTY. For example, S_CONTACT is an extension table of S_PARTY. Because S_CONTACT is of type Extension (Siebel), you cannot use it as a parent table for an extension table. You must use S_PARTY. For a business component based on your new table to show data from S_CONTACT, you must create a Join object that references S_CONTACT and has a Join Specification child object with a Source Field property set to Parent Id and Destination Column property set to ROW_ID. The row ID of an S_CONTACT record will be the same as the row ID of the corresponding S_PARTY record.

3. Base Table. Identifies the base table if the table in the object definition is an extension table. If the table in the object definition is a base table, this property is blank. An extension table always identifies a base table.

4. **User Name**. A longer, descriptive name that aids in identifying the table when used in configuration.

5. **Alias**. A name that can be used as a synonym for the table name to make the name more understandable. For example, an alias such as S_Organization_External could be specified for the S_ORG_EXT table.

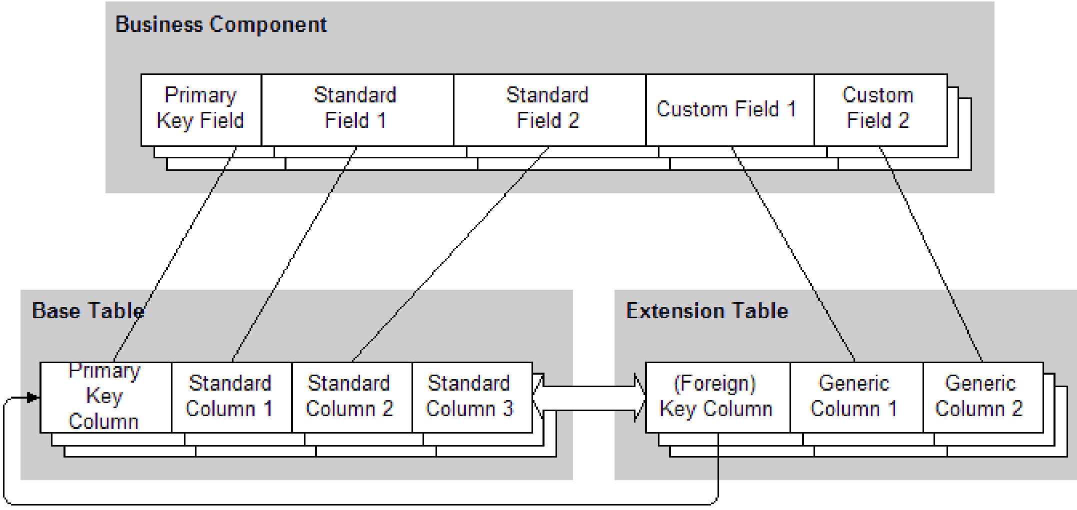
Data Tables :

Data tables comprise most of the tables in Siebel applications. They serve as base tables for business components, and their columns provide the data for fields. Data tables can be public or private.

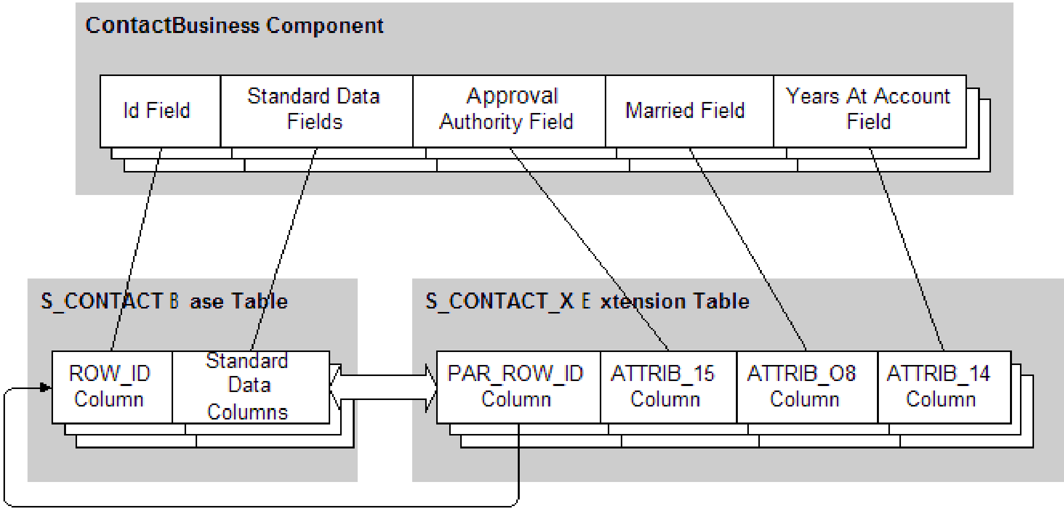
4. Explain in detail about Extension Tables:

An extension table provides additional columns to a data table that cannot be directly added to the original table because the underlying DBMS may support only a limited number of columns, or will not allow adding a column to a table once it is populated with data. An extension table allows you to provide additional columns for use as fields in a business component without violating DBMS or Siebel application restrictions.

An extension table is a logical augmentation of an existing table. Its columns are provided mostly for developers, and are generally not used by standard Siebel applications. An extension table extends a base table in the sense that it effectively adds additional columns. These columns are not physically part of the base table, but are available for use in a business component alongside the base table columns as if they were. When columns in a base table are updated, the time stamps of its extension tables are not updated unless columns in those extension tables are also updated. When records in an extension table are changed, system columns in a parent table are updated. This is done because the associated record in an extension table is considered by the object manager to be logically a part of its parent record. The relationships between a base table, an extension table, and the business component that uses them are given below in the figure.



Business Component, Base Table and Extension Table



Extension Table Example

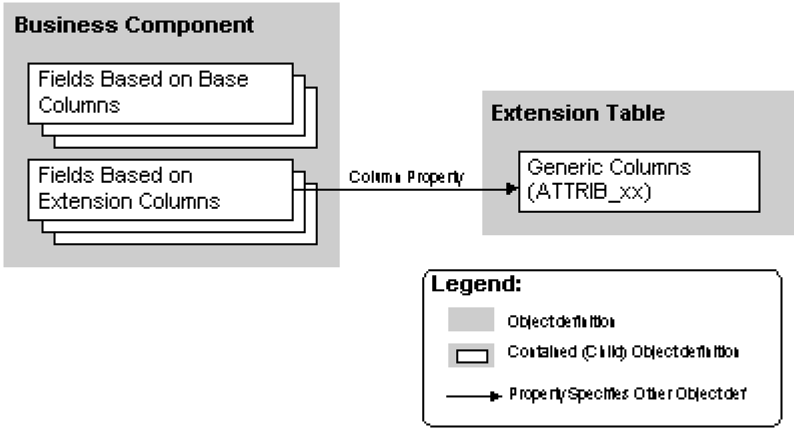
Extension tables can be of the one-to-one or one-to-many style:

Rows in *one-to-one* extension table have a one-to-one relationship with corresponding rows in the base table. A one-to-one extension table extends the base table horizontally

In a ***one-to-many*** extension table, there are multiple extension table rows for each base table row. There are standard one-to-many extension tables for certain of the major business components, including Opportunity, Contact and Account. These are used primarily to create multi-value groups based on user-created business components.

One-to-One Extension Tables

One-to-one extension tables have the _X suffix on their names (with the exception of TAS tables, which have the suffix _T). The details of the object definition relationships (excluding the implied join) are given in the figure.



Extension Table Details (Excluding Implied Join)

The object definitions in the above figure are:

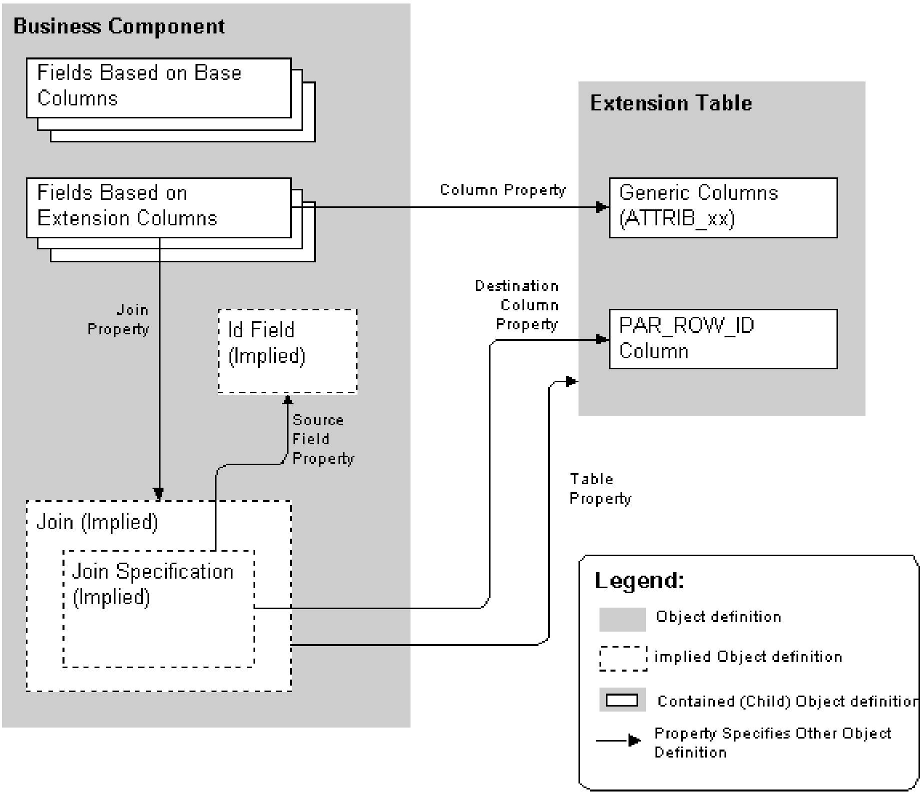
- **Business component.** Business component being extended.
- **Fields based on base columns.** Fields that represent data from columns in the business component's base table. They are unaffected by the extension table.
- **Fields based on extension columns.** Represent data from columns in the extension table.
- **Extension table.** Provides columns that may be used to add developer-defined fields to the business component.

Implied Joins

Underlying the one-to-one extension table's relationships with the base table and business component is a set of hidden relationships called an *implied* or *implicit join*. The implied join makes the extension table rows available on a one-to-one basis to the business component that uses the extension table. Every extension table has an implied join with the business component it extends. This join always has the name of the extension table.

An implied join is different from joins defined as object definitions. Data can be updated through an implied join. Data can be displayed only through other joins. This update capability is important for extension table functionality.

When a field in the business component is based on a column in the extension table, the Column property of the Field object is set to the name of the column, and the Join property is set to the name of the extension table. For example, the Birthday field in the Contact business component has a Column property value of ATTRIB_26 and a Join property value of S_CONTACT_X.



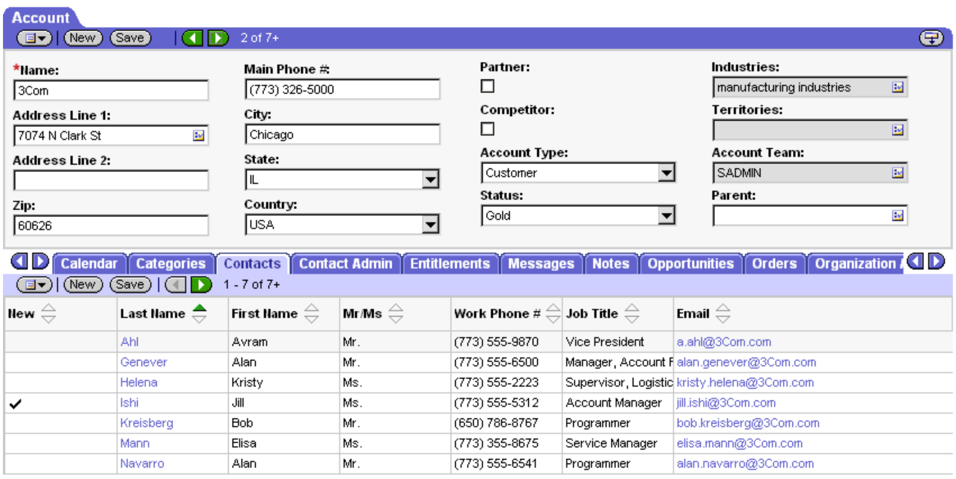
Extension Table Details with Implied Join

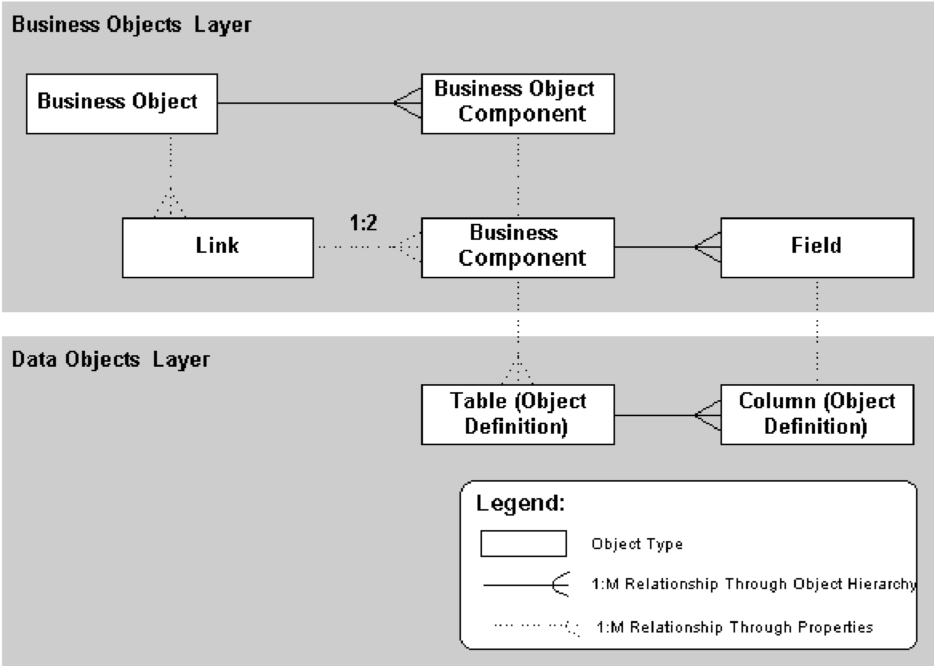
5. Explain Intersection Table with neat architecture.

An *intersection table* implements a many-to-many relationship between two business components.

A *many-to-many relationship* is one in which there is a one-to-many relationship from either direction. For example, there is a many-to-many relationship between Opportunities and Contacts. One Opportunity can be associated with many Contact people, and one Contact person can be associated with many Opportunities.

The below figure shows the Account Detail - Contacts View, in which one account is displayed with multiple detail contacts.





Intersection Table Architecture

The following are descriptions of the object definitions:

Business object. The business object references the link (indirectly through the business object's child business object component) that uses the intersection table. It also contains the two business components included in the link.

Business object components. Business Object Component object definitions are used to include business components in the business object. Business Object Component is a child object type of Business Object. The detail business object component references both the detail business component, by means of the Business Component property, and the link, by means of the Link property. The master business object component only references its corresponding business component.

Link. The link object definition establishes a one-to-many relationship between the two business components in a particular direction. That is, the property settings in the link specify that one business component is the master and the other is the detail in the master-detail relationship.

Master and detail business components. The two business components are specified in the link. They provide data to the user interface object definitions that display the master-detail relationship. The base table of each business component contains the ROW_ID column referenced by the Inter Child Column (detail) and Inter Parent Column (master) properties of the Link object type.

Intersection table. The intersection table holds the associations between rows in the base tables of the master and detail business components. Each row in the intersection table represents one association between the two business components. Two columns in the intersection table serve as foreign keys to the base tables of the two business components.

These columns are identified in the Inter Parent Column and Inter Child Column properties of the link.

Inter Parent column. This column in the intersection table holds the pointer to the associated row in the master business component's base table. It is identified in the Inter Parent Column property of the Link object.

Inter Child column. This column in the intersection table holds the pointer to the associated row in the detail business component's base table. It is identified in the Inter Child Column property of the Link object.

ROW_ID columns. The base table of each business component has a unique identifier column for the rows in that table. This is the ROW_ID column

Column Objects

A Column object definition is the direct representation of a database column in a DBMS. The name, data type, length, primary key and foreign key status, alias, and other properties of the database column are recorded as properties in the corresponding column object definition.

A column has one of several styles based on the value in the Type property. These styles include Data, Extension, IFMGR, System, and others.

Column Object Type

The column object corresponds to one database column in the database table that is represented by the parent table object definition. Each database column in the database table needs to have a corresponding column object definition. The important properties of the Column object type are as follows:

Name. Provides the name of the database column in the database table.

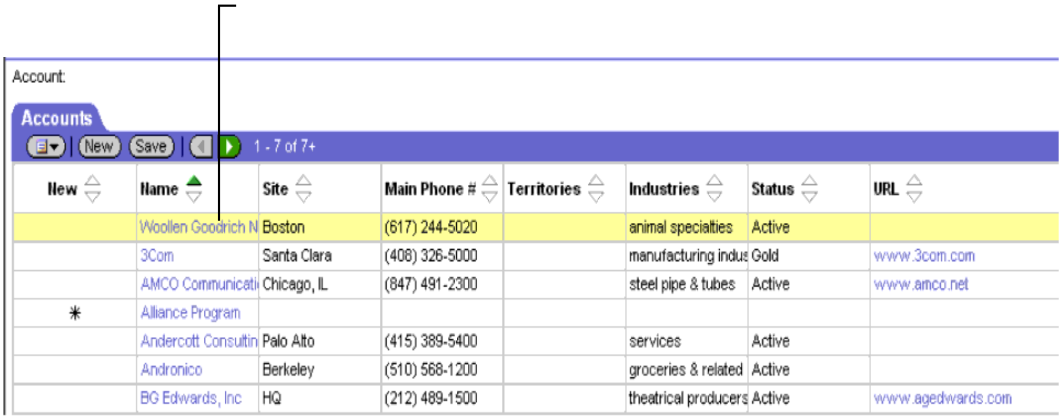
Default. Provides a default value when new rows of this table are added.

Physical Type (Physical Type Name). Identifies the data type of the column in the database. The following data types are supported:

- **Character.** Used for fixed-length text. Also used for Boolean columns, which are character columns with a length of 1.
- **Long.** Long text. You can store approximately 16K worth of data in long columns. By default, you cannot have char greater than 1. If you want, you need to set the preference under Options/Database.
- **Varchar.** Variable-length text. Used for memo-type fields and to store row-ID and foreign key values.
- **Number.** Any numeric data. Typical numeric columns in Siebel applications are 22,7 for general-purpose numbers, and 10,0 for integers.

6. Describe Business Components in detail.

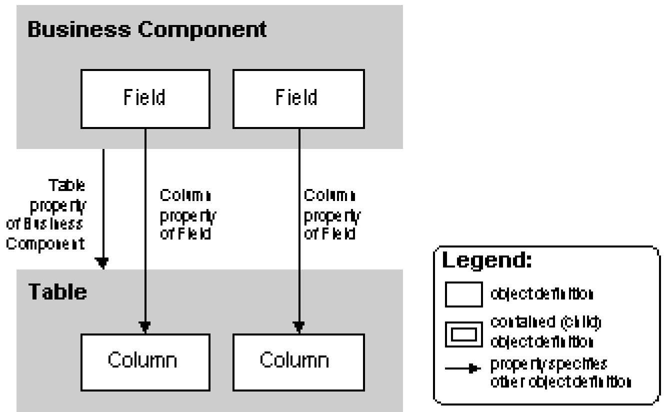
A *business component* is a logical entity that associates columns from one or more tables into a single structure. Business components provide a layer of wrapping over tables, so that applets reference business components rather than the underlying tables. This creates convenience (all associated columns are together in one bundle), developer-friendly naming, and the separation of the developer role from the database administrator role. A business component can also have a default sort or search specification, providing records to applets in a predetermined sort order and according to a selection criterion. When instantiated in a Siebel application, a business component is comparable to a recordset. Its definition in Siebel Tools provides the foundation for controlling how data is selected from, inserted, and updated within the tables it references.



New	Name	Site	Main Phone #	Territories	Industries	Status	URL
	Woollen Goodrich N	Boston	(617) 244-5020		animal specialties	Active	
	3Com	Santa Clara	(408) 326-5000		manufacturing indus	Gold	www.3com.com
	AMCO Communicati	Chicago, IL	(847) 491-2300		steel pipe & tubes	Active	www.amco.net
*	Alliance Program						
	Andercott Consultin	Palo Alto	(415) 389-5400		services	Active	
	Andronico	Berkeley	(510) 568-1200		groceries & related	Active	
	BG Edwards, Inc	HQ	(212) 489-1500		theatrical producers	Active	www.agedwards.com

Business Component Records in a List Applet

A business component contains fields. Each column whose data is included in a business component is represented with a field. The Field object type is a child of the Business Component object type. The Column object type is a child of the Table object type. The set of field child object definitions of a business component maps a corresponding set of columns into the business component.



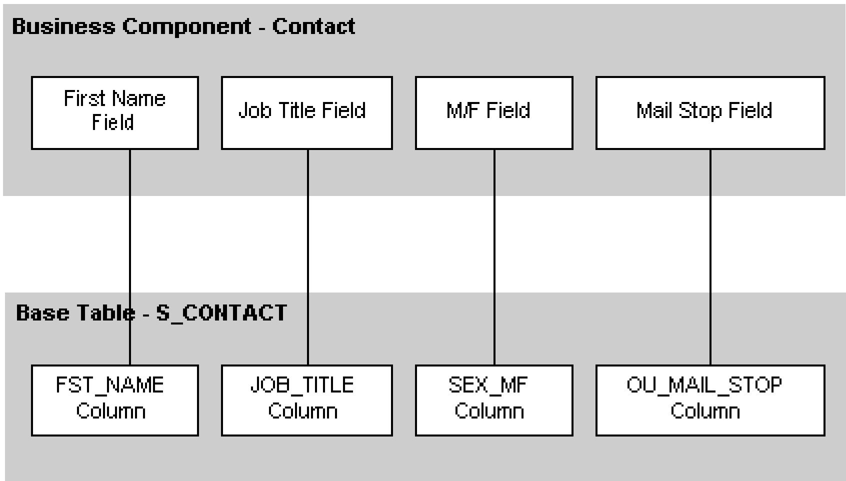
Business Component Field and Column Relationships

In addition to connecting a column to the business component, the Field object type also allows you to specify a meaningful name that refers to that column. Columns in tables are often cryptically named to match the names in the DBMS, whereas fields can have more meaningful and longer names than the columns they represent (75 characters long as opposed to, typically, 30 characters).

Base Tables of Business Components

A *base table* of a business component is assigned to the business component to provide the most important columns for use as fields in the business component. Fields built on the base table can be edited, whereas fields built on joined tables can only be displayed.

The base table is assigned to the business component with the Table property in the Business Component's object definition. The below an example of some fields in the Contact business component that map corresponding columns from the business component's base table, S_CONTACT.



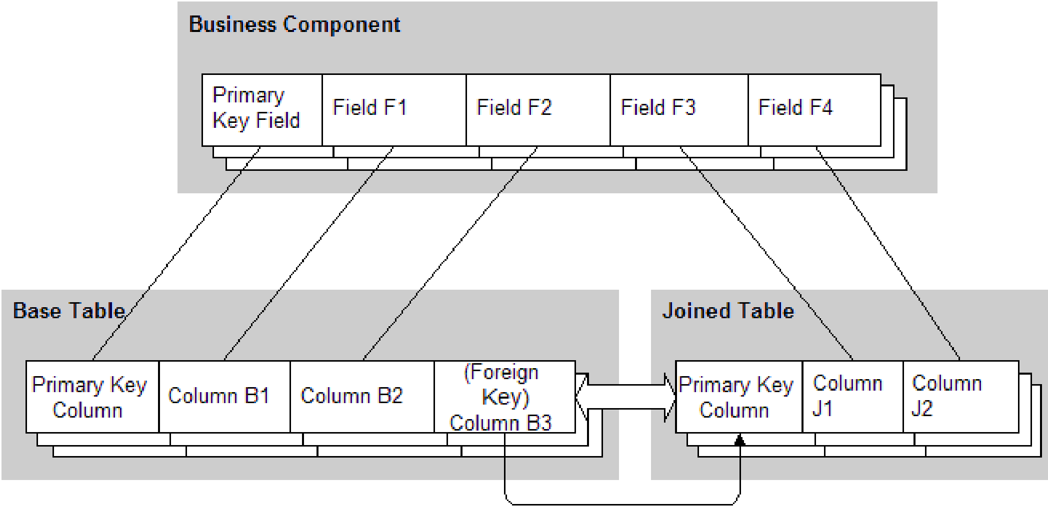
Examples of Fields Representing Columns

Every business component has a base table assigned to it. It is not essential that the business component include all of the columns in the base table, although typically it will include most of them. In particular, system columns in the base table such as ROW_ID, CREATED_BY and LAST_UPD_BY are automatically represented in the business component through implied fields. System columns do not require field object definitions in the business component.

Joined Tables

A *joined table* provides rows on a one-to-one basis to the business component as a result of a foreign key relationship between the joined table and the business component's base table. That is, for every record in the business component (which corresponds to a row in the base table) there can be a corresponding row in the joined table. However, not every record in the base table will have a record in the joined table.

The data obtained by a business component through a join (other than to an extension table) is read-only in that business component.



Fields from the Base Table and a Joined Table

Extension Tables

Extension tables are a special kind of joined table. Like other joined tables, extension tables provide rows on a one-to-one basis in parallel with base table rows. Extension tables are identified by the _X suffix in the table name, such as S_ORG_EXT_X, which extends S_ORG_EXT.

Extension tables are provided specifically to allow columns to be virtually added to a base table rather than physically added. This provides the means to expand base tables without violating DBMS or database design restrictions, and without the need to perform complicated database restructuring operations. Extension table data, unlike the data in other joined tables, can be updated in the business component.

Intersection Business Components

An *intersection business component* is a business component based on an intersection table. It provides the means to display all of the combinations of data in a many-to-many relationship, instead of only one or the other one-to-many relationship of which it is composed.

Intersection tables implement many-to-many relationships. Some (such as S_OPTY_CON and S_ACCNT_POSTN) also provide intersection data through a join to one or the other master business component that uses the intersection table. Intersection data is data that resides in columns other than the two required foreign key columns in the intersection table, and is specific to the intersection of the two master business components.

7. Describe in detail about Virtual Business Components:

Business components based on external data are called *virtual business components*. Virtual business components are used when the business component has to obtain data from a location other than a database table in the Siebel database, but the information has to be presented in the standard Siebel user interface (applets and views). This is typically real-time information from another database, such as from the Report Encyclopedia in Actuate, or from an SAP table, although anything that can supply data in response to a SQL query is a candidate.

Virtual business components allow you to:

- Represent external data (for example, data in an SAP R/3 database) as a virtual business component within a Siebel application—the business component configuration specifies the DLL to use to access the data
- Use business services to transfer data

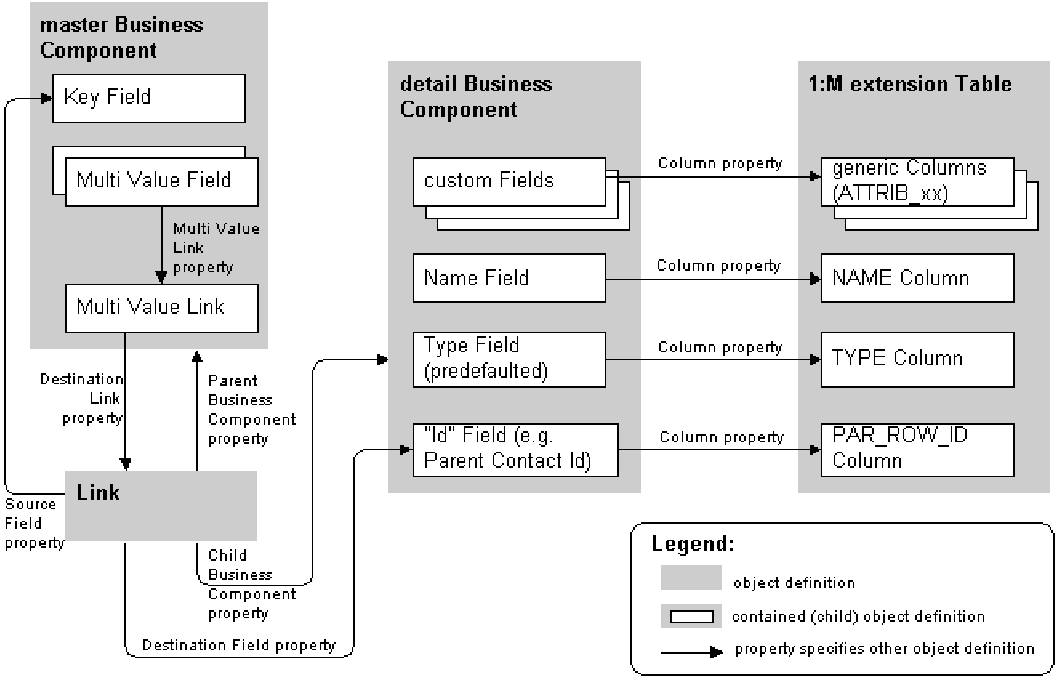
Virtual business components support properties such as:

- Single-value fields
- Field-level validation
- Standard business component event model (for example, PreNewRecord, PreDelete, and so on)
- Insert, delete, query, and update operations

Additional information about virtual business components:

- Applets can be based on virtual business components.
- Virtual business components can be accessed through object interfaces.
- All business component events are available for scripting.
- Virtual business components cannot be docked.
- Virtual business components can be used as stand-alone or children business components in a business object.
- Virtual business components support dynamic applet toggles. For more information about applet toggles.

8. Explain Master-Detail Business Components in detail.



One-to-Many Extension Table Details

Master Business Components

The master business components will hold the new multi-value fields. Master business component contain the following important object definitions:

- **Key field.** This is the key field in the master business component; it is used to reference individual records. Typically, it is named Id. The Source Field property of the Link object definition points to this field. The property value may be blank because, by default, a blank Source Field value refers to the Id field.

Multi-value field. The multi-value field provides access to a corresponding field value in the current record of the detail business component. The control or list column that displays the multi-value field normally will be able to invoke a multi-value group applet for display and maintenance of the detail records.

- **Multi-value link.** The multi-value link provides access to the set of records in the detail business component. One multi-value link is created for each multi-value group that is created using the one-to-many extension table.

Link

The link object definition creates the one-to-many relationship between the master and detail business components. There are no special link configuration issues related to one-to-many extension tables.

Detail Business Component

The detail business component represents the one-to-many extension table used by the multi-value link and multi-value group applet. Multiple custom business components can be created using the same one-to-many extension table. Each custom business component presents a different type of data for use in a different multi-value group.

The detail business component contains custom fields that represent generic (ATTRIBxx) extension columns, and hold whatever data is required for the application. For example, an Area Of Expertise business component might have a Subject Area field, a Years Of Experience field, and a Licensed field. Each field is a mapping of a different generic extension column.

The following three fields are part of the User Key (U1), which uniquely identifies a row for EIM:

Name field. The name field represents the NAME column from the extension table. It provides the means for the user to enter an identifying value in each record. For example, in a Hobbies business component, the name field might be called Hobby. The user would enter the name of a hobby into each record in this field.

Type field. The type field is usually named Type, and represents the TYPE column. It contains the same value for all records in one multi-value group, and distinguishes the records of that multi-value group from others. It should be set in the Predefault property to some identifying word or phrase, such as HOBBY, EXPERTISE or PRIOR JOB, and should not be exposed in the user interface.

Parent ID field. The parent ID field represents the PAR_ROW_ID column. Generally it is named Parent Contact Id, Parent Account Id, or something similar. It identifies the row ID of the base table row corresponding to the parent record in the master business component. The parent ID field is specified in the Destination Field property of the Link object.

The detail business component contains one important property for use with a one- to-many extension table:

- **Search Specification.** The Search Specification property should be set to restrict the records retrieved to only those with a specific value in the Type field. This is the same value that is specified in the Pre Default Value property for that field.

9. Explain Business Objects in Detail.

Business Objects A business object represents a major functional area of the enterprise—every major entity has a business object. Examples of business objects are Opportunity, Account, and Contact. A business object is a collection of related business components; for example (as shown in Figure 6), the Opportunity business object consists of Opportunities plus related Contacts, Activities, Products, and Issues. Each business object has one business component that serves as the master or driving business component. This master business component provides focus for the business object, and they both have the same name (the name is Opportunity).

Business Object Types:

Business Component. A business component is a logical entity that associates columns from one or more tables into a single structure. Business components provide a layer of wrapping over tables, causing applets to reference business components rather than the underlying tables. This design creates convenience (all associated columns together in one bundle), developer-friendly naming, and the isolation of the developer role from the database administrator role. Multiple users can instantiate copies of the same business component. Data changes made by any one user are reflected in all instances of the business component.

Field. A field object definition associates a column to a business component. This is how columns from tables are assigned to a business component and provided with meaningful names that the customer developer can easily change. Alternately, a field's values can be calculated from the values in other fields in the business component. Fields supply data to controls and list columns in the Web interface.

Business Object. A business object implements a business model (logical database diagram), tying together a set of interrelated business components using links. The links provide the one-to-many relationships that govern how the business components interrelate in the context of this business object.

Business Object Component. A business object component object definition is used to include a business component and, generally, a link in the business object. The link specifies how the business component is related to another business component in the context of the same business object. **Link.** A link implements a one-to-many relationship between business components. The Link object type makes master-detail views possible. A masterdetail view displays one record of the master business component with many detail business component records corresponding to the master. A pair of links also may be used to implement a many-to-many relationship.

Multi-Value Link. A multi-value link is used in the implementation of a multi-value group. A multi-value group is a user-maintainable list of detail records associated with a master record. The user invokes the list of detail records from the master record when it is displayed in a list or form applet. For example, in an applet displaying the Account business component, the user can click the Select button to the right of the Address text box to see a pop-up window displaying multiple Address records associated with the currently displayed account.

Join. A join object definition creates a relationship between a business component and a table that is not the business component's base table. The join allows the business component to build fields using columns from the non-base (joined) table. The join uses a foreign key in the business component to obtain rows on a one-to-one basis from the joined table, even though the two do not necessarily have a one-to-one relationship.

Join Specification. A join specification is a child object type of join that provides details about how the join is implemented within the business component.

User Property. A user property is a temporary storage field in a business component that is not tied to the database. When a user sets a value in a business component user property, it is not visible to other users of the same business component. User properties are not saved across sessions.

Business Service. A business service is a reusable module containing a set of methods. It provides the ability to call its C++ or script methods from customer-defined scripts and object interface logic, through the invoke-method mechanism.

10. Explain Applet and types of applet?

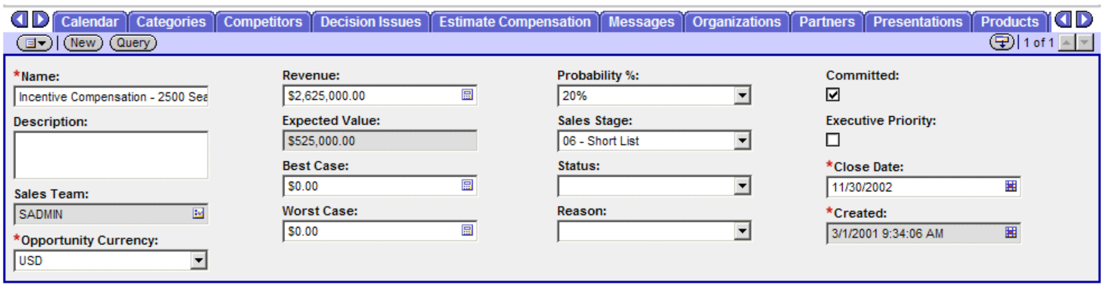
An *applet* is a data entry form, composed of controls, that occupies some portion of the Siebe application user interface. An applet can be configured to allow data entry, provide a table of data rows, or display business graphics, or a navigation tree. It provides viewing, entry modification, and navigation capabilities for data in one business component.

An applet is always associated with a business component. Although the same business component can be associated with multiple applets, an applet is associated with only one business component.

Applets are associated with one or more Siebel Web templates. Web templates are files that contain HTML and proprietary Siebel tags that define the layout and format of the applet in the user interface.

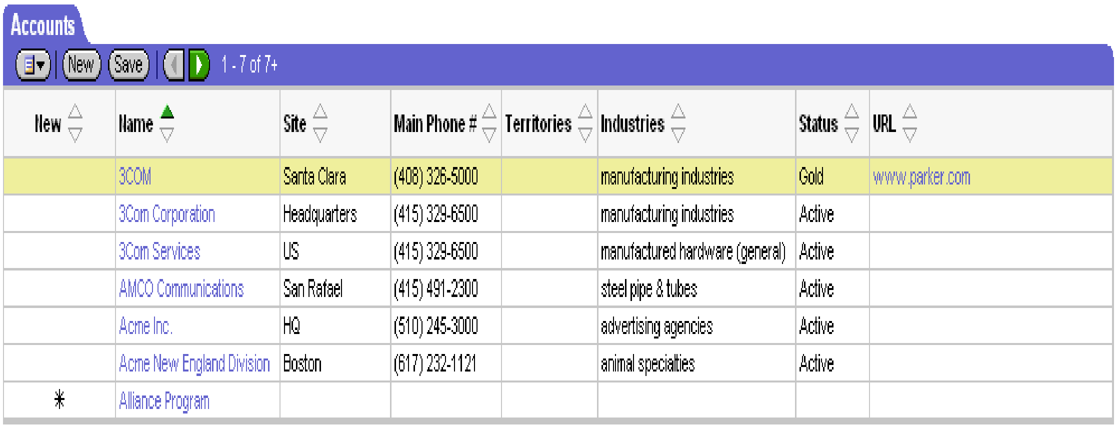
Types of Applets:

Form applet. A *form applet* displays data in a data entry form. Fields in the business component appear on the form applet as text boxes, check boxes, and other standard controls. A form applet appears in below figure.



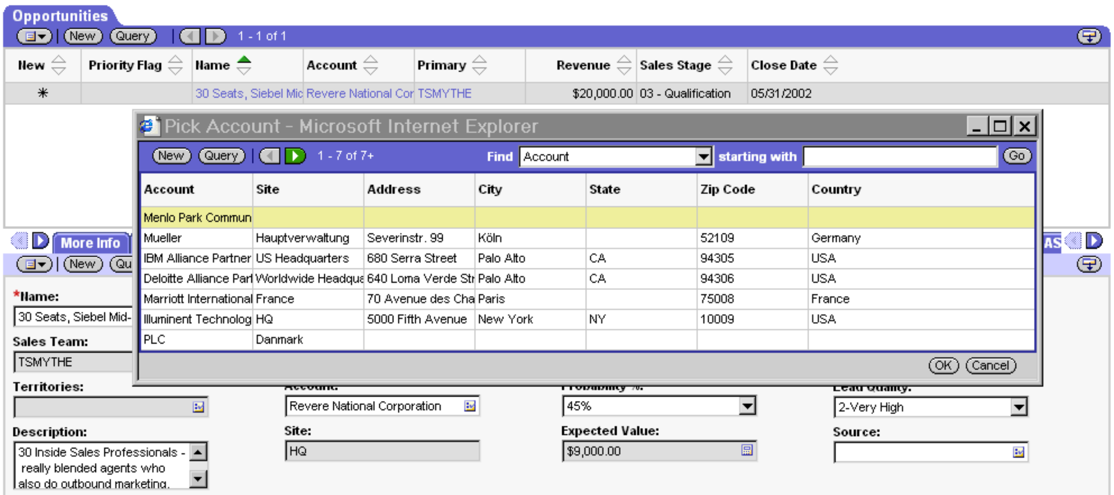
Form Applet

List applet. A *list applet* allows the simultaneous display of data from multiple records. A list applet displays data in a list table format, much like a spreadsheet or word processor table. Rows in the list applet correspond to records in the business component; list columns in the list applet correspond to fields in the business component. In addition to textual data, lists also support images in JPEG and GIF formats and edit controls such as check boxes, drop-down lists, MVGs, and text fields. A list applet appears in below figure.



List Applet

Pick applet. A *pick applet* is a dialog box window that appears when a selection is to be made in a control or list column that has the check mark icon to its right. The pick applet provides a list or table of selection values, from which the user selects a value or record. A pick applet displays data that has a M:1 relationship to the data in the parent applet. A pick applet appears in below figure.

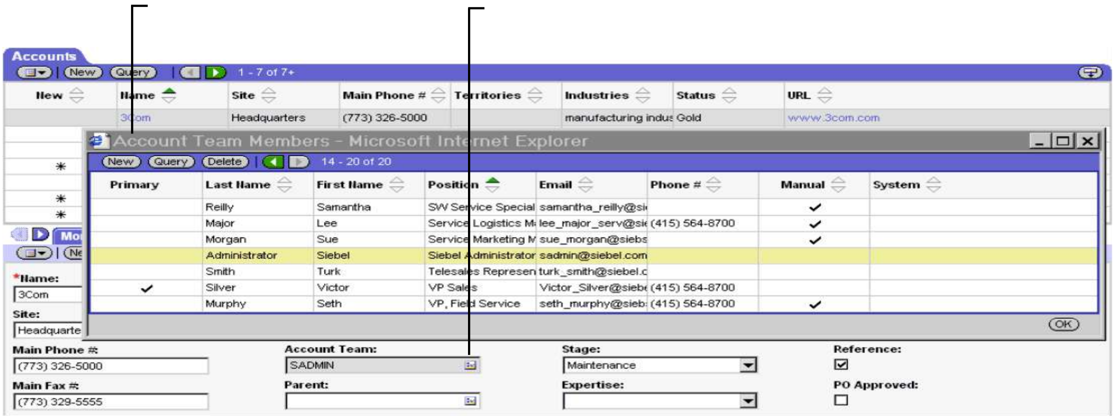


Pick Applet

Multi-value group applet. A *multi-value group applet* is used for entry, maintenance, and viewing of a list of detail records associated with one or more fields in the currently displayed master record. MVGs allow the user to associate multiple records to a single field in a form or list and provide a way of representing one-to-many relationships within a single record of data.

There are two ways in which users can add data to the MVG:

- Inputting data
- Selecting from existing database records



Multi-Value Group Applet

Chart applet. A *chart applet* graphically displays data from a business component in a bar chart, line graph, pie chart, scatter diagram or other format. It summarizes and illustrates data relationships. Charts are usually accessed through a tab in the third-level navigation level and contain a number of sub- category views (multiple charts of data). These are displayed in an overview of miniature chart graphics (.gif images) with title text. Both the mini-graphics and the title text for the chart are hyperlinked to the detailed version of the chart. A chart applet appears below.

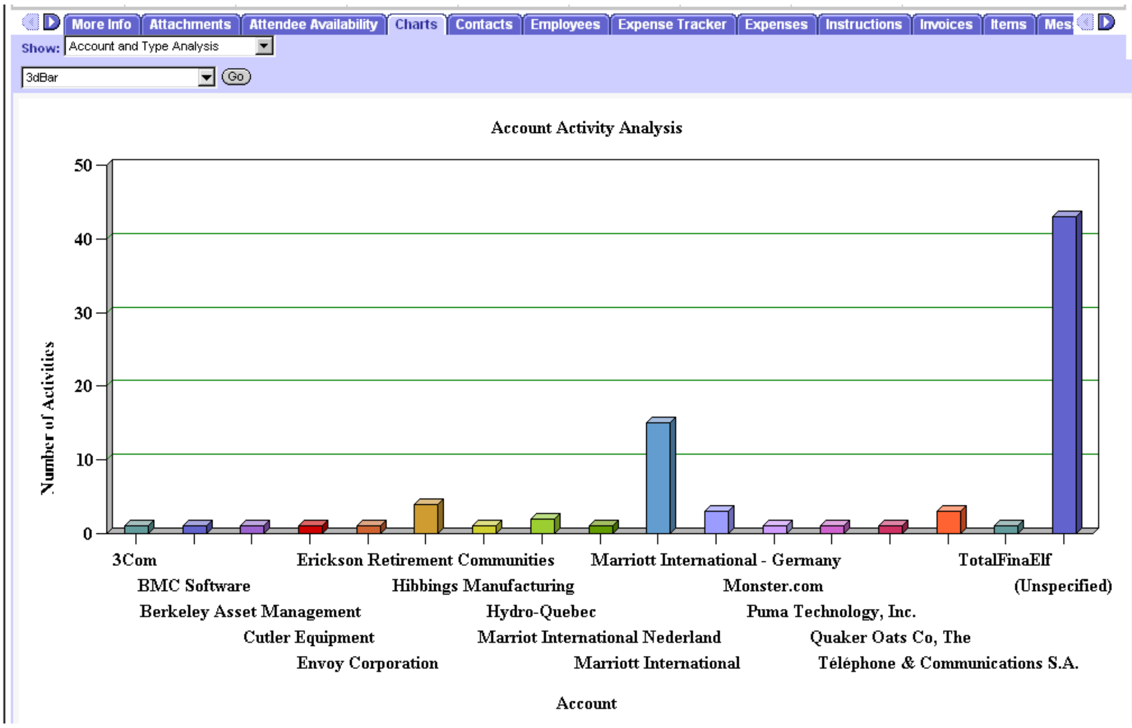
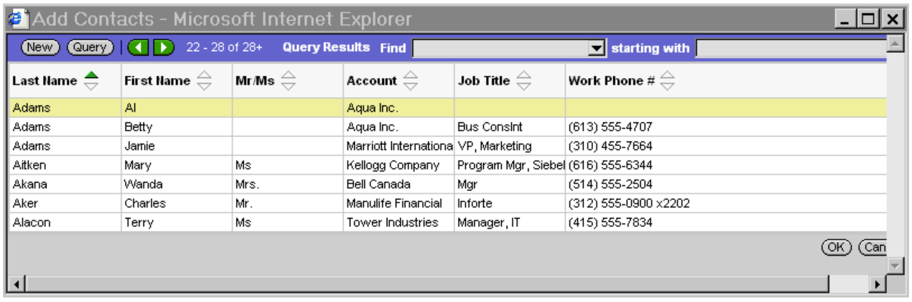


Chart Applet

Association applet. An *association applet* provides the user with the ability to associate records of two business components that have a many-to-many relationship. It is invoked from the check mark icon in a multi-value group applet. An association applet appear below.

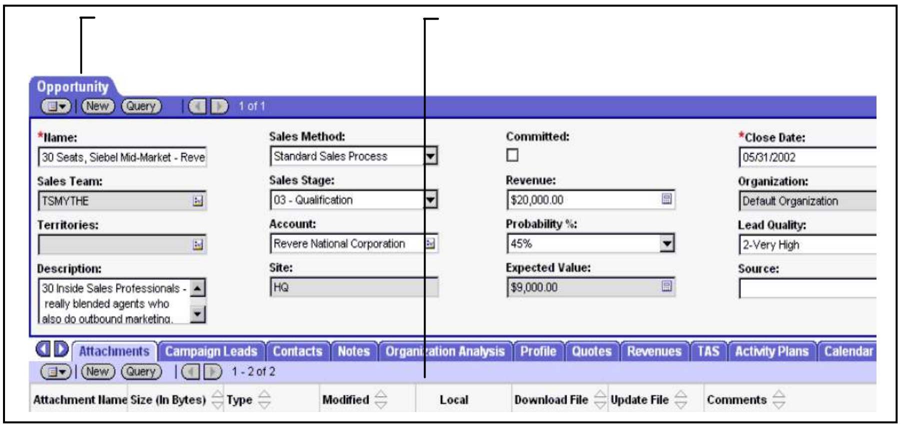


Association Applet

Explorer or Tree applet. A *tree applet* is used to create an explorer view that allows the user to navigate hierarchically through a structured list of object instances. A tree applet appears below.

Explorer or Tree Applet

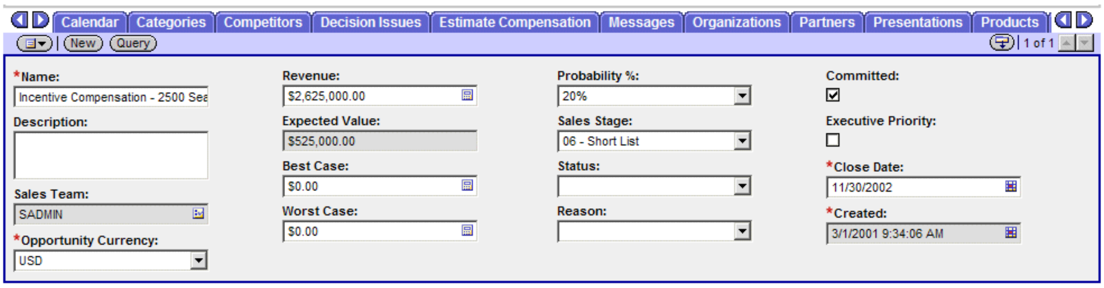
File attachment applet. *File attachment applets* provide access to external documents, such as spreadsheets, word processing documents, and presentations, that have been imported in compressed format into records in a Siebel application. A file attachment applet appears below.



File Attachment Applet

Form Applets

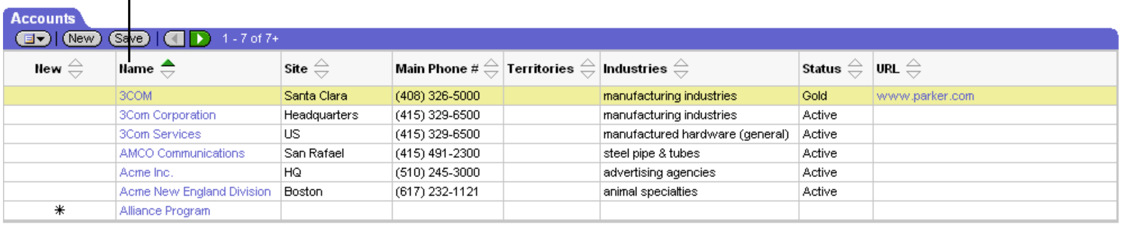
A *form applet* presents business component information in a data entry form layout. An example of a form applet in Siebel Call Center appears in below figure.



Form Applet in Siebel Call Center

List Applets

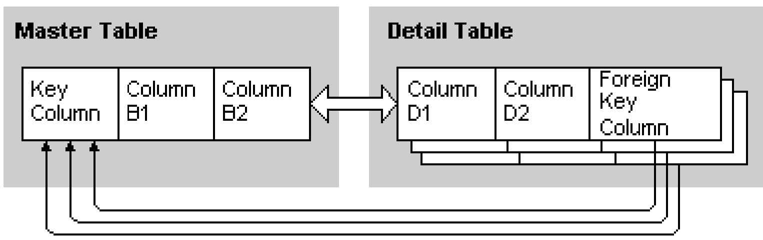
A *list applet* allows simultaneous display of data from multiple records and presents business component information in a list table format with multicolumn layout with each record of data represented in a row. In addition to textual data, lists also support images in JPEG and GIF formats and edits control such as check boxes, drop-down lists, noneditable MVGs, and text fields.



List Applet in Siebel Sales

11. Explain in detail about Joins?

A Join object definition creates a relationship between a business component and a table other than its base table. The join allows the business component to use columns from that table. The join uses a foreign key in the business component to obtain rows on a one-to-one basis from the joined table, even though the two do not necessarily have a one-to-one relationship.



Master-Detail Relationship in a Join

Contacts

Jenny Frank	Acct 1
Fred Carlson	Acct 2
Sam Hudson	Acct 3
Sally Van Doren	Acct 1
Mike Fernside	Acct 2
Hans Baker	Acct 3

Accounts

Acct 1	Smith's Dry Goods	Cambridge	MA
Acct 2	Jones Cleaners	Berkeley	CA
Acct 3	The Sports Zone	Minneapolis	MN

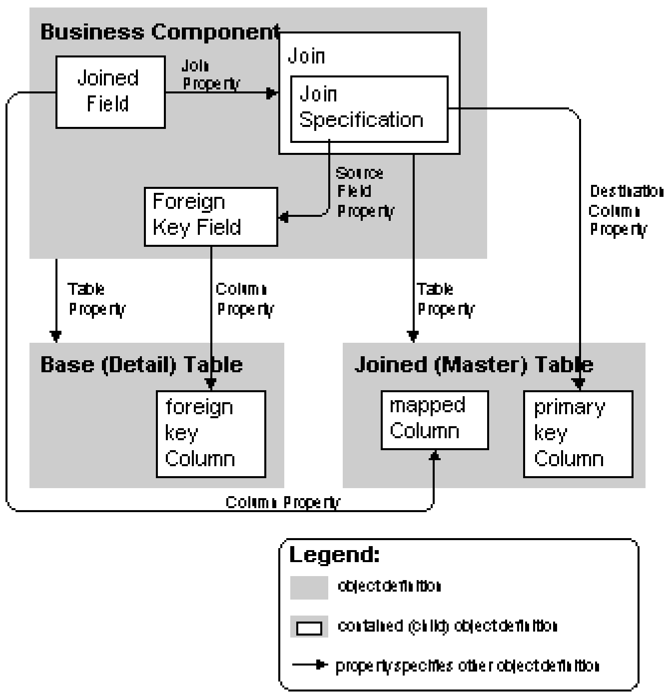
join

Jenny Frank	Acct 1	Smith's Dry Goods	Cambridge	MA
Fred Carlson	Acct 2	Jones Cleaners	Berkeley	CA
Sam Hudson	Acct 3	The Sports Zone	Minneapolis	MN
Sally Van Doren	Acct 1	Smith's Dry Goods	Cambridge	MA
Mike Fernside	Acct 2	Jones Cleaners	Berkeley	CA
Hans Baker	Acct 3	The Sports Zone	Minneapolis	MN

Set of Rows Resulting from a Join

A join is always one-to-one and it is always between a business component and a table. Once a join is created, you can create additional fields in the business component based on columns in the joined table. In the diagram, the account name, city and state are fields that can be added to the Contact business component because of this join.

How a Join is constructed:



Join Relationships

The roles of the object definitions in the diagram are summarized as follows:

Business Component object type. The business component is the parent object definition of the join. Because of the join, fields in the business component (called joined fields) can represent columns from the joined table.

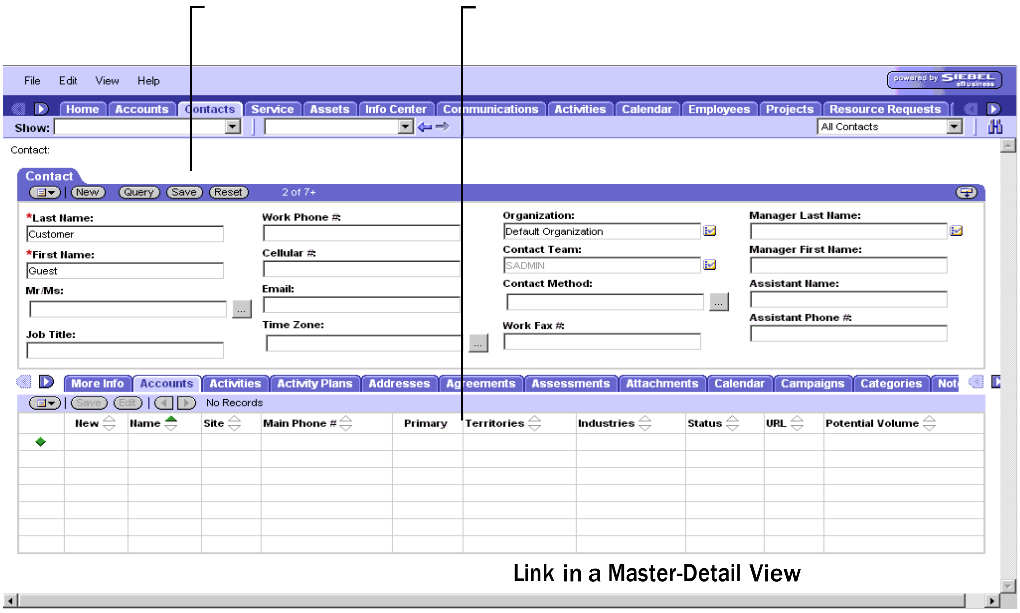
Joined field. A joined field in the business component represents a column from a table other than the business component's base table. Therefore, a joined field must obtain its values through a join. A joined field has the name of the join in its Join property. Together the Join property and Column property identify the column and how to access it. When creating a joined field in a business component, you can change the Type property from the default DTYPE_TEXT to a more appropriate type. For example, if you are joining a table column that contains phone numbers, you can change the Type field to DTYPE_PHONE.

Join object type. Join is a child object type of the Business Component object type. The Join object definition uniquely identifies a join relationship for the parent business component and provides the name of the destination (joined) table. The join object definition identifies the joined table in the Table property. The name of the base table is already known to the business component. Typically, a join object definition is given the same name as the joined table.

Join Specification object type. The join specification object definition is a child of the join object definition. It identifies the foreign key field in the business component and the primary key column in the joined table.

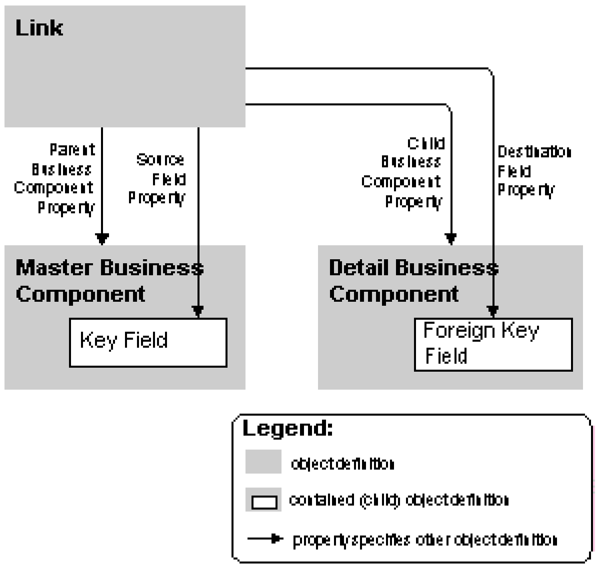
12. Describe Links in detail.

A link implements a one-to-many (or master-detail) relationship between business components based on their base tables. The Link object type makes master-detail views possible, in which one record of the master business component displays with many detail business component records that correspond to the master.



Link in a Master-Detail View

How a Link is constructed:



Link. The Link object definition specifies a master-detail relationship between two business components. It identifies the master and detail business components, the key field in the master business component, and the foreign key field in the detail business component.

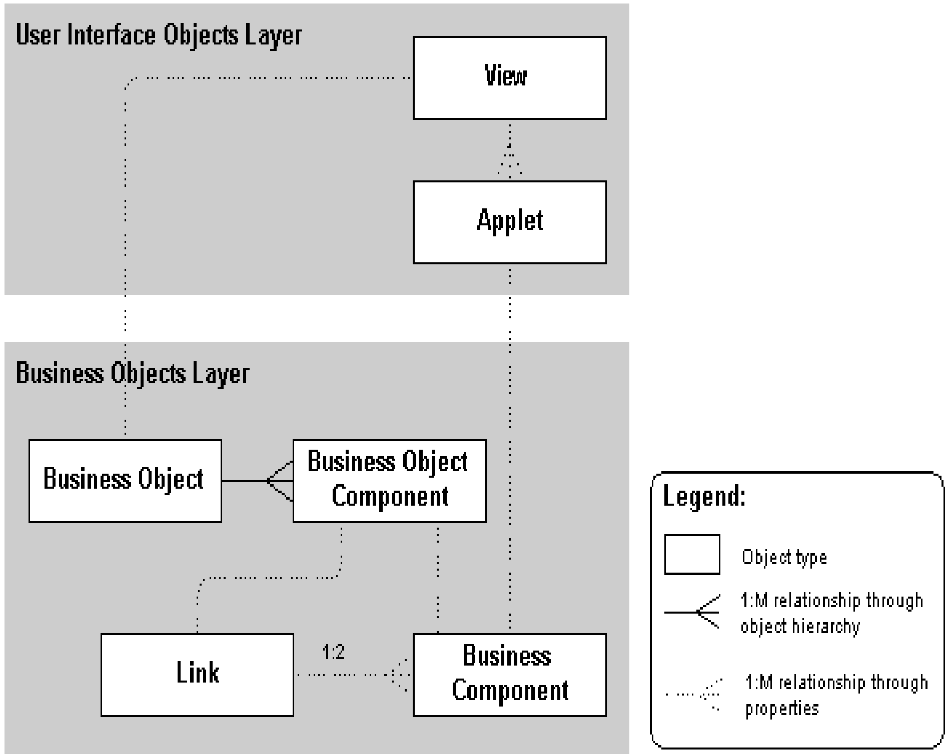
Master business component. The master business component is the "one" in the one-to-many relationship. The name of this object definition is specified in the Parent Business Component property in the Link object definition.

Detail business component. The detail business component is the "many" in the one-to-many relationship. The name of this object definition is specified in the Child Business Component property in the Link object definition.

Source (primary key) field. The source field, also known as the primary key field, is a field in the master business component that uniquely identifies records in the business component. It represents the ROW_ID column from the business component's base table. The name of this field is specified in the Source Field property in the Link object definition. Source field typically, but not necessarily, represents the row id column from the business component's base table.

Destination (foreign key) field. The destination field, also known as the foreign key field, is a field in the detail business component that points back to the master record in the business component. Account Id and Opportunity Id are typical foreign key fields.

Link Architecture



In a master-detail view, a Link object definition is incorporated into a business object (by means of a Business Object Component object definition) to establish the master-detail relationship. This relationship applies to any use of the two business components together within the context of the business object. Each view specifies the business object it uses in its Business Object property. This forces the view to operate as a master-detail view, as specified in the link, without any additional configuration of the view.

A *multi-value link* implements a special use of the Link object type, which is the maintenance by the user of a list of records attached to a control or list column in an applet. The group of attached detail records is called a *multi-value group*.

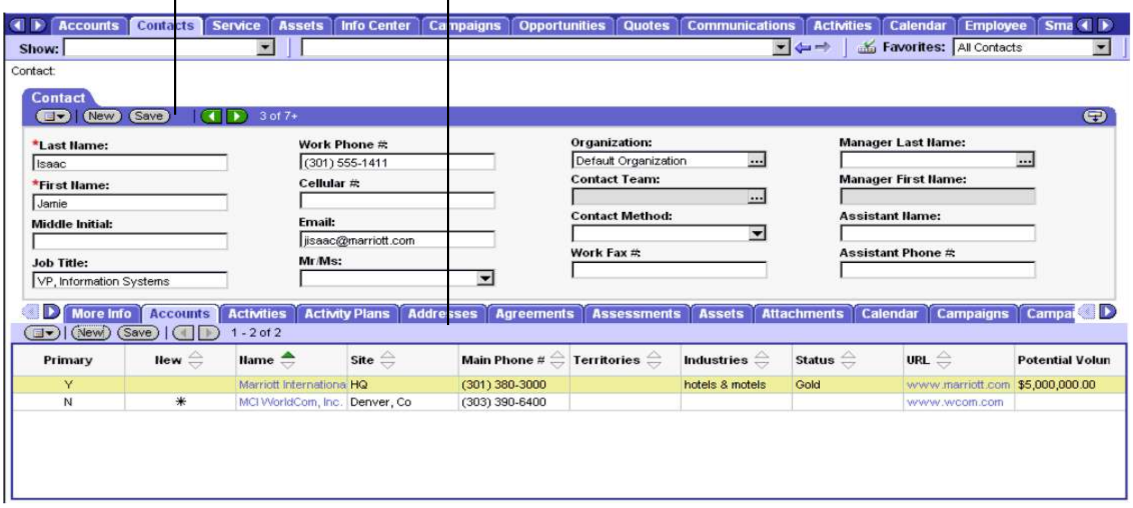
13. Describe Views in detail.

Views are typically of the following styles:

List-form view. In a list-form view, a list applet and a form applet display data from the same business component. The list applet appears above the form applet. The form applet presents the same information as the currently selected record in the list applet, with a different arrangement that may include more fields.

Master-detail view:

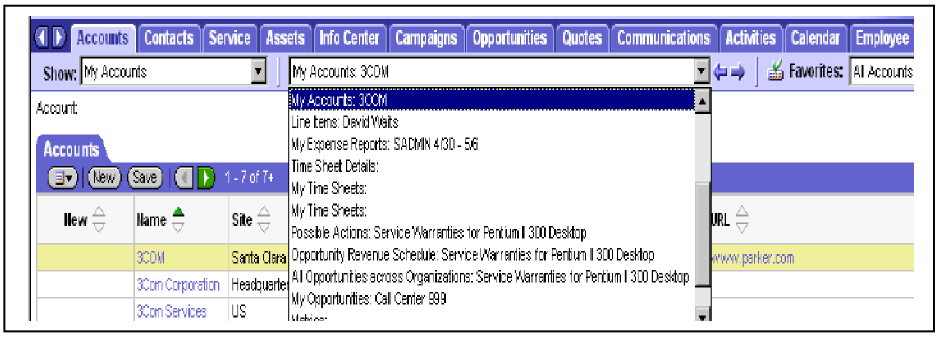
In a *master-detail view,* typically a form applet and a list applet display data from two business components related by a link. The form applet appears above the list applet. The form applet displays one record from the master business component in the master-detail relationship. The list applet displays all of the records from the detail business component that have as their master record the record currently displayed in the form applet.



Master-Detail View in a Siebel Application

Thread bar

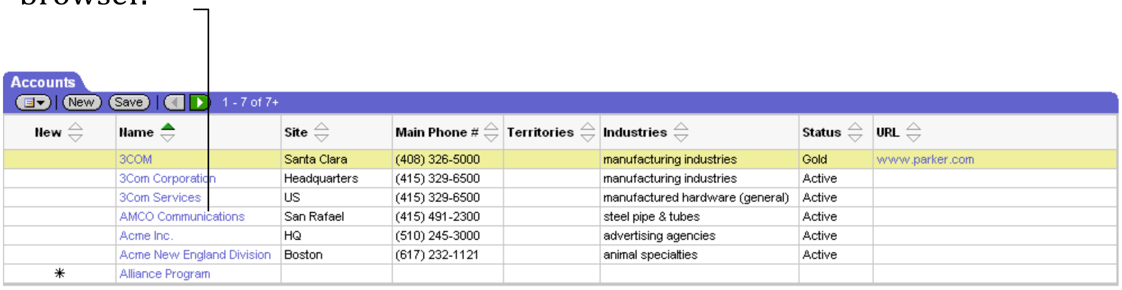
The *thread bar* is a navigational tool for the user. It provides the means to navigate from view to view among the views previously visited in the current screen.



Thread Bar in a Siebel Application

Drilldown Behavior in a View

The *Drilldown Object* object type is a child of Applet, used primarily in list applets. It allows the user to drill down from a cell in a list applet (or using a pop-up menu in either a form or list applet) to a particular view. Drilldown controls or list columns in a list applet in Siebel applications consist of colored, underlined text, much like a hypertext link in a Web browser.

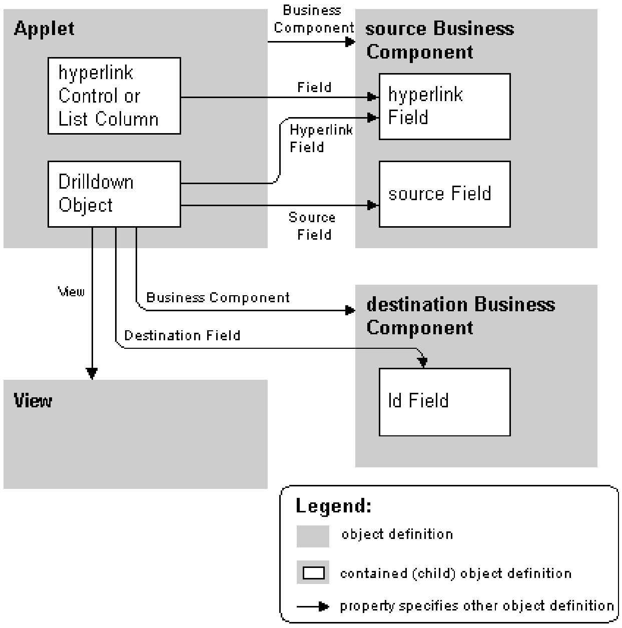


Drilldown List Columns in a List-Form View

Types of Drilldown Behavior:

1. Static Drilldown Behavior:

In the example, underlined account name appears in the list column labeled Name. If the user clicks the account in the Name list column, a master-detail view appears, with the selected account in a form applet above an applet displaying the corresponding list of contacts.

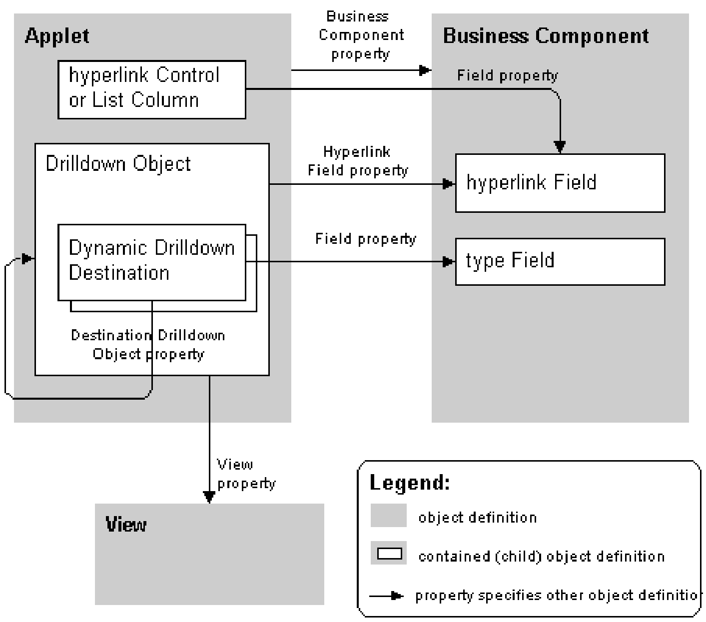


Static Drilldown Configuration

Dynamic drilldown enables hyperlink navigation to multiple views from the same hyperlink field, depending on the value of a field in the applet's current record.

This is useful in the situation where special processing is desired for various types of contacts, opportunities, accounts, and so on. The business component may have a field that indicates a classification, such as the Lead Quality for an opportunity or the primary Industry for an account. The drilldown behavior can be to check this field in the current record, and navigate to different views for different values found there.

Dynamic drilldown behavior for a hyperlink field (and the corresponding list column or control) is configured with one or more Dynamic Drilldown Destination child object definitions of the Drilldown Object.

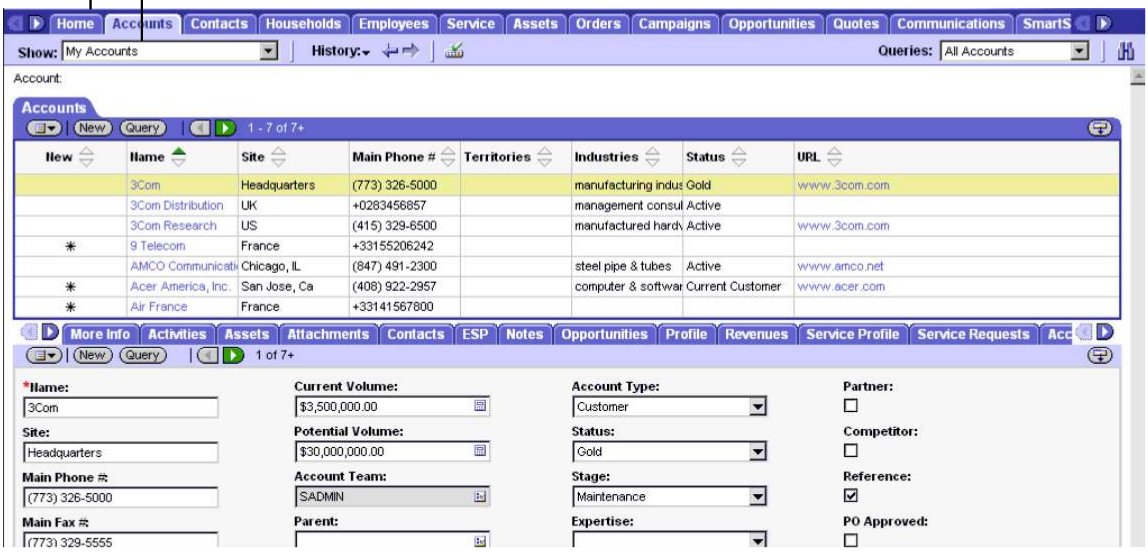


Dynamic Drilldown Configuration

14. Describe Screens in detail.

A *screen* is a logical collection of views. It is not a visual construct in itself; rather, it associates views so that other visual constructs, such as the Site Map and tab bar, can reflect the list of views contained in the currently active screen.

A screen does not have a direct relationship with a business object in the same way that a view does. No property in the Screen object type specifies a business object. However, a screen normally contains only views relating to the same business object; this is good design practice. In this sense, it can be loosely said that a screen corresponds to one business object.



Applications of a Screen:

Siebel applications are primarily a collection of screens that users can invoke from the desktop by double-clicking an icon or by pointing a browser to a server running the application. Each combination of screens that is appropriate to a specific class of users can be provided as an application. Siebel Sales, Siebel Service, and Siebel eMarketing are examples of applications. Custom applications can be configured as well, uniquely combining user interface object definitions to meet particular requirements of the organization.

In addition to collecting a group of screens and their views, an application object definition includes the following:

Find object definitions that configure the Find dialog box.

Scripts written in Siebel VB or Siebel eScript and browser JavaScript that can be implemented as event procedures on startup, prior to closing, and so on. These are implemented through Application Script child object definitions, and created and maintained in the Siebel VB or Siebel eScript Editor.

Custom menu options for Siebel-provided methods. These are implemented through the application method menu item object definitions, and created in the Applet Method Menu Item Wizard.

Configuring Application:

When configuring applications for deployment in High Interactivity mode, consider the following:

- Browser scripting is fully supported in High Interactivity mode.
- For fields to interpret and display custom HTML, such as a URL entered by the user, the field's Type property must be set to URL. If it is not set to URL, the HTML is presented and interpreted as plain text. For example, if a user typed a URL in a field of type TEXT, the URL would not be recognized as a link to a Web page.
- You cannot modify the appearance of the rich text editor.
- You cannot modify the background and text color of list applets.
- You cannot place method-invoking controls, such as the delete function, on every row in a list. Instead place a button that calls the method on the applet itself. The function will act on the selected record.

There are cases when an application's configuration file is set to run in High Interactivity mode and all the applets in a view are configured to support High Interactivity, but the view appears in Standard Interactivity mode. Reasons this might occur are:

- One of the applets is in the Query mode. Because High Interactivity implicitly supports query operations from the user interface, it does not support the explicit use of the Query mode.
- One of the applets is in the New mode and uses a New template that is different from the Edit template used in its default mode. This can be avoided by inactivating New templates associated with the applets used in High Interactivity applications. The framework will then default to using the Edit template itself to create new records.
- One of the list applets has multi-row edits or multi-row select enabled.
- One of the list applets is a hierarchical list applet.
- The view uses a template that shows applets in a catalog-style layout. None of the employee applications should be using this layout.
- A combo box picklist uses Long Lists or has an associated pick applet. For example, if you perform an action from a High Interactivity applet that causes a pick applet to be displayed, the pick applet will not be in High Interactivity mode.

Unit-I – ERP AND ITS RELATED TECHNOLOGIES

QUESTION BANK

Introduction : ERP - Definition – Concept – Fundamentals – Need for ERP - Advantages of ERP – Implementation of ERP – Key issues and Characteristics of ERP Typical architecture components of ERP – ERP system Architecture. **ERP and related technologies**: Business Process RE-engineering – Management Information System– Decision Support System - Executive Support System – On-Line Analytical Processing, Supply Chain Management, Customer Relationship Management.

PART A

1. Define ERP.
2. What is enterprise solutions?
3. Describe about Business Intelligence.
4. List out the Characteristics of ERP.
5. What is the need of ERP?
6. List the advantages of ERP.
7. What are the fundamental components of ERP?
8. List out the elements of ERP.
9. Why companies are forced to have ERP suite?
10. In what way ERP is used in real world?
11. List the implementation methodology phases of accelerated ERP approach.
12. What are the capabilities of effective co-ordination management?
13. Mention the elements required in ERP to achieve flexibility.
14. What is Two Tier architecture?
15. What is Business Process Reengineering (BPR)?
16. Define Management Information System.
17. Depict the neat diagram of Business Process Reengineering cycle.
18. List out the advantages of BPR.
19. Write a note on Decision Support System (DSS).
20. What are the various phases are available in DSS.
21. Define Executive Support System.
22. What is On-Line Analytical Processing (OLAP)?
23. Define Multidimensional OLAP.

22. Define MOVE statement and write the syntax for MOVE statement.
23. What is the purpose of CLEAR statement?
24. Write a sample program for CLEAR statement.
25. Describe WHILE statement and show the sample program using WHILE statement.
26. Show the structure of ABAP program.
27. Mention the different types of ABAP programs.
28. What are the statements are used to declare variable statically in ABAP?
29. Write the syntax for CONSTANT statement.
30. List out the different types of FORMAT statements available in ABAP and write its syntax.

PART – B

1. Define SAP and brief about the history of SAP.
2. Explain in detail about architecture of SAP R/3.
3. Detail about Customer Relationship Management, Business Warehouse and Advanced Planner and Optimizer.
4. Brief about ABAP/4 workbench tools.
5. Detail about ABAP syntax and datatypes.
6. Explicate about constants and variables.
7. Sketch and explain in detail about structure of ABAP/4 program
8. List out and explain the ABAP statements.

UNIT III

Oracle Suite : Oracle Apps 11i - Application Framework - File System - Workflow Analysis - SQL / PLSQLfundamentals - Creating Forms - Oracle Reports.Oracle Electronic Data Interchange – functions of EDI – Data File Structure - Oracle Data, Oracle Database - Oracle Database - DW vs OLTP - DW Connectors.

2 Marks

1. What is Oracle Application Architecture?
2. What are the three tiers in an Oracle E-Business Suite installation?
3. What is Web server?
4. What is Oracle Application Framework?
5. What is Discoverer Server?
6. What are operations from Admin server?
7. What is Daily Business Intelligence (DBI)?
8. What is oracle overflow?
9. What is oracle workflow builder?
10. What is the Oracle Workflow Business Event System?
11. What is the Oracle Workflow Event Manager?
12. What are the features in Oracle E-Business Suite Release 11i?
14. What is the use cost-based optimization (CBO)?
15. What are Materialized Views?
16. What is Real Application Clusters?
17. Define Structured Query Language (SQL)?
18. What is Data Definition Language (DDL) Statements?
19. What is Data Manipulation Language (DML) Statements?
20. What is PL/SQL?
21. List advantages of PL/SQL subprograms?
23. Advantages of PL/SQL Packages?
24. What is Cursor?
25. What is Electronic Data Interchange (EDI)?
26. What are the functions performed by EDI?
27. Write about Data File Structure?

28. List some unstructured data?

29. What is LOB?

30. Write about external LOBs?

31. What is Oracle Spatial ?

32. What is Oracle multimedia ?

33. Write advantages of Oracle Text?

34. What is DBMS?

35. What is RDBMS?

36. What is transaction management?

37. What is meant by Data warehousing?

38. List Characteristics of Data warehouse?

39. Write Data warehouse Vs OLTP?

11 Marks

1. Explain PL/SQL?

2. Write about Oracle workflow analysis?

3. Write about Electronic Data Interchange (EDI)?

4. Write about Oracle Data?

5.Write Data Warehousing Vs OLTP?

6.Write Oracle 11i File System

7. Explain Oracle Database?

8. Explain Data File Structure?

9. Explain SQL?

10. Explain Oracle Database 11i Feature?

11. How to generate Report?

12. How to develop forms in Oracle?

13. Explain any two Data warehouse Connectors?

Unit-IV – PEOPLESOFT

QUESTION BANK

PeopleSoft: Basic PeopleSoft Functionality – Opening Multiple Windows - Database structure – Understanding People Soft Data Mover – Records - Pages vs. Forms. **PeopleSoft HRMS:** Introduction to PeopleSoft HRMS database - PeopleSoft products - Functional PeopleSoft - financial management system - PeopleSoft Enterprise HRMS.

PART A (2 MARKS)

1. Define people soft.
2. When PeopleSoft is founded?
3. Draft a note on evolution of PeopleSoft.
4. Which corporation control PeopleSoft?
5. List out the applications of PeopleSoft.
6. Mention the benefits of PeopleSoft.
7. Define Multiple Windows.
8. What is People soft Database structure?
9. Write a note on PeopleSoft Database Layer.
10. Define PeopleSoft Forms.
11. Write a short notes on Data Mover
12. Delineate Records.
13. Define PAGES.
14. What are the types of records?
15. Write a note on Multiple Occurrences of Data.
16. Define Sub records.
17. How a Form is created?
18. List the types of fields in forms.

19. Define PeopleSoft HRMS.
20. Mention the types of PeopleSoft Enterprise HRMS.
21. What is PeopleSoft Enterprise HRMS Integration?
22. Write a note on PeopleSoft Enterprise HRMS Implementation.
23. Draft a note on Service Automation.
24. Define Enterprise Performance Management.
25. What is meant by Supplier Relationship Management?
26. Delineate Supply Chain Management.
27. List out the FMS Modules.
28. Draw the diagram of PeopleSoft workflow process.
29. What is Business Reports?
30. List out the products of PeopleSoft.
31. What is meant by PeopleSoft accelerator?
32. Define Financial Management Systems.

PART – B (11 MARKS)

1. Explain in detail about basic functionality of PeopleSoft.
2. Sketch and explain in detail about PeopleSoft database structure.
3. Describe in detail about Data Mover.
4. Explicate in detail about Records and its types.
5. Discuss in detail about PeopleSoft pages and forms.
6. Elucidate about PeopleSoft HRMS database.
7. Explain in brief about PeopleSoft Products.
8. Detail about Functional management system.
9. Draft a note on PeopleSoft enterprise HRMS.

UNIT – V

Siebel Enterprise Applications - Siebel eBusiness Applications – Siebel Tools – Tables and Columns – Business Component – Business Objects – Applets – Joins – Links – Views – Screens – Configuring applications.

2 MARKS

1. What is Siebel?
2. Name some customers of Siebel?
3. What is Siebel Enterprise Application Integration?
4. What are the Features of Siebel EAI?
5. What are Siebel e-Business applications?
6. What is CRM?
7. Write the three levels of CRM?
8. What is Siebel Tool?
9. Whether Siebel Tools is a programming environment?
10. What are the two windows of Navigation in Siebel Tools ?
11. Define Tables?
12. Define Base Table?
13. Write the Properties of the Table Object Type?
14. Define Data tables?
15. Define an Extension table?
16. Define an Interaction Table?
17. What is Column object?
18. What are the types of Column Objects ?
19. What are the types of Columns?
20. Define Data Columns?
21. Define Extension Columns?
22. Define System Column?
23. What is meant by Business Component?
24. Define Business Object?
25. Define Applets?
26. Write the types of Applets?
27. Define Form Applet?
28. Define List Applet?
29. Define Joins?
30. Define Links?

31. What is Multi value Link?

32. Define Views?

33. Define List Form View?

34. Define Master-detail view?

35. Define Thread Bar?

36. What is Drilldown Behavior in a View?

37. Give the types of Drilldown Behavior?

38. Define screens?

39. Write any Configuring applications for deployment in High Interactivity mode.

11 MARKS

1. Explain CRM Processes in detail.
2. Explain Siebel Tool in detail?
3. Explain in details about Tables.
4. Explain in detail about Extension Tables:
5. Explain Intersection Table with neat architecture.
6. Describe Business Components in detail.
7. Describe in detail about Virtual Business Components
8. Explain Master-Detail Business Components in detail.
9. Explain Business Objects in Detail.
10. Explain Applet and types of applet?
11. Explain in detail about Joins?
12. Describe Links in detail.
13. Describe Views in detail.
14. Describe Screens in detail.

5636177

B.Tech. DEGREE EXAMINATION,
APRIL/MAY 2016.

Sixth Semester

Computer Science and Engineering

ENTERPRISE SOLUTIONS

Time : Three hours — Maximum : 75 marks

PART A — ($10 \times 2 = 20$ marks)

Answer ALL questions.

All questions carry equal marks.

1. List the characteristics of ERP.
2. Cite the manufacturing roots of ERP.
3. How would you take order in SAP R/3?
4. Define Data Dictionary.
5. What is SQL profile parameter?
6. State the function of workflow instance monitor view.
7. List out the benefits of data mover.
8. What are checklists?
9. Cite the responsibilities in Siebel Applications.
10. What does Enterprise Integration Manager use to prevent duplication of records during processing?

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. How ERP has changed the nature of Business process Reengineering? Explain. (11)

Or

12. (a) ERP's seem like a good idea, so why is return on investment so low? (6)

(b) List and explain the ERP Implementation challenges. (5)

UNIT II

13. (a) Difference between ABAP/4 and ABAP/5. (5)

(b) Summarize the functions of Wokbench tools. (6)

Or

14. Explain the Architecture of SA R/3. (11)

UNIT III

15. Draw and explain the Architecture of ORACLE Apps 11i.

Or

16. (a) Acquaint the different functions of EDI. (6)

(b) Compare OLTP and OLAP. (5)

UNIT IV

17. Discuss the different types of tables in people HRMS.

Or

18. (a) Compare pages vs. forms.

(b) Summarize the uses of finan management system.

UNIT V

19. (a) Illustrate the concept of Siebel Tools.

(b) Distinguish between form and list applets

Or

20. (a) Describe the best practices for configu siebel applications.

(b) Write short notes on Siebel objects.

5636177

B.Tech. DEGREE EXAMINATION,
NOVEMBER/DECEMBER 2016.

Sixth Semester

Computer Science and Engineering

ENTERPRISE SOLUTIONS

(2013-2014 onwards)

Time : Three hours — Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

1. Define ERP.
2. Write down any two key issues of ERP.
3. Mention any two differences between SAP R/2 and SAP R/3.
4. Give the structure of ABAP/4 system.
5. Mention any two functions of EDI.
6. What is use of use DW connector?
7. Mention any two differences between pages and forms.

8. Name any two PeopleSoft products.

9. Mention any two Siebel eBusiness applications.

10. What is business object? Give an example.

PART B — $(5 \times 11 = 55$ marks)

Answer ALL questions, ONE from each unit.

All questions carry equal marks.

UNIT I

11. Explain about typical architecture components of ERP in detail.

Or

12. Write short notes on :

(a) Business process re-engineering.

(b) Customer relationship management.

UNIT II

13. Discuss about the architecture of SAP/3 and its modules.

Or

14. Explain the following :

(a) ABAP/4 data dictionary.

(b) Move and Move-Corresponding.

UNIT III

15. Discuss the application framework and file sy... of Oracle Apps 11i briefly.

Or

16. Write and explain the steps involved in creatin... forms and generating reports in Oracle 11i.

UNIT IV

17. Explain how multiple windows are opened i... PeopleSoft.

Or

18. Write down the steps involved in creating HRM... database in PeopleSoft.

UNIT V

19. Discuss about Siebel tools and configuring Siebe... eBusiness applications.

Or

20. Write short notes on :

(a) Business component.

(b) Applets.

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B.Tech. DEGREE EXAMINATION, APRIL/MAY 2017.

Sixth Semester

Computer Science and Engineering

ENTERPRISE SOLUTIONS

(2013 – 14 onwards)

Time : Three hours | Maximum : 75 marks

PART A — ($10 \times 2 = 20$ marks)

Answer ALL questions.

All questions carry equal marks.

1. Describe Enterprise Resource Planning (ERP).

2. What is a Decision Support System?

3. What is Net Weaver?

4. Give the syntax for MOVE and MOVE-CORRESPONDING in ABAP/4.

5. What is workflow analysis in Oracle?

6. What are Oracle Reports?

7. How does PeopleSoft fit in financial management?

8. List any two basic PeopleSoft functionality.

9. Describe an Applet in Siebel.

10. State any two Siebel eBusiness Applications.

PART B — (5 × 11 = 55 marks)

Answer ALL questions, choosing ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Discuss ERP system architecture.

Or

12. Briefly describe Business Process Re-engineering.

UNIT II

13. Discuss the architecture of SAP R/3.

Or

14. Discuss the structure of ABAP/4 program.

UNIT III

15. Discuss Oracle Application framework in detail.

Or

16. Describe the functions of Oracle Electronic Data Interchange.

UNIT IV

17. Discuss in detail PeopleSoft HRMS.

Or

18. Describe and differentiate Pages and Fo People Soft.

UNIT V

19. Discuss Siebel tools in brief.

Or

20. Describe and differentiate Business Com and Business Objects in Siebel En Applications.

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B.Tech. DEGREE EXAMINATION, NOVEMBER 2017.

Sixth Semester

Computer Science and Engineering

ENTERPRISE SOLUTIONS

(2013-2014 onwards)

Time : Three hours — Maximum : 75 marks

PART A — ($10 \times 2 = 20$ marks)

Answer ALL questions.

All questions carry equal marks.

1. Define ERP.

2. What is BPR?

3. What is ABAP 4?

4. What are the characteristics of SAP R/3?

5. What is meant by workflow analysis?

6. What are DW connectors?

7. List out the PeopleSoft products.

8. Distinguish between pages and forms in PeopleSoft.

9. What is Siebel?

10. What are Applets?

PART B — ($5 \times 11 = 55$ marks)

Answer ALL questions.

All questions carry equal marks.

UNIT I

11. (a) Discuss the advantages of ERP. (5)

(b) Explain the ERP system architecture. (6)

Or

12. Explain the following :

(a) Supply Chain Management. (6)

(b) Executive Support System. (5)

UNIT II

13. Explain the architecture of SAP R3.

Or

14. Explain in brief the various statements in ABAP/4.

UNIT III

15. Explain with an example creating forms us PL/SQL.

Or

16. Discuss on Oracle Database.

UNIT IV

17. Explain the basic PeopleSoft functionality.

Or

18. Write a detailed note on PeopleSoft FIRMS.

UNIT V

19. Discuss on Siebel eBusiness Applications.

Or

20. Write a note on Siebel tools.

RAJIV GANDHI COLLEGE OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
CS T61 – ENTERPRISE SOLUTIONS
INTERNAL I

YEAR/SEM/SEC: Y3/S6 **MAX. MARKS: 75**

Y

PART –A

(Answer all the Questions) **10*2=20**

1. What is ERP and how it is related to SAP? L1, CO1
2. What is Decision Support System? L1, CO1
3. What is OLAP? L1, CO1
4. Sate the various business modules in ERP system. L2, CO1
5. What are ERP packages? L1, CO1
6. What is Net Weaver? L1, CO1
7. List out the general SAP R/3 modules. L1, CO1
8. What is meant by business warehouse? L1, CO1
9. State the use of CRM. L2, CO2
10. What are the characteristics of SAP R/3? L1, CO1

PART-B **5* 11 = 55**

ANSWER ALL THE QUESTIONS

11. Discuss the need issues and characteristics of ERP. L2, CO2

Or

12. Explain about typical architecture components of ERP in detail. L1, CO2

13. Explain the following L1, CO1
    A. Business Process Re-Engineering
    B. Supply Chain Management

Or

14. Explain the following L1, CO1
    A. Customer Relationship Management
    B. OLAP and its types

15. Discuss the architecture of SAP R/3. L2, CO2

Or

16. Discuss the importance of SAP R/3. How is it carried out? L2, CO2

RAJIV GANDHI COLLEGE OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
CS T61 – ENTERPRISE SOLUTIONS
INTERNAL I

YEAR/SEM/SEC: Y3/S6 **MAX. MARKS: 75**

PART –A

(Answer all the Questions) **10*2=20**

1. What is ERP and how it is related to SAP? L1, CO1
2. What is Decision Support System? L1, CO1
3. What is OLAP? L1, CO1
4. Sate the various business modules in ERP system. L2, CO1
5. What are ERP packages? L1, CO1
6. What is Net Weaver? L1, CO1
7. List out the general SAP R/3 modules. L1, CO1
8. What is meant by business warehouse? L1, CO1
9. State the use of CRM. L2, CO2
10. What are the characteristics of SAP R/3? L1, CO1

PART-B **5* 11 = 55**

ANSWER ALL THE QUESTIONS

11. Discuss the need issues and characteristics of ERP. L2, CO2

Or

12. Explain about typical architecture components of ERP in detail. L1, CO2

13. Explain the following L1, CO1
    A. Business Process Re-Engineering
    B. Supply Chain Management

Or

14. Explain the following L1, CO1
    A. Customer Relationship Management
    B. OLAP and its types

15. Discuss the architecture of SAP R/3. L2, CO2

Or

16. Discuss the importance of SAP R/3. How is it carried out? L2, CO2

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CS T61 – ENTERPRISE SOLUTIONS

INTERNAL II

YEAR/SEM/SEC: Y3/S6 **MAX. MARKS: 75** **Y**

PART –A

(Answer all the Questions) **10*2=20**

1. Describe ABAP/4 data dictionary. L2, CO2
2. Give the syntax for MOVE and MOVE- CORRESPONDING in ABAP/4 L1, CO1
3. Mention any two differences between SAP R/2 and SAP R/3. L1, CO1
4. Give the structure of ABAP/4 system. L1, CO1
5. What are domains and data elements? L1, CO1
6. Define SQL Loader? L1, CO1
7. List oracle report forms. L1, CO1
8. What is DW Connectors? L1, CO1
9. What is OLTP?Give example. L1, CO1
10. Explain the Functions of EDI? L2, CO2

PART-B

5* 11 = 55

ANSWER ALL THE QUESTIONS

11. Discuss in detail Architecture of SAP R/3. L2, CO2

Or

12. Elaborate in detail various components of ABAP/4 Workbench. L2, CO2

13. Explain the following L2, CO2
    A. Disclose the ABAP/4 data dictionary.
    B. Illustrate the structure of ABAP/4 program.

Or

14. Discuss the functions of EDI. L2, CO2

15. Draw and explain the Architecture of ORACLE Apps 11i. L2, CO2

Or

16. Explain with an example creating forms using PL/SQL. L2, CO2

RAJIV GANDHI COLLEGE OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CS T61– ENTERPRISE SOLUTIONS

MODEL EXAM

YEAR/SEM/SEC: Y3/S6/A&B MAX. MARKS: 75 TIMES: 3 HRS

PART – A பகுதி A (2*10 = 20)

Answer all the questions:

அனைத்து கேள்விகளுக்கும் பதிலளிக்கவும்

1. Define ERP. L1, CO1
   ஈஆர்பியை வரையறுக்கவும்.
2. List out the characteristics of ERP. L1, CO1
   ஈஆர்பியின் பண்புகளை பட்டியலிடுங்கள்.
3. What is ABAP/4 data dictionary? L1, CO1
   ABAP/4 தரவு அகராதி என்றால் என்ன?
4. Write the SAP modules? L1, CO2
   SAP தொகுதிகளை எழுதவா? L1, CO2
5. What is meant by workflow analysis? L1, CO1
   பணிப்பாய்வு பகுப்பாய்வு என்றால் என்ன?
6. Explain the function of EDI. L2, CO2
   EDI இன் செயல்பாட்டை விளக்குங்கள்
7. List out the PeopleSoft Products. L1, CO1
   PeopleSoft தயாரிப்புகளை பட்டியலிடுங்கள்.
8. What are the records field events available in PeopleSoft? L1, CO1
   பீப்பிள்சாஃப்டில் என்ன பதிவுகள் கள நிகழ்வுகள் உள்ளன?
9. What are Applets? L1, CO1
   ஆப்பிள்கள் என்றால் என்ன?
10. What are Siebel tools? L1, CO1
    சீபல் கருவிகள் என்றால் என்ன?

PART -B பகுதி-B (5*11=55)

UNIT-I

11. Explain about typical architecture components of ERP.
    ஈஆர்பியின் வழக்கமான கட்டிடக்கலை கூறுகள் பற்றி விளக்கவும். L2, CO2

(Or)

12. a) Explain the ERP system architecture. L2, CO2
    ஈஆர்பி அமைப்பு கட்டமைப்பை விளக்குங்கள்
    b) Explain Business Process Re-Engineering.
    வணிக செயல்முறை மறு-பொறியியலை விளக்குங்கள்

UNIT-II

13. Discuss the architecture of SAP R/3. L1, CO1

SAP R/3 இன் கட்டமைப்பைப் பற்றி விவாதிக்கவும்.

(Or)

14. Describe in detail Oracle suite's Applications Framework. L2, CO2

ஆரக்கிள் தொகுப்பின் பயன்பாடுகள் கட்டமைப்பை விரிவாக விவரிக்கவும்.

UNIT-III

15. Describe in detail Oracle suite's Applications Framework. L2, CO2

ஆரக்கிள் தொகுப்பின் பயன்பாடுகள் கட்டமைப்பை விரிவாக விவரிக்கவும்.

(Or)

16. Discuss various functions of EDI. L2, CO2

EDI இன் பல்வேறு செயல்பாடுகளைப் பற்றி விவாதிக்கவும்.

UNIT-IV

17. Discuss in detail PeopleSoft HRMS. L2, CO2

PeopleSoft HRMS பற்றி விரிவாக விவாதிக்கவும்.

(Or)

18. Describe and differentiate pages and forms in PeopleSoft. L2, CO2

PeopleSoft இல் பக்கங்கள் மற்றும் படிவங்களை விவரிக்கவும் மற்றும் வேறுபடுத்தவும்.

UNIT-V

19. Write a note on Siebel tools. L1, CO2

சீபல் கருவிகள் பற்றிய குறிப்பை எழுதவும்.

(Or)

20. Describe Business Component in detail L2, CO2

வணிகக் கூறுகளை விரிவாக விவரிக்கவும்

RAJIV GANDHI COLLEGE OF ENGINEERING AND TECHNOLOGY
Pondy Cuddalore Main Road, SH 49,
Kirumampakkam, Puducherry 607403

REMEDIAL CLASS ATTENDANCE SHEET

Date of Remedial Class: 15·05·2024 Time: 3·00 to 4·00

Year and Department: Y3 / CSE

Name of the Class Teacher: S. Suria

Name of the Subject: Enterprise solutions

Topic Covered: SAP, Oracle, PL/SQL

Sl. No.	Reg. No.	Name of the Student	Signature of the student	Remarks (Test marks/ Reasons for Absence)
1.	21TD0002	Abishek · S	S. Abishek	19
2.	21TD0048	Mohammed Nehal · T	Nehal	18
3.	21TD0051	Muhammed Rafin Ashraf · T	Rafin	21
4.	21TD0054	Oviya · B	B. Oviya	22
5.	21TD0072	Sathiya · A	Sathiya	18
6.	21TD0075	Shanmugadasan M	Shan	19
7.	21TD0089	Vetrivelan · R.K	VRK	20
8.	21TD2011	Vignesh · S	Vignesh S	18
9.				
10.				
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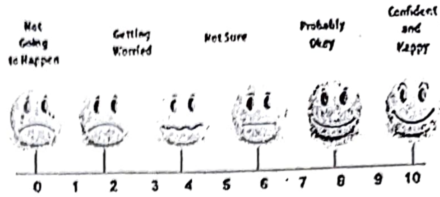
19.				
20.				
21.				
22.				
23.				
24.				
25.				

No of Students Present: 8

Suria.
Remedial Class Teacher

SIGNATURE OF HOD



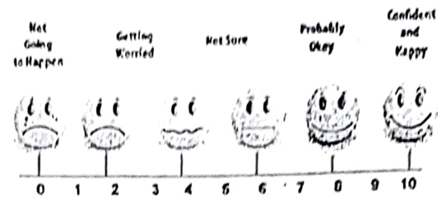
Students Feedback form after examination

Name of the Subject : Enterprise Solutions

Date / Session : 23.5.2024

Name of the Faculty Member : Suria . S

		0	1	2	3	4	5	6	7	8	9	10
1.	What was the preparation level for this examination										✓	
2.	Will you pass the exam									✓		
3.	Will you achieve good grades									✓		
4.	Was the lessons taught by your teacher were useful for you to pass the subject									✓		
5.	Were the notes/materials provided by the subject teacher was useful for you to write this examination										✓	
6.	Do you feel that attending the classes regularly was useful for doing well in the examinations									✓		
7.	Were the 2 marks questions asked in the Class test/internal examinations has appeared in the Question paper									✓		
8.	Were the 11 marks questions asked in the Class test/internal examinations has appeared in the Question paper										✓	
9.	How do you rate your performance in your internal examination									✓		
10.	How do you rate the teaching performance of your teacher									✓		
11.	How do you rate your attention towards regular classes									✓		
12.	If you are regular to college, how do you rate your attendance percentage in the scale of 10 for this subject										✓	
13.	How do you rate the question paper in the scale of 10 (tough 0 to very easy 10)									✓		
14.	Did you follow other college materials or notes	✓										



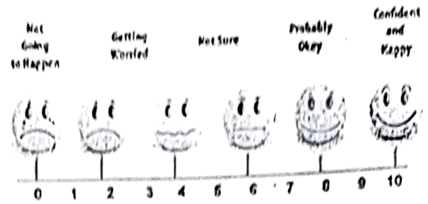
Students Feedback form after examination

Name of the Subject : Enterprise Solutions

Date / Session : 23.5.2024

Name of the Faculty Member : Sruthi . S

		0	1	2	3	4	5	6	7	8	9	10
1.	What was the preparation level for this examination							✓				
2.	Will you pass the exam							✓				
3.	Will you achieve good grades								✓			
4.	Was the lessons taught by your teacher were useful for you to pass the subject										✓	
5.	Were the notes/materials provided by the subject teacher was useful for you to write this examination									✓		
6.	Do you feel that attending the classes regularly was useful for doing well in the examinations								✓			
7.	Were the 2 marks questions asked in the Class test/internal examinations has appeared in the Question paper								✓			
8.	Were the 11 marks questions asked in the Class test/internal examinations has appeared in the Question paper									✓		
9.	How do you rate your performance in your internal examination								✓			
10.	How do you rate the teaching performance of your teacher									✓		
11.	How do you rate your attention towards regular classes								✓			
12.	If you are regular to college, how do you rate your attendance percentage in the scale of 10 for this subject									✓		
13.	How do you rate the question paper in the scale of 10 (tough 0 to very easy 10)								✓			
14.	Did you follow other college materials or notes	✓										



Students Feedback form after examination

Name of the Subject : Enterprise Solutions

Date / Session : 23.5.2024

Name of the Faculty Member : Suria. S

		0	1	2	3	4	5	6	7	8	9	10
1.	What was the preparation level for this examination										✓	
2.	Will you pass the exam										✓	
3.	Will you achieve good grades									✓		
4.	Was the lessons taught by your teacher were useful for you to pass the subject									✓		
5.	Were the notes/materials provided by the subject teacher was useful for you to write this examination										✓	
6.	Do you feel that attending the classes regularly was useful for doing well in the examinations									✓		
7.	Were the 2 marks questions asked in the Class test/internal examinations has appeared in the Question paper									✓		
8.	Were the 11 marks questions asked in the Class test/internal examinations has appeared in the Question paper										✓	
9.	How do you rate your performance in your internal examination									✓		
10.	How do you rate the teaching performance of your teacher									✓		
11.	How do you rate your attention towards regular classes										✓	
12.	If you are regular to college, how do you rate your attendance percentage in the scale of 10 for this subject									✓	✓	
13.	How do you rate the question paper in the scale of 10 (tough 0 to very easy 10)										✓	
14.	Did you follow other college materials or notes	✓										